

Summaries on Research Articles

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1 A squeezed mechanical oscillator with millisecond quantum decoherence

Challenges presented :

1. Engineering strong coupling of mechanical oscillators to an auxiliary degree of freedom.
2. Good mechanical isolation from environment i.e low quantum decoherence (thermal decoherence and dephasing).

What they do to overcome challenges :

1. Design a superconducting circuit based optomechanical platform.
2. Achieve low quantum decoherence and large optomechanical coupling that allows preparing the lowest possible quantum energy ground level and squeezed states with high fidelity.
3. They report a thermal decoherence rate of 20.50Hz (corresponding to 130 quanta/s–1 motional heating rate) and a pure dephasing rate of 0.09 Hz, showing that quantum decoherence is dominated by thermal decoherence.

Quantum decoherence can be characterized with two independent rates:

- **Thermal decoherence rate:**

$$\Gamma_{\text{th}} = (n_m^{\text{th}} + 1)\Gamma_m$$

where n_m^{th} is the thermal bath occupation, and Γ_m is the bare mechanical damping rate. This rate describes how quickly phonons are exchanged with the thermal environment.

- **Pure dephasing rate:**

$$\Gamma_{\phi}$$

which arises from mechanical frequency fluctuations and represents phonon-number-conserving interactions with the environment.

Resolved-Sideband Regime (“Good-Cavity Limit”)

In the resolved-sideband regime, the cavity linewidth (decay rate, κ) is much smaller than the mechanical oscillation frequency Ω_m :

$$\Omega_m > \kappa$$

This condition allows the system to resolve individual sidebands in the frequency spectrum, corresponding to processes where photons add or remove vibrational quanta (phonons). Sideband-resolved systems enable efficient cooling of the mechanical oscillator to its quantum ground state by selectively driving the anti-Stokes sideband (cooling transition) and suppressing the Stokes sideband (heating transition). This is the optimal regime for quantum control, including ground-state cooling, squeezing, and coherent state manipulation.

Non-Resolved-Sideband Regime

In the non-resolved-sideband regime, the cavity linewidth is comparable to or larger than the mechanical frequency:

$$\kappa \gtrsim \Omega_m \quad \text{or} \quad \kappa \gg \Omega_m$$

Here, the optical sidebands overlap due to the broad cavity linewidth, making it difficult to selectively address only the cooling sideband. As a result, both cooling and heating processes occur simultaneously with similar probability. This limits the minimum achievable phonon occupation and reduces the fidelity of quantum state preparation. This regime is common in systems with low mechanical frequencies or low-quality optical/microwave cavities. While some protocols are designed for this regime, it generally cannot reach the same level of quantum control as the resolved-sideband regime.

Cavity-Free Regime

In standard cavity optomechanics, the mechanical element is coupled to a resonant optical or microwave cavity, which enhances the interaction strength via resonance. In the cavity-free regime, no such cavity is present. The mechanical oscillator interacts directly with free-space light or other quantum systems without resonance enhancement. These systems may offer new opportunities for integration and miniaturization, but the optomechanical coupling is typically much weaker and less tunable. As a result, implementing quantum protocols like ground-state cooling or squeezing is much more challenging in cavity-free systems.

Device Design :

1. Vacuum-gap capacitor: A capacitor where the dielectric (insulating) layer between the plates is vacuum (or air), not a solid material. This reduces dielectric losses. The vertical separation between the top (moving) plate and the bottom (fixed) plate is 180 nanometers.
2. Top plate suspended: The upper plate of the capacitor is free to move—it is not rigidly fixed. This allows it to act as a mechanical oscillator.
3. Circular trench: A shallow ring-shaped depression etched into the substrate. The top plate (a drumhead-like structure) is suspended over this trench.

4. The capacitor is connected in parallel (shunted) with a spiral-shaped superconducting inductor. Together, they form an LC circuit. It resonates at frequency $\omega_c/2\pi$ 5.55 Hz and has a decay rate $\kappa/2\pi=250\text{kHz}$. This is inductively coupled to a waveguide.

Measurements :

1. *Mechanical Ring-down Measurement and Key Parameters*

- *Mechanical ring-down measurement:*

- The mechanical oscillator is initially excited.
- It is then allowed to freely decay.
- The decay time reveals the mechanical energy loss.

- *Mechanical Quality Factor:*

$$Q_m = \frac{\Omega_m}{\Gamma_m} = 40 \times 10^6$$

- Extremely high mechanical quality factor.
- Indicates the oscillator can vibrate for *40 million cycles* before losing significant energy.

- *Parameters of the Fundamental Drumhead Mode:*

- Frequency: $\frac{\Omega_m}{2\pi} = 1.8\text{MHz}$
- Bare dissipation rate: $\frac{\Gamma_m}{2\pi} = 45\text{mHz}$

- *Vacuum Optomechanical Coupling Rate g_0 :*

- Represents the cavity frequency shift due to zero-point motion of the mechanical oscillator.
- Value: $\frac{g_0}{2\pi} = 13.4\text{Hz}$
- Although small, this shift per phonon is measurable.
- Smaller gap sizes lead to larger g_0 values.

- *Frequency Fluctuation and Dephasing:*

- Observed frequency fluctuation: $< 0.1\text{Hz}$
- Sets an upper bound for mechanical dephasing.
- Inferred from the power spectral density (PSD) of the thermomechanical sideband that was averaged over an hour.
- Subtracted the resolution bandwidth of 1 Hz to isolate the fluctuation.

2. High mechanical Q factor and strong optomechanical coupling enables effective sideband cooling, a technique that removes phonons (vibrational quanta) by using red-detuned light to extract energy. The goal is to bring the oscillator to its quantum ground state, i.e., nearly zero phonons. The final phonon number under sideband cooling is given by:

$$n_m = \frac{n_m^{\text{th}}}{1 + \mathcal{C}} + \frac{1}{1 + \mathcal{C}} n_c$$

Where:

- n_m : Final phonon occupation number (after cooling)

- n_m^{th} : Initial thermal phonon occupation (before cooling)
- $\mathcal{C}$: Optomechanical co-operativity (dimensionless cooling strength)
- n_c : Cavity photon number due to thermal heating
- $\frac{n_m^{\text{th}}}{1 + \mathcal{C}}$: Cooling of the mechanical mode
- $\frac{1}{1 + \mathcal{C}} n_c$: Residual heating from the cavity itself

About Co-operativity ($\mathcal{C}$)

- $\mathcal{C} = \frac{4n_p g_0^2}{\kappa \Gamma_m}$, where:
 - n_p : Number of pump photons in the cavity
 - g_0 : Single-photon optomechanical coupling strength
 - κ : Cavity energy decay rate (linewidth)
 - Γ_m : Bare damping rate
- Higher $\mathcal{C}$ means:
 - Stronger cooling
 - Lower n_m

Cavity Heating and Photon Number (n_c)

- n_c : Number of thermal photons in the cavity (typically microwave photons)
- Cavity couples to a thermal bath:
 - Intrinsic cavity loss rate: κ_0
 - Thermal photon number in the bath: n_c^{th}
- More $\kappa_0 \Rightarrow$ more unwanted photons $\Rightarrow$ more heating

Pump-Induced Heating

- While increasing $\mathcal{C}$ (by stronger pump) improves cooling,
- Too strong a pump causes heating due to:
 - Energy dissipation in resistive parts
 - Electromagnetic heating or local hotspots
- This increases n_c , which limits the lowest achievable n_m
- In the high- $\mathcal{C}$ limit ($\mathcal{C} \rightarrow \infty$):

$$n_m \approx n_c$$

Cavity Heating Mechanism and Solution

- Dominant source: Thin native oxide layer at galvanic connections between circuit layers
- This oxide introduces resistance, which:
 - Dissipates energy when the strong pump current flows

- Heats the cavity
- Solution: Oxide removal (e.g., via etching or better fabrication)
- Result: Achieved $n_c \approx 0.05$ photons $\Rightarrow$ nearly no heating, even under strong cooling
- This enabled cooling the mechanical oscillator close to its quantum ground state
- Goal: Measure the mechanical phonon occupation number n_m near its quantum ground state.
- *Method: Sideband Asymmetry*
 - Quantum effect: In the ground state, anti-Stokes sideband (phonon annihilation) is stronger than Stokes (phonon creation).
 - Allows determination of n_m based on spectral asymmetry.
- *Out-of-loop calibration:*
 - Independent of the main feedback/control loop.
 - Ensures accurate, unbiased measurement.
- *Experimental Setup:*
 - A strong cooling tone (red-detuned by Ω_m) cools the mechanical mode.
 - Two weak probe tones:
 - * Blue-detuned probe $\rightarrow$ generates **Stokes** sideband (adds phonons).
 - * Red-detuned probe $\rightarrow$ generates **anti-Stokes** sideband (removes phonons).
 - Probe tones are spaced by a few kHz around cavity resonance to isolate their sidebands.
- *Signal Amplification:*
 - Uses a Josephson Traveling-Wave Parametric Amplifier (JTWPA).
 - Adds ~ 0.3 quanta of thermal noise: $n_{T,\text{add}} \approx 0.3$.
 - Total added noise: ≈ 0.9 quanta ($\approx \frac{1}{2} + 0.3$).
 - Gain: $G_T = 25$ dB $\Rightarrow$ Signal boosted by $\sim 300\times$.
 - This suppresses noise from the HEMT amplifier (adds ~ 8 quanta).
- *Spectral Measurements:*
 - Fit the power spectral density (PSD) of each signal with Lorentzian functions.
 - Extract the following:
 - * P_c : Thermal cavity noise power.
 - * P_b : Power in the blue-sideband (Stokes).
 - * P_r : Power in the red-sideband (anti-Stokes).
- *Ideal Sideband Relations (without cavity heating):*

$$P_b \propto n_m + 1, \quad P_r \propto n_m$$

- *Problem: Cavity Heating (n_c)*
 - Cavity has residual thermal photons n_c due to heating.
 - Breaks clean sideband asymmetry: need to know n_c to extract n_m .
- *Normalization and Analytical Model:*

- Normalize sidebands by P_c to cancel unknown gain:

$$\frac{P_b}{P_c} = \frac{\Gamma_b}{\kappa} \cdot \frac{n_c}{n_m + 1 + 2n_c}, \quad \frac{P_r}{P_c} = \frac{\Gamma_r}{\kappa} \cdot \frac{n_c}{n_m - 2n_c}$$

- Where:
 - * Γ_b, Γ_r : Optomechanical damping/amplification rates from blue/red probes.
 - * κ : Total cavity decay rate.
- These equations can be solved to extract both n_m and n_c .

- *Calibration-Free Operation:*

- Once calibrated, n_m and n_c can be found using:
 - * PSD from the cooling pump sideband.
 - * PSD of the cavity thermal emission (probes off).
- Avoids backaction from the blue probe (which adds phonons).
- Leads to more accurate, undisturbed measurements.

- *Results:*

- By increasing co-operativity $\mathcal{C}$, they map:

$$n_m(\mathcal{C}), \quad n_c(\mathcal{C})$$

- Achieved minimum phonon occupation:

$$n_m^{\min} \approx 0.068 \Rightarrow \text{Ground state probability: 93\%}$$

- Ground state energy suppression: ≈ -8.7 dB
- Limiting factor: Residual cavity heating (not technical noise)

3. Thermal Decoherence Measurement in a Mechanical Oscillator

- They study how a mechanical oscillator loses quantum coherence over time due to thermalization.
- *Protocol:*
 - Cool the mechanical oscillator close to its quantum ground state (*very low phonon number*).
 - Turn off control and let it evolve freely for a time τ_{ev} .
 - During this time, it thermalizes by absorbing phonons from the environment.
- *Detection method:*
 - Use a *blue-detuned pump* (above the cavity frequency) to read out the mechanical motion.
 - This induces *parametric amplification*, increasing signal strength by 50 dB.
 - Motion quadratures are encoded in sidebands near the cavity resonance.
- *Quantum state tomography:*
 - Perform *prepare–evolve–measure* cycles, each time measuring different quadratures.
 - Reconstruct the full quantum state of the oscillator from repeated measurements.
- *Calibration:*

- Use known input states (with calibrated phonon number) based on sideband asymmetry.
- Extract the optical added noise: $n_{\text{add}}^{\text{opt}} \approx 0.8$ phonons.
- Confirms the readout measurement adds less than 1 phonon of uncertainty — *very low noise*.
- *Thermalization fit:*
 - Plot phonon number vs evolution time τ_{ev} .
 - Fit to an exponential curve to extract the mechanical bare damping rate.
 - Measured *bare damping rate*: 80 mHz, matching ring-down experiments. This agrees with ring down measurements
- *Thermal decoherence rate:*
 - From short-time behavior, extract decoherence rate including interaction with thermal bath. This is faster than bare damping rate because it accounts both energy loss and gain due to interaction with thermal environment.

$$\Gamma_{\text{th}} = 20.5 \text{ Hz}$$

- Quantum coherence decays on a timescale of:

$$T_1 = \frac{1}{\Gamma_{\text{th}}} \approx 7.7 \text{ ms}$$

- Represents extremely long coherence for a mechanical quantum system.

4. Dissipative Optomechanical Squeezing of a Mechanical Oscillator

- *Quantum squeezed state generation:*
 - Achieved using dissipative optomechanical squeezing.
 - Method involves driving the cavity with two laser tones, symmetrically detuned around the cavity resonance.
- *Squeezing performance:*
 - One mechanical quadrature is squeezed to -2.7 dB below the vacuum noise level.
 - The orthogonal quadrature is anti-squeezed to $+8.1 \text{ dB}$, preserving Heisenberg uncertainty. These are signatures of a genuine quantum squeezed state.
- *Time evolution of the squeezed state:*
 - The squeezed variance remains below the quantum zero-point level for up to 2 milliseconds.
 - Demonstrates very slow decoherence and high stability.
- *Dephasing rate:*
 - Inferred mechanical dephasing rate of 0.09 Hz.
 - Indicates extremely low phase noise and oscillator stability.

Doubts:

1. In the resolved-sideband regime, the quantum backaction (unavoidable disturbance added to the mechanical system by quantum measurement) does not influence the final phonon occupation. Why?

2 Direct observation of deterministic macroscopic entanglement

The challenge:

Maintaining and verifying fragile quantum entanglement in comparatively massive mechanical objects, which are subject to decoherence, thermal noise, and experimental imperfections.

The solution:

1. Pulsed electromechanical coupling for deterministic entanglement generation. Instead of continuous-wave driving, they use short, controlled microwave pulses to manipulate the mechanical oscillators states. Pulsed control reduces the interaction time with the environment, limiting decoherence during the entanglement process. Their results demonstrate pulsed, time-domain control of three important building blocks for CV quantum information processing and quantum communication: state initialization, entanglement, and measurement.
2. And they perform quantum state tomography to directly observe entanglement between two mechanical drumheads (70 picograms) and directly observe their state. Quantum-limited measurement of the position and momentum quadratures of both mechanical oscillators. By repeating these measurements, they characterize their joint covariance matrix. This tomography (Tomography means reconstructing the quantum state based on measurement data. It's like a quantum 3D scan of the system's full state) demonstrates clear evidence of continuous-variable (CV) entanglement (Continuous-variable refers to observables like position and momentum) in the measurement signals, without noise subtraction. The covariance matrix is a complete description of a Gaussian continuous-variable quantum state. It tells you how the two systems are statistically related, including whether they are entangled. Let the quadrature operators for oscillator A be $\hat{x}_A$, $\hat{p}_A$, and for oscillator B be $\hat{x}_B$, $\hat{p}_B$.

Define the operator vector:

$$\hat{R} = (\hat{x}_A, \hat{p}_A, \hat{x}_B, \hat{p}_B)^\top$$

The covariance matrix V is a 4×4 real symmetric matrix with elements:

$$V_{ij} = \frac{1}{2} \langle \Delta \hat{R}_i \Delta \hat{R}_j + \Delta \hat{R}_j \Delta \hat{R}_i \rangle$$

where

$$\Delta \hat{R}_i = \hat{R}_i - \langle \hat{R}_i \rangle$$

The covariance matrix can be written in block form as:

$$V = \begin{pmatrix} V_A & C \\ C^\top & V_B \end{pmatrix}$$

where V_A and V_B are 2×2 covariance matrices for subsystems A and B , and C encodes correlations between them.

Expanded matrix form:

$$V = \begin{pmatrix} \langle \Delta \hat{x}_A^2 \rangle & \frac{1}{2} \langle \{ \Delta \hat{x}_A, \Delta \hat{p}_A \} \rangle & \langle \Delta \hat{x}_A \Delta \hat{x}_B \rangle & \langle \Delta \hat{x}_A \Delta \hat{p}_B \rangle \\ \frac{1}{2} \langle \{ \Delta \hat{x}_A, \Delta \hat{p}_A \} \rangle & \langle \Delta \hat{p}_A^2 \rangle & \langle \Delta \hat{p}_A \Delta \hat{x}_B \rangle & \langle \Delta \hat{p}_A \Delta \hat{p}_B \rangle \\ \langle \Delta \hat{x}_B \Delta \hat{x}_A \rangle & \langle \Delta \hat{x}_B \Delta \hat{p}_A \rangle & \langle \Delta \hat{x}_B^2 \rangle & \frac{1}{2} \langle \{ \Delta \hat{x}_B, \Delta \hat{p}_B \} \rangle \\ \langle \Delta \hat{p}_B \Delta \hat{x}_A \rangle & \langle \Delta \hat{p}_B \Delta \hat{p}_A \rangle & \frac{1}{2} \langle \{ \Delta \hat{x}_B, \Delta \hat{p}_B \} \rangle & \langle \Delta \hat{p}_B^2 \rangle \end{pmatrix}$$

Example: Two-Mode Squeezed Vacuum State

The covariance matrix for a two-mode squeezed vacuum state is:

$$V = \begin{pmatrix} \cosh(2r) & 0 & \sinh(2r) & 0 \\ 0 & \cosh(2r) & 0 & -\sinh(2r) \\ \sinh(2r) & 0 & \cosh(2r) & 0 \\ 0 & -\sinh(2r) & 0 & \cosh(2r) \end{pmatrix}$$

where r is the squeezing parameter.

- Diagonal terms represent variances of each quadrature.
- Off-diagonal terms represent quantum correlations (entanglement).
- The entangled state observed in the experiment demonstrates strong correlations between two systems that appear quite different in nature. The system is composed of two harmonic oscillators, and its state can be characterized using the first and second moments (means and variances) of the quadrature operators.
- The quadratures are denoted as X_j and P_j for each oscillator ($j = 1, 2$), which are analogous to position and momentum operators. These quadratures are dimensionless and satisfy the canonical commutation relations:

$$[X_j, P_j] = i$$

The quadratures are derived from the mechanical position and momentum operators and serve as normalized observables in the phase space description.

- After the entanglement is generated, the X -quadrature of each oscillator follows a Gaussian probability distribution. This distribution has a variance significantly larger than the zero-point variance (i.e., it is "hot" in terms of uncertainty compared to the ground state).
- However, when comparing X_1 and X_2 , the values are found to be highly correlated. In contrast, the P -quadratures (P_1 and P_2) are found to be strongly anticorrelated. Although these correlations are striking, they alone could still be explained by classical (non-quantum) correlations. To confirm that these correlations arise due to quantum entanglement, the Simon-Duan inseparability criterion is used.
- This criterion uses the covariance matrix of the system to determine whether the correlations are quantum in nature. The covariance matrix is defined for the vector of quadrature operators:

$$\vec{S} = (X_1, P_1, X_2, P_2)$$

- The covariance matrix C is a 4×4 matrix with elements defined as:

$$C_{jk} = \frac{1}{2} \langle (S_j - \langle S_j \rangle)(S_k - \langle S_k \rangle) + (S_k - \langle S_k \rangle)(S_j - \langle S_j \rangle) \rangle$$

where $\langle \cdot \rangle$ denotes the quantum expectation value.

- This definition ensures that C captures both the variances and covariances of the quadrature observables.

- To quantify the entanglement, the smallest symplectic eigenvalue $\tilde{v}$ of the partially transposed covariance matrix is computed.
- The partial transpose is performed with respect to one of the oscillators and modifies the sign of one quadrature (typically $P_2 \rightarrow -P_2$).
- According to the Simon-Duan criterion, the two-mode state is entangled if:

$$\tilde{v} < \frac{1}{2}$$

- This threshold value corresponds to the vacuum (zero-point) fluctuation level, where the variance of each quadrature in the ground state is $1/2$.
- A value of $\tilde{v} < 1/2$ cannot be explained by any classical correlations and therefore confirms genuine quantum entanglement between the two oscillators.
- To extract the full covariance matrix of the two-oscillator system with minimal assumptions, all four quadratures of motion are measured in each single experimental run.
- These quadratures are:

$$\{X_1, P_1, X_2, P_2\}$$

which fully describe the state of both oscillators in phase space.

- The measurement method is analogous to heterodyne detection used in quantum optics, where both quadratures of a mode are measured simultaneously. Compared to homodyne detection (which measures only one quadrature at a time), this heterodyne-like method allows more accurate estimation of the covariance matrix.
- The inefficiencies in the measurement setup can be modeled using an effective beam splitter model. In this model, each measured quadrature s_j (where $S_j \in \{X_1, P_1, X_2, P_2\}$) is a mixture of the true quadrature S_j and added vacuum noise x_j :

$$s_j = \sqrt{\eta_j} S_j + \sqrt{1 - \eta_j} x_j$$

- Here, η_j is the measurement efficiency for oscillator j , and x_j is a Gaussian random variable with:

$$\langle x_j \rangle = 0, \quad \langle x_j^2 \rangle = \frac{1}{2}$$

which corresponds to vacuum noise. The random variables x_j and x_k are statistically independent for $j \neq k$.

- Once the measurement chain is calibrated (i.e., the efficiencies η_j are known), the measured variables s_j can be used to reconstruct the covariance matrix of the system.
- From the reconstructed covariance matrix, one can compute the minimal symplectic eigenvalue $\tilde{v}_{\text{meas}}$ associated with the measured state.

- For the quantum states considered here, it is known that:

$$\tilde{v}_{\text{meas}} < \frac{1}{2} \iff \tilde{v} < \frac{1}{2}$$

so either condition confirms entanglement. This equivalence holds more directly if the quantum state is symmetric under the exchange of the two oscillators and experiences symmetric loss.

- In that case, the measured symplectic eigenvalue is related to the true eigenvalue $\tilde{v}$ by:

$$\tilde{v}_{\text{meas}} = \eta \tilde{v} + (1 - \eta) \frac{1}{2}$$

- This expression reflects that vacuum noise degrades the measurement in proportion to the inefficiency $1 - \eta$. As the efficiency η decreases, $\tilde{v}_{\text{meas}}$ approaches $\frac{1}{2}$, making it harder to confirm that $\tilde{v} < \frac{1}{2}$ with statistical confidence.
- In other words, low measurement efficiency leads to poor entanglement certification because the added vacuum noise obscures quantum correlations.

Device Design and Coupling technique:

- Two mechanical oscillators are fabricated from lithographically patterned thin-film aluminum membranes shaped like drums, each with a mass of approximately 70 picograms, suspended above a sapphire substrate.
- The fundamental mechanical modes of these drums: the left drum vibrates at frequency $f_{m,1} = 10.9\text{MHz}$, and the right drum at $f_{m,2} = 15.9\text{MHz}$. The motion of these drums is manipulated and measured using electromechanical techniques. Both drums are embedded in a single microwave resonator, called the cavity, whose resonance frequency $f_c = 6.0806\text{GHz}$ shifts depending on the drums' motion.
- A microwave pulse reflected off the cavity applies forces to the drums and encodes the amplitudes of their motion quadratures into Doppler-shifted sidebands. The carrier frequency of the incoming microwave pulse determines whether the drums are cooled, entangled, or measured.
- To entangle the two drums, they simultaneously irradiate the cavity with two microwave pulses. One pulse has a carrier frequency $f_c + f_{m,1}$, which induces a two-mode squeezing (TMS) interaction that entangles the cavity mode with drum 1 by generating correlated photon-phonon pairs—photons at frequency f_c and phonons at $f_{m,1}$.
- The other pulse has a carrier frequency $f_c - f_{m,2}$, which induces a beam-splitter (BS) interaction that swaps photons in the cavity at frequency f_c with phonons in drum 2 at frequency $f_{m,2}$.
- If the TMS and BS interactions were applied separately, the TMS pulse would energize (excite) drum 1, and the BS pulse would cool drum 2. However, since these two pulses are applied simultaneously, energy flows to both drums. Thus, the cavity acts as a mediator of interaction between the two drums. As a result, strong correlations develop between the quadratures of motion of the two drums.

Measurements:

1. Measurement Technique and Efficiency:

- *Measurement mechanism:* A microwave readout pulse is sent into the system and then reflected after interacting with the drums. This reflected signal encodes information about the drum motion.
- *Use of TMS as preamplifier:* The two-mode squeezing (TMS) interaction, intrinsic to the device, is utilized as a preamplifier to enhance readout, improving the signal-to-noise ratio.
- *Extension of previous techniques:* Techniques previously developed for single-drum readout are extended here using frequency multiplexing (Frequency multiplexing means encoding or addressing different physical systems or signals (like oscillators, drums, or communication channels) using distinct frequencies, so that they can be accessed or measured simultaneously but independently) to simultaneously read out both drums.
- *Improved measurement efficiencies:*

$$\eta_1 = 0.26 \pm 0.02, \quad \eta_2 = 0.153 \pm 0.003$$

- *Heisenberg limit on efficiency:* The Heisenberg uncertainty principle imposes a fundamental limit:

$$\eta_j \leq \frac{1}{2}$$

for concurrent measurement of both quadratures x_j and p_j .

Fiducial Cold State Preparation:

- *Pulse sequence:* A sequence of ground-state cooling followed by readout is applied, allowing simultaneous measurement of

$$x_1, \quad p_1, \quad x_2, \quad p_2$$

- *Observations:*
 - Two-dimensional histograms show no correlations between the measured variables.
 - The covariance matrix is nearly diagonal, indicating statistical independence.
 - The measured variances are:

$$V_1 = 0.75 \pm 0.01, \quad V_2 = 0.63 \pm 0.01$$

where the variance is defined as:

$$V_j = \frac{1}{2} (\langle x_j^2 \rangle + \langle p_j^2 \rangle), \quad \text{for } j = 1, 2$$

Entangled State Protocol:

- *Pulse sequence:* Apply sequential pulses for:
 - (a) Ground-state cooling
 - (b) Entanglement
 - (c) Readout
- *Histogram observations:*

- Quadrature histograms remain Gaussian but with increased variances.
- Drum 1 (undergoing TMS interaction) shows:

$$V_1 = 10.9 \pm 0.1$$

- Drum 2 (undergoing BS interaction) shows:

$$V_2 = 4.63 \pm 0.05$$

- *Correlations:*
 - Positive correlation between x_1 and x_2
 - Negative correlation (anticorrelation) between p_1 and p_2
- *Covariance matrix structure:*
 - Dominant diagonal elements
 - Significant off-diagonal terms:

$$C_{1,3} \approx -C_{2,4}, \quad C_{3,1} \approx -C_{4,2}$$

2. • The experiment studies the evolution of the two-drum quantum state under different entangling pulse durations, all kept below 100 ms to avoid thermal decoherence.

- **First focus:** Individual variances
 - For short pulse durations (< 1 ms), the drums evolve independently:
 - * Drum 2 is cooled.
 - * Drum 1 becomes energized.
 - This behavior matches what would be expected if the drums did not interact.
 - For longer entangling pulses (> 1 ms), both drums deviate from this behavior:
 - * Their variances grow together at similar rates, indicating interaction.
- **Second focus:** Angle of the (x_1, x_2) distribution
 - The theory predicts how the major axis angle of the elliptical (x_1, x_2) distribution changes over time.
- **Third focus:** Measured symplectic eigenvalue n_{meas}
 - Imperfect cooling causes residual **thermal motion** in the initial state.
 - The entangling pulse must overcome classical noise to generate genuine quantum entanglement.
 - For pulse durations $\lesssim 4$ ms:

$$n_{\text{meas}} > \frac{1}{2}$$

which indicates only classical correlations.

- For longer pulses, true quantum behavior emerges:

$$n_{\text{meas}} < \frac{1}{2}$$

indicating entanglement.

- At 16.8 ms interaction time:

$$n_{\text{meas}} = 0.44 \pm 0.004 (\text{stat})^{+0.022}_{-0.021} (\text{sys})$$

- “stat” = statistical uncertainty, estimated using *bootstrapping* (Bootstrapping is a powerful statistical method used to estimate the uncertainty) of a measurement or a model by resampling the data..
- “sys” = systematic uncertainty, dominated by uncertainty in measurement efficiencies.
- This is a direct measurement of entanglement between two macroscopic quantum objects. It quantifies the amount of entanglement after noise and decoherence have affected the system.
- **Estimating the pre-noise entanglement:**
 - A semi-definite program (SDP) is used to find the closest physically valid covariance matrix (in ℓ_2 distance) consistent with the data after noise.
 - This yields an estimate for the true entanglement criterion:

$$n = 0.18 \pm 0.03 (\text{stat})^{+0.13}_{-0.11} (\text{sys})$$

- Such a high degree of entanglement in a macroscopic system could be used to explore:
 - Einstein-Podolsky-Rosen (EPR) nonlocality at the mesoscopic scale (Einstein-Podolsky-Rosen (EPR) nonlocality refers to the paradox and deep quantum phenomenon that reveals the nonlocal nature of quantum entanglement — where two or more quantum systems can exhibit strong correlations even when separated by large distances).
 - Fundamental tests of quantum mechanics.

Results:

A highly entangled state of two macroscopic mechanical oscillators, surpassing the entanglement threshold by

$$4.43^{+0.6}_{-0.7} (\text{stat})^{+4.4}_{-2.3} (\text{sys}) \text{ dB.}$$

What is the entanglement threshold ?

The entanglement threshold is the minimum condition under which two systems must be entangled, based on measurable physical quantities. In continuous-variable (CV) quantum systems — such as mechanical oscillators, optical modes, or position-momentum pairs — the entanglement threshold is most often defined using:

Duan-Simon Criterion (for Gaussian states): For two systems A and B, define collective quadratures:

$$\hat{u} = \hat{x}_A + \hat{x}_B, \quad \hat{v} = \hat{p}_A - \hat{p}_B$$

Then compute the total variance:

$$\langle (\Delta \hat{u})^2 \rangle + \langle (\Delta \hat{v})^2 \rangle$$

For separable (i.e., non-entangled) states, the Duan-Simon criterion gives:

$$\langle (\Delta \hat{u})^2 \rangle + \langle (\Delta \hat{v})^2 \rangle \geq 4$$

(assuming $\hbar = 1$).

If the measured sum falls **below 4**, the inequality is violated, indicating the presence of entanglement.

2. Entanglement Level often, the degree of entanglement is reported in decibels (dB) as:

$$\text{Entanglement Level (dB)} = -10 \log_{10} \left(\frac{\langle (\Delta \hat{u})^2 \rangle + \langle (\Delta \hat{v})^2 \rangle}{4} \right)$$

- 0dB means exactly at the entanglement threshold.
- > 0 dB means confirmed entanglement.
- Higher values (e.g., 4 dB, 6 dB, 10 dB) indicate stronger entanglement.

What is a Symplectic eigenvalue?

Symplectic eigenvalue is a special type of eigenvalue that arises in *symplectic geometry* and *quantum mechanics*, particularly when analyzing Gaussian states in continuous-variable quantum systems like quantum optics or optomechanics. It is fundamentally different from the usual eigenvalue of a matrix — it is tied to the *symplectic structure* that governs phase space in both classical and quantum physics. In quantum mechanics, especially for systems described by *quadratures* (like position x and momentum p), the *covariance matrix* V describes all second-order moments of the system. But not all real symmetric matrices can represent physical quantum states — they must obey the *Heisenberg uncertainty principle*, which requires a certain mathematical structure. To check that structure and extract physical properties (like entanglement), we need symplectic eigenvalues.

1. Covariance Matrix:

For a system of n bosonic modes, define the quadrature vector:

$$\vec{R} = (x_1, p_1, x_2, p_2, \dots, x_n, p_n)^T$$

Then, the covariance matrix V is a real symmetric $2n \times 2n$ matrix with elements:

$$V_{jk} = \frac{1}{2} \langle R_j R_k + R_k R_j \rangle - \langle R_j \rangle \langle R_k \rangle$$

2. Symplectic Form:

The symplectic matrix Ω encodes the canonical commutation relations:

$$\Omega = \bigoplus_{i=1}^n \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

So for $n = 2$, it is a 4×4 matrix:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

3. Symplectic Eigenvalues:

To define *symplectic eigenvalues* of V , compute the eigenvalues $\{\pm i v_k\}$ of the matrix:

$$i\Omega V$$

The positive numbers v_k (for $k = 1, 2, \dots, n$) are called the *symplectic eigenvalues* of V . These are always *real and positive* due to the physical properties of V .

Thus,

$$\text{Spectrum}(i\Omega V) = \{\pm v_1, \pm v_2, \dots, \pm v_n\}$$

Why Are They Important?

The symplectic eigenvalues tell us:

- Whether a state satisfies the *uncertainty principle*:

$$V + i\frac{\hbar}{2}\Omega \geq 0 \iff v_k \geq \frac{\hbar}{2}$$

- How “quantum” the state is. If $v_k = \frac{1}{2}$, the state is a pure vacuum or coherent state.
- Whether a bipartite Gaussian state is entangled, via partial transposition (e.g., *Peres-Horodecki criterion*): if the *smallest symplectic eigenvalue* $\tilde{v}$ of the *partially transposed* covariance matrix is less than $\frac{1}{2}$, the state is entangled.

Consider a bipartite Gaussian state described by the covariance matrix V . The **Peres-Horodecki criterion** (also called the PPT criterion) states that: A bipartite quantum state is separable (i.e., not entangled) if and only if its partial transpose corresponds to a valid physical quantum state. For Gaussian states, the partial transpose operation translates into a specific transformation on the covariance matrix, resulting in the partially transposed covariance matrix V^{PT} . To check whether V^{PT} describes a physical quantum state, it must satisfy the generalized uncertainty principle:

$$V^{PT} + i\frac{\hbar}{2}\Omega \geq 0,$$

where Ω is the symplectic form. This condition is equivalent to requiring that all the symplectic eigenvalues $\tilde{v}_k$ of V^{PT} fulfill:

$$\tilde{v}_k \geq \frac{\hbar}{2}.$$

If there exists at least one symplectic eigenvalue $\tilde{v}_k$ such that

$$\tilde{v}_k < \frac{\hbar}{2},$$

then V^{PT} does not correspond to a physical state, implying that the original state V is entangled. In the common convention where $\hbar = 1$, this threshold is $\frac{1}{2}$, so the criterion becomes:

The bipartite Gaussian state is entangled if $\min_k \tilde{v}_k < \frac{1}{2}$.

This test is powerful and widely used to detect entanglement in continuous-variable systems, such as those involving modes of light or mechanical oscillators.

Doubts:

1. For entanglement to be useful, it must be prepared efficiently, followed by measurement and control with precision that is inversely proportional to the square root of the masses of the objects involved. Why?
2. Is the Peres-Horodecki criterion an ideal way to identify entangled states or not? I have read contrasting opinions on this.

3 Coherent coupling between a ferromagnetic magnon and a superconducting qubit

Challenge :

- The coherent manipulation of a magnon at the single-quantum level remains a significant challenge due to the following reason:

$$\text{Lack of Anharmonicity: } \Delta E_{n+1} - \Delta E_n = 0 \quad (1)$$

where ΔE_n represents the energy difference between successive energy levels. In systems lacking anharmonicity, the energy levels are equally spaced.

- With an increased spin density needed for a stronger coupling, the spin-spin interactions within the ensemble drastically degrade the coherence of the system, leading to a trade-off.

What they present :

- In materials with an organized structure (like crystals), disturbances can cause waves or vibrations that spread over large areas. A magnon is a quantized packet of energy representing these waves in systems where electron spins are aligned in an ordered way.
- The authors show how a single magnon in a small magnetic sphere interacts coherently with a superconducting qubit, using virtual photons in a microwave cavity to mediate the interaction.
- They achieve a strong interaction between the magnon and qubit, much stronger than energy losses (damping), leading to a "strong coupling regime." A parametric drive is applied to adjust and control the strength of the magnon-qubit coupling.
- A novel approach is introduced where, contrary to intuition, strong exchange interactions between neighboring spins are used to achieve a ferromagnetically ordered state.
- These ferromagnetic systems generally have a much higher spin density compared to paramagnetic spin crystals previously mentioned. In such systems, strong exchange and dipolar interactions significantly influence the behavior (dynamics) of the spins. As a result, the collective spin excitations (magnetostatic modes) exhibit very narrow linewidths, indicating low energy dissipation.
- The simplest of these modes is called the Kittel mode. The Kittel mode in ferromagnetism refers to a uniform precession (Precession is a change in the orientation of the rotational axis of a rotating body) of magnetization in a ferromagnetic material. Named after physicist Charles Kittel, this mode describes how all the magnetic moments in the material precess coherently around an external magnetic field at the same frequency.
- This study demonstrates a hybrid quantum system that integrates two very different types of collective excitations: the Kittel mode in a ferromagnet and a superconducting qubit.
- Over the past decade, significant advancements have been made in superconducting qubit technologies and their circuit integration, making them leading platforms for quantum information processing.

- These existing techniques are directly applicable to the quantum manipulation of magnons (quantized spin waves). The nonlinearity (anharmonicity) from the superconducting qubit is essential for enabling quantum control in the hybrid system.

Device Design and Characterization :

- A transmon qubit and a single-crystalline yttrium iron garnet (YIG) sphere are placed inside a microwave cavity. The transmon qubit, which includes a 0.7-mm-long dipole antenna, has a resonant frequency $\omega_q/2\pi = 8.136$ GHz.
- Specifically, the coupling strength between the qubit and the TE_{102} (transverse electric) cavity mode with frequency $\omega_{102}/2\pi = 8.488$ GHz is $g_q/2\pi = 121$ MHz.
- A YIG sphere with a diameter of 0.5 mm is attached to an aluminum oxide rod and positioned near the magnetic-field antinode of the TE_{102} cavity mode.
- A static magnetic field $B_{\text{static}} \approx 0.29$ T is applied locally to magnetize the YIG sphere into a single-domain ferromagnetic state. As a result, the YIG sphere exhibits a large collective magnetic dipole moment, $N\mu_B$, due to the alignment of a large number $N = 1.4 \times 10^{18}$ of electron spins. This strong collective moment allows the sphere to couple effectively to the magnetic component of the cavity mode.
- The large N provides a significant enhancement over paramagnetic systems previously studied, due to YIG's high spin density. To isolate and study the magnon-photon coupling, the qubit is tuned far away from resonance during these measurements.
- They show a clear normal-mode splitting between the cavity mode (TE_{102}) and the Kittel mode (uniform magnon mode) in the YIG sphere. The observed anticrossing behavior provides strong evidence for coherent coupling between the photon mode and the magnon mode.
- The coupling strength between the magnon and cavity photon is found to be $g_m/2\pi = 21.0$ MHz. The linewidths of the TE_{102} mode and the Kittel mode are measured to be $\kappa_{102}/2\pi = 2.5$ MHz and $\gamma_m/2\pi = 1.4$ MHz, respectively. An additional splitting in the upper frequency branch is attributed to another magnetostatic mode that is off-resonance (detuned) from the Kittel mode.

Static coupling scheme between qubit and magnon:

- The qubit couples to the cavity mode via its electric dipole, while the magnon couples to the cavity mode via its magnetic dipole. Thus the qubit and the magnon do not directly interact with each other due to their different coupling mechanisms and spatial separation. To establish interaction between them, an indirect (static) coupling scheme is implemented, using the cavity mode as a mediator.
- Both the qubit and the magnon frequencies, ω_q and ω_{FMR} , are tuned far away from the cavity resonance frequency $\omega_c \equiv \omega_{102}$. When the frequencies of the qubit and the magnon are made nearly equal ($\omega_q \simeq \omega_{\text{FMR}}$), they can coherently exchange excitations.
- This exchange occurs via virtual photon excitations in the cavity, i.e., without actual population of cavity photons. The effective interaction between the qubit and the magnon is described by a Jaynes-Cummings-type Hamiltonian:

$$\hat{H}_{qm,s} = \hbar g_{qm,s} (\hat{s}_+ \hat{c} + \hat{s}_- \hat{c}^\dagger)$$

- Here, $g_{qm,s} = \frac{g_q g_m}{\Delta}$ is the effective coupling strength between the qubit and the magnon. $\hat{s}_+$ and $\hat{s}_-$ are the raising and lowering operators of the qubit, while $\hat{c}$ is the annihilation operator of the magnon excitation. The detuning Δ is defined as the difference between the bare frequencies of the qubit and the cavity: $\Delta = \omega_q - \omega_c$.
- In this hybridized system, the first two excited states form bonding and anti-bonding superpositions of the qubit and magnon excitations.

Qubit Excitation Spectroscopy:

- The state of the qubit is read out indirectly via its influence on the cavity's TE₁₀₃ mode.
- Although the TE₁₀₃ mode (at frequency $\omega_{103}/2\pi = 10.461$ GHz) is detuned from the qubit, it still experiences a dispersive frequency shift depending on the qubit state. This dispersive shift alters the cavity reflection coefficient, denoted by $\text{Re}(\Delta r)$, which is measurable at ω_{103} .
- The plot displays an anticrossing pattern, indicating coherent coupling between the qubit and the Kittel mode of the magnon.
- This anticrossing represents the magnon-vacuum-induced Rabi splitting, a hallmark of strong coupling.
- From the observed splitting, the effective qubit-magnon coupling strength is extracted as $g_{qm,s}/2\pi = 10.0$ MHz.
- This coupling strength is significantly greater than the linewidths of the individual qubit and magnon: $g_q/2\pi = 1.2$ MHz and $g_m/2\pi = 1.3$ MHz, respectively.

Dynamic coupling scheme:

- For dynamic control over the magnon's quantum state, it's beneficial to have fast switching (on and off) of its interaction with the qubit. In the static coupling approach, however, it is technically difficult to change the magnetic field fast enough to enable dynamic switching at the scale of the coupling rate. To overcome this, a parametrically induced interaction is implemented. To suppress the unwanted static coupling, the qubit and the magnon (Kittel mode) are detuned by 274 MHz (and $\omega_{\text{FMR}}/2\pi = 8.410$ GHz).
- A microwave drive is applied at a frequency equal to the average of the qubit and magnon frequencies:

$$\omega_d = \frac{\omega_q + \omega_{\text{FMR}}}{2}$$

to induce the parametric interaction.

- This parametric coupling is a third-order nonlinear process, where two drive photons are absorbed to simultaneously excite the qubit and a magnon. The required third-order nonlinearity arises from the qubit's intrinsic anharmonicity and the large coupling constants g_q and g_m .
- The effective Hamiltonian describing this parametrically induced interaction is:

$$\hat{H}_{qm,p} = \hbar g_{qm,p} (\hat{s}_+ \hat{c}^\dagger + \hat{s}_- \hat{c})$$

where $g_{qm,p}$ is the parametric coupling strength. The effective coupling strength $g_{qm,p}$ scales with the applied microwave drive power P_d .

- The qubit basis states $\{|g\rangle, |e\rangle\}$ and the Kittel-mode magnon-number states $\{|n_m\rangle = |0\rangle, |1\rangle, \dots\}$.
- The two-photon drive induces transitions between $|g, n_m\rangle$ and $|e, n_m + 1\rangle$, with a Rabi splitting of:

$$2g_{qm,p}\sqrt{n_m + 1}$$

- To observe this effect, a probe ω_{FMR} is used to monitor changes in the magnon excitation spectrum under the drive. When the drive frequency satisfies the two-photon resonance condition, the single spectral dip splits into two. As the drive power increases, this spectral splitting becomes more prominent, indicating stronger coupling.
- Additionally, a shift in the magnon resonance frequency is observed—from 8.410 GHz to 8.405 GHz—marking a transition in the system dynamics. This frequency shift is attributed to a qubit-state-dependent Stark shift. At low drive powers, the system primarily undergoes the transition $|g, 0\rangle \leftrightarrow |g, 1\rangle$. At higher drive powers (iii to vi), the transition $|e, 0\rangle \leftrightarrow |e, 1\rangle$ becomes dominant, due to population transfer from $|g, 0\rangle$ to $|e, 0\rangle$.
- This population transfer is caused by both the high drive and the residual static coupling between the qubit and the magnon. The anticrossing in the $|e, 0\rangle \leftrightarrow |e, 1\rangle$ transition clearly demonstrates the presence of parametrically induced coherent coupling. This confirms that the parametric coupling strength can be dynamically tuned by adjusting the drive power.

Results:

- The maximum observed parametric coupling strength is:

$$g_{qm,p}/2\pi = 3.4 \text{ MHz}$$

- This value is larger than both the qubit and magnon decoherence rates, confirming that coherent interaction is achieved.
- Importantly, the linewidths of the spectral dips match the expected value:

$$\frac{g_q + g_m}{2}$$

and do not show any significant broadening from the microwave drive. This study shows that quantum control over magnons in a macroscopic ferromagnetic crystal is now feasible.

4 Black-Box Superconducting Circuit Quantization

Challenge Presented:

Current theoretical models are reaching their limits and to further ability to design, optimize, and manipulate these Superconducting electronic circuits, developing models beyond these limits becomes necessary.

- Standard models (e.g., Jaynes–Cummings) struggle with strong coupling and multimode effects.
- Non-computational states require high-order perturbation theory.
- Circuit parameters (E_J , E_C) are hard to extract without iterative fitting.
- Off-resonant modes affect qubit-qubit coupling and coherence (multimode Purcell effect).
- Including many modes leads to divergent series and QED renormalization issues.
- Physical geometry (e.g., dipole antennas) alters mode structure — needs many modes to model accurately.

Introduction:

Superconducting qubits based on Josephson junctions (JJs) exhibit nonlinear and coherent quantum dynamics, making them highly suitable for quantum information processing. The following key points summarize the foundational concepts and motivation behind the transmon qubit architecture:

- An isolated ideal Josephson junction (JJ) has a single collective degree of freedom: the superconducting phase difference ϕ across the junction, which is related to the current and voltage via the Josephson relations.
- At zero temperature, and within the subgap regime (no quasiparticle excitations), the physics of the JJ is governed by two energy scales:
 - E_J : the Josephson energy, associated with Cooper pair tunneling.
 - $E_C = e^2/2C$: the charging energy, determined by the junction capacitance.
- The quantum dynamics of such a system are described by the Cooper-pair box (CPB) Hamiltonian:

$$H_{\text{CPB}} = 4E_C(\hat{N} - N_g)^2 - E_J \cos(\hat{\phi})$$

where $\hat{N}$ is the Cooper-pair number operator, $\hat{\phi}$ is the conjugate phase operator, and N_g is an externally controlled offset charge.

- The variables $\hat{N}$ and $\hat{\phi}$ obey the canonical commutation relation:

$$[\hat{\phi}, \hat{N}] = i$$

- The CPB Hamiltonian is exactly solvable in terms of Mathieu functions, revealing key features of its energy spectrum.
- A central insight is that the *charge dispersion*—the variation of energy levels with N_g —is exponentially suppressed as E_J/E_C increases. This results in reduced sensitivity to charge noise.

- Meanwhile, the *anharmonicity* of the spectrum, which is essential for distinguishing qubit states, decreases only algebraically with E_J/E_C . Thus, it remains sufficient for qubit operation even at large E_J/E_C .
- This leads to the so-called *transmon regime*, defined by $E_J \gg E_C$, where:
 - Charge dispersion is strongly suppressed.
 - The qubit retains enough anharmonicity for selective addressing of quantum states.
 - The energy level linewidths remain narrow, ensuring long coherence.
- This work focuses on theoretical modeling and analysis of circuits operating in the transmon regime, where the benefits of reduced noise sensitivity and controlled nonlinearity are exploited.

Derivation: Cooper-pair Box Hamiltonian to Mathieu Equation

Begin with the Cooper-pair box (CPB) Hamiltonian:

$$\hat{H} = 4E_C(\hat{N} - N_g)^2 - E_J \cos(\hat{\phi}),$$

where:

- $\hat{N}$: number operator for Cooper pairs (integer-valued)
- N_g : offset gate charge (dimensionless) = $C_g V_g / 2e$
- $\hat{\phi}$: superconducting phase difference operator
- $[\hat{\phi}, \hat{N}] = i$
- E_C : charging energy
- E_J : Josephson energy

They work in the phase basis where:

$$\hat{N} = -i \frac{d}{d\phi}, \quad \hat{\phi} \rightarrow \phi,$$

and the wavefunction $\psi(\phi)$ is 2π -periodic:

$$\psi(\phi + 2\pi) = \psi(\phi).$$

The time-independent Schrödinger equation is:

$$\left[4E_C \left(-i \frac{d}{d\phi} - N_g \right)^2 - E_J \cos(\phi) \right] \psi(\phi) = E \psi(\phi).$$

Change of Variables to eliminate the offset charge N_g , define:

$$\psi(\phi) = e^{iN_g\phi} f(\phi).$$

Now compute the action of the kinetic term:

$$\left(-i\frac{d}{d\phi} - N_g\right)^2 \psi(\phi) = e^{iN_g\phi} \left(-i\frac{d}{d\phi}\right)^2 f(\phi).$$

So the Hamiltonian becomes:

$$\hat{H} = -4E_C \frac{d^2}{d\phi^2} - E_J \cos(\phi),$$

and the Schrödinger equation reduces to:

$$\left[-4E_C \frac{d^2}{d\phi^2} - E_J \cos(\phi)\right] f(\phi) = E f(\phi).$$

Mathieu Form: We rewrite the equation in standard form. Divide both sides by $4E_C$:

$$\left[-\frac{d^2}{d\phi^2} - \frac{E_J}{4E_C} \cos(\phi)\right] f(\phi) = \frac{E}{4E_C} f(\phi).$$

Define:

$$a = \frac{E}{E_C}, \quad q = -\frac{E_J}{2E_C},$$

and multiply both sides by 1:

$$\frac{d^2 f}{d\phi^2} + [a - 2q \cos(\phi)] f(\phi) = 0.$$

This is the **Mathieu equation**:

$$\frac{d^2 f}{d\phi^2} + [a - 2q \cos(\phi)] f(\phi) = 0,$$

where $f(\phi)$ is 2π -periodic. The solutions to this equation are known as Mathieu functions and describe the eigenstates of the Cooper-pair box.

The **Jaynes–Cummings (JC) model** describes the coherent interaction between a single mode of the quantized electromagnetic field (e.g., in a cavity) and a two-level atom (e.g., a qubit). It is a cornerstone of quantum optics and cavity/circuit quantum electrodynamics (QED). The total Hamiltonian is given by:

$$\hat{H} = \hat{H}_{\text{field}} + \hat{H}_{\text{atom}} + \hat{H}_{\text{int}},$$

where each term is defined as follows:

- Field Hamiltonian (quantized harmonic oscillator):

$$\hat{H}_{\text{field}} = \hbar\omega_r \hat{a}^\dagger \hat{a}$$

where $\hat{a}^\dagger, \hat{a}$ are photon creation and annihilation operators, and ω_r is the resonator frequency.

- Atom Hamiltonian (two-level system):

$$\hat{H}_{\text{atom}} = \frac{1}{2} \hbar \omega_q \hat{\sigma}_z$$

where ω_q is the qubit (atom) transition frequency, and $\hat{\sigma}_z$ is the Pauli z -matrix:

$$\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|.$$

- Interaction Hamiltonian (under rotating wave approximation, RWA):

$$\hat{H}_{\text{int}} = \hbar g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$$

where g is the coupling strength, and $\hat{\sigma}_\pm$ are atomic raising and lowering operators:

$$\hat{\sigma}_+ = |e\rangle\langle g|, \quad \hat{\sigma}_- = |g\rangle\langle e|.$$

Total Hamiltonian (in RWA). Combining all terms, we write:

$$\hat{H}_{\text{JC}} = \hbar \omega_r \hat{a}^\dagger \hat{a} + \frac{1}{2} \hbar \omega_q \hat{\sigma}_z + \hbar g (\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$$

- The JC model describes coherent exchange of excitation between the atom and the field.

Purcell Effect (Short Summary)

- The Purcell effect describes the modification of a quantum emitter's spontaneous emission rate due to its electromagnetic environment (e.g., a cavity).
- In circuit QED, it leads to enhanced qubit decay when the qubit is coupled to a lossy cavity. When the resonator decay rate (i.e., how quickly it loses energy to the environment) is increased, the resonator becomes more leaky.
- The decay rate is given by:

$$\Gamma_{\text{Purcell}} = \frac{g^2}{\kappa} \cdot \frac{\kappa^2}{\Delta^2 + \kappa^2}$$

where:

- g : qubit–cavity coupling strength,
- κ : cavity decay rate,
- $\Delta = \omega_q - \omega_c$: detuning between qubit and cavity.
- The effect becomes more pronounced when the qubit is near-resonant with a lossy cavity mode.
- In multimode systems, multiple off-resonant modes can contribute to decoherence (multimode Purcell effect).

What they do : The authors propose a semiclassical quantization method:

1. Step 1: Linearize the circuit, solve it classically to find eigenmodes.

2. Step 2: Use the linear response to determine the quantum Hamiltonian.
3. Step 3: Add anharmonicity perturbatively to recover the full behavior.

- We consider a single Josephson junction (JJ) with Josephson energy E_J and charging energy E_C , embedded in an arbitrary linear electromagnetic environment.
- Neglecting dissipation and external bias, the JJ is governed by the Cooper-pair box Hamiltonian:

$$H = 4E_C(\hat{N} - N_g)^2 - E_J \cos(\hat{\phi})$$

- In the transmon regime ($E_J \gg E_C$), phase fluctuations ϕ are small and quantum tunneling between potential wells is negligible.
- The cosine potential is expanded as a Taylor series in ϕ :

$$\cos(\phi) \approx 1 - \frac{\phi^2}{2} + \frac{\phi^4}{24} - \dots$$

leading to a decomposition into linear and nonlinear components.

- The linear part of the JJ behaves like an LC oscillator with:

$$L_J = \frac{\phi_0^2}{E_J}, \quad C_J = \frac{e^2}{2E_C}, \quad \text{where } \phi_0 = \frac{\hbar}{2e}$$

- The leading nonlinear correction contributes energy:

$$E_{\text{nl}} = \frac{\phi_0^2 \phi^4}{24L_J}$$

- The impedance $Z(\omega)$ of the linear circuit is a complex meromorphic function (A meromorphic function is a complex function that is analytic everywhere except at a discrete set of isolated poles (points where it goes to infinity in a controlled way)) and plays a central role in the quantization of the circuit.
- According to Foster's theorem, $Z(\omega)$ can be synthesized as a series of parallel LCR resonators, each corresponding to a mode of the environment.
- This formulation enables quantization in terms of normal modes and allows treating the qubit and cavity on equal footing.
- The impedance of the linearized circuit is modeled as a sum of M parallel LCR resonators:

$$Z(\omega) = \sum_{p=1}^M \left(\frac{1}{j\omega C_p} + j\omega L_p + \frac{1}{R_p} \right)^{-1}$$

where $j = -i$ (imaginary unit) as per electrical engineering convention.

- This equivalent circuit corresponds to diagonalizing a system of coupled harmonic oscillators.

- Resonance frequencies correspond to the real parts of the poles of $Z(\omega)$, or equivalently, to the zeros of the admittance:

$$Y(\omega) = \frac{1}{Z(\omega)}$$

- For weak dissipation (i.e., $R_p \gg \sqrt{L_p/C_p}$), the resonance frequency of mode p is approximately:

$$\omega_p = \frac{1}{\sqrt{L_p C_p}}$$

- The imaginary part of the resonance frequency determines the linewidth (finite width) of the mode:

$$\text{Width} \sim \frac{1}{2R_p C_p}$$

- The effective resistance at resonance is given by:

$$R_p = \frac{1}{\text{Re} Y(\omega_p)}$$

- The effective capacitance is given by the derivative of the imaginary part of the admittance:

$$C_p = \frac{1}{2} \text{Im} Y'(\omega_p)$$

- Here, the prime (') denotes derivative with respect to frequency.
- It is a general property of passive systems that:

$$\text{Im} Y'(\omega) > 0$$

- The quality factor Q_p of mode p is given compactly by:

$$Q_p = \frac{\omega_p^2 \text{Im} Y'(\omega_p)}{\text{Re} Y(\omega_p)}$$

- When applied to the qubit mode, this formula estimates the Purcell-limited qubit lifetime:

$$T_1 = \frac{Q_{\text{qb}}}{\omega_{\text{qb}}}$$

where T_1 is the energy relaxation time due to spontaneous emission (Purcell effect).

Deriving the Effective Low-Energy Quantum Hamiltonian:

- To derive a low-energy quantum description of the circuit, they:
 - Neglect dissipation, i.e., take $R_p \rightarrow \infty$.
 - Focus on the linearized equivalent circuit composed of LC oscillators.

- Introduce the normal flux coordinates for each oscillator:

$$\phi_p(t) = f_p e^{j\omega_p t} + f_p^* e^{-j\omega_p t}$$

where f_p is the complex amplitude of mode p , and ω_p is its resonance frequency.

- The classical Hamiltonian of the full linear circuit becomes:

$$H_0 = 2 \sum_{p=1}^M (f_p)^* \frac{1}{L_p} f_p$$

where M is the number of modes and the subscript 0 denotes the linear part.

- Kirchhoff's voltage law implies the total flux across the junction is the sum of all modal contributions:

$$\phi(t) = \sum_{p=1}^M \phi_p(t)$$

with $\phi(t) = \int_{-\infty}^t V(\tau) d\tau$, the flux coordinate.

- By the second Josephson relation:

$$\varphi(t) = \frac{\phi(t)}{\phi_0} \mod 2\pi$$

where $\phi_0 = \hbar/2e$ is the reduced flux quantum, and $\varphi(t)$ is the superconducting phase difference across the junction.

- To quantize the system:

- Promote classical amplitudes f_p to operators:

$$f_p \rightarrow \sqrt{\frac{\hbar Z_p^{\text{eff}}}{2}} a_p, \quad f_p^* \rightarrow \sqrt{\frac{\hbar Z_p^{\text{eff}}}{2}} a_p^\dagger$$

- Here, $a_p, a_p^\dagger$ are bosonic annihilation and creation operators.

- The effective impedance of each mode is given by:

$$Z_p^{\text{eff}} = \frac{2}{\omega_p \text{Im} Y'(\omega_p)}$$

where $Y(\omega) = 1/Z(\omega)$ is the admittance of the linear circuit.

- Canonical quantization is applied by promoting the classical flux amplitudes f_p to quantum operators.
- The replacement rule is:

$$f_p \rightarrow \sqrt{\frac{\hbar}{2Z_p^{\text{eff}}}} a_p, \quad f_p^* \rightarrow \sqrt{\frac{\hbar}{2Z_p^{\text{eff}}}} a_p^\dagger$$

where:

- $a_p, a_p^\dagger$ are bosonic annihilation and creation operators satisfying $[a_p, a_q^\dagger] = \delta_{pq}$,
- Z_p^{eff} is the effective impedance of mode p , given by:

$$Z_p^{\text{eff}} = \frac{2}{\omega_p \text{Im} Y'(\omega_p)}$$

- This quantization transforms each normal mode of the linear circuit into a quantum harmonic oscillator.
- The quantized Hamiltonian becomes:

$$\hat{H}_0 = \sum_{p=1}^M \hbar \omega_p a_p^\dagger a_p$$

representing M uncoupled harmonic oscillator modes.

- The quantum operator for total flux across the junction is:

$$\hat{\phi} = \sum_{p=1}^M \sqrt{\frac{\hbar Z_p^{\text{eff}}}{2}} (a_p + a_p^\dagger)$$

- These modes a_p are collective excitations of the entire linear environment.
 - Their frequencies ω_p include the effects of coupling.
 - No rotating wave approximation is used in this construction.
- The total Hamiltonian of the circuit including the Josephson junction (JJ) is given by:

$$H = H_0 + H_{\text{nl}}$$

where:

- H_0 : the linear Hamiltonian (sum of harmonic oscillator modes).
- $H_{\text{nl}} = \frac{\hat{\phi}^4}{24\phi_0^2 L_J} + \mathcal{O}\left(\left(\frac{\hat{\phi}}{\phi_0}\right)^6\right)$: the leading-order nonlinear contribution from the cosine potential of the Josephson junction.
- They treat H_{nl} as a small perturbation to H_0 , assuming that the eigenstates:

$$|n_1, n_2, \dots, n_M\rangle$$

of H_0 are non-degenerate, with unperturbed energies:

$$E_{n_1, n_2, \dots, n_M}^{(0)} = \sum_i n_i \hbar \omega_i$$

- Keeping only the leading order $\hat{\phi}^4$ term, the reduced effective Hamiltonian becomes:

$$H^{(4)} = H_0' + \frac{1}{2} \sum_{pp'} \chi_{pp'} \hat{n}_p \hat{n}_{p'}$$

where:

- $\hat{n}_p = a_p^\dagger a_p$: photon number operator for mode p .
- $H'_0 = H_0 + \sum_p \Delta_p \hat{n}_p$: includes a small correction to the mode frequencies (Lamb shift).
- The Lamb shift correction κ_p is given by:

$$\Delta_p = -\frac{e^2}{2L_J} \left(Z_p^{\text{eff}} \sum_q Z_q^{\text{eff}} - \frac{(Z_p^{\text{eff}})^2}{2} \right)$$

- The interaction coefficients $\chi_{pp'}$ represent:
 - χ_{pp} : the self-Kerr nonlinearity (anharmonicity) of mode p .
 - $\chi_{pp'}$ for $p \neq p'$: the cross-Kerr shift — frequency shift of mode p due to one excitation in mode p' .
- Explicitly, the Kerr coefficients are:

$$\chi_{pp} = \frac{L_p}{L_J} \cdot \frac{C_J}{C_p} E_C, \quad \chi_{pp'} = 2\sqrt{\chi_{pp}\chi_{p'p'}}$$

where:

- L_J : Josephson inductance.
- C_J : junction capacitance.
- L_p, C_p : inductance and capacitance of mode p .
- E_C : charging energy of the junction.

Lamb Shift in Superconducting Circuits

- The Lamb shift refers to a small correction to the energy levels of quantum systems due to interactions with the electromagnetic vacuum or environment.
- In superconducting circuits, the Lamb shift arises when a Josephson junction (JJ) is coupled to a multimode linear environment (e.g., LC resonators).
- Physically, this shift is due to the fact that each mode “feels” the presence of other modes through the nonlinear coupling, even in the absence of real excitations.
- All modes acquire some anharmonicity due to the nonlinearity introduced by the Josephson junction (JJ).
- As a result, there is no longer a strict separation between **qubit modes** (strongly anharmonic) and **cavity modes** (ideally harmonic).
- Terminology:
 - A mode with strong anharmonicity is referred to as *qubitlike*.
 - A mode with weak anharmonicity is referred to as *cavitylike*.
- In the lowest-order approximation, the anharmonicity of mode p is:

- Proportional to the *inductive participation ratio*: $i_p = \frac{L_p}{L_J}$
- Inversely proportional to the *capacitive participation ratio*: $c_p = \frac{C_p}{C_J}$
- In the case of a transmon qubit capacitively coupled to a cavity:
 - There is no galvanic (direct) short of the JJ, so from the sum rule:

$$\lim_{\omega \rightarrow 0} \frac{Z(\omega)}{j\omega} = \sum_p L_p = L_J \Rightarrow i_p \leq 1$$

- With no capacitance in series with C_J , the capacitive sum rule implies:

$$\lim_{\omega \rightarrow 0} j\omega Z(\omega) = \sum_p \frac{1}{C_p} = \frac{1}{C}, \quad \text{where } C = C_J + C_k \Rightarrow c_p \leq 1$$

where C_k is the total capacitance in parallel with C_J .

- Therefore, the effective anharmonicity of a qubitlike mode is always smaller than that of a bare qubit (given by E_C).
- The cross-Kerr shift between two modes p and p' is:

$$\chi_{pp'} = 2\sqrt{\alpha_p \alpha_{p'}}$$

where α_p is the self-Kerr (anharmonicity) of mode p .

- Limitations: These analytical expressions do not account for higher-order effects such as:
 - The *sign reversal* of the cross-Kerr shift in the straddling regime.
 - Straddling Regime: The straddling regime occurs when the qubit transition frequency ω_q lies between two resonator modes:

$$\omega_{r1} < \omega_q < \omega_{r2}$$

In this regime, both modes contribute to the dispersive shift, possibly with opposite signs, leading to nontrivial effects such as:

- * Enhanced or suppressed qubit-resonator coupling,
- * Sign reversal of the cross-Kerr shift,
- * Necessity of a full multimode treatment of the Hamiltonian.

- Such effects are fully captured by the complete model:

$$H = H_0 + H_{nl}$$

which can be numerically solved.

- The dressed modes (A dressed mode (or dressed state) refers to a modified version of a system's natural mode after accounting for interactions or couplings — typically with another quantum system) already include all coupling between bare modes, so:

- Only a few dressed modes ($M^* \ll M$) are needed for accurate simulations.
- This reduces the size of the effective Hilbert space, which scales as:

$$\prod_{p=1}^{M^*} (N_p + 1)$$

where N_p is the maximum number of excitations in mode p (e.g., $N_p = 1$ in a two-level approximation).

Charge Dispersion in a JJ-Cavity System

- **Assumption:** The main analysis neglects *charge dispersion effects* — i.e., the variation in energy levels due to different offset charges on the superconducting island.
- **Question:** What happens to charge dispersion when a Josephson Junction (JJ) is coupled to a cavity?
- **Analogy:** As in the *Caldeira-Leggett model*, coupling the JJ to harmonic oscillator modes (cavity) suppresses quantum tunneling of the phase, which in turn suppresses charge dispersion.
- **Mechanism:** The suppression occurs because a flux tunneling event through the JJ (a ϕ slip) excites the environment (the linear modes), and the probability that no excitations are created in this process becomes very small. Therefore, the tunneling is suppressed, and charge dispersion is reduced.
- **Suppression Estimate:** The probability P_0 of the circuit remaining in its ground state after a 2π flux slip is given by:

$$P_0 = \exp \left(-\frac{1}{2} \sum_p \frac{\delta q^2}{2C_p \hbar \omega_p} \right)$$

where:

- $\delta q = \frac{C_J \Phi_0}{\tau}$ is the momentum (charge) kick from a flux slip of duration τ ,
- $C_p = \frac{1}{2} \text{Im}[Y'(\omega_p)]$ is an effective mode capacitance,
- The sum excludes the qubit mode (only cavity/environmental modes considered).
- **Interpretation:** As P_0 becomes exponentially small, flux tunneling is strongly suppressed. Hence, charge dispersion is significantly reduced, justifying its neglect in the Hamiltonian.
- **Important Note:** Despite suppression in the qubit, each eigenmode of the coupled system still inherits some small charge dispersion — a result of hybridization with the nonlinear JJ. This residual charge dispersion is especially relevant for systems aiming to store quantum information in high-Q (low-loss) cavity modes coupled to JJs.

Generalization to N Junctions

- The method is extended to circuits with N Josephson junctions (JJs), all connected in parallel to a shared linear environment.

- The resonance frequencies of the linear part of the circuit are obtained from the zeros of the admittance:

$$Y_k(\omega) = Z_{kk}^{-1}(\omega),$$

where $k=1,2,\dots,N$ and Z is the $N \times N$ impedance matrix, and each junction defines a *port* in this matrix.

- The flux operator at junction l (using reference port k where $l=1,2,\dots,N$) is:

$$\hat{\phi}_l^{(k)} = \sum_{p=1}^M \frac{Z_{lk}(\omega_p)}{Z_{kk}(\omega_p)} \sqrt{\frac{\hbar Z_{kp}^{\text{eff}}}{2}} (a_p + a_p^\dagger),$$

where the effective impedance is given by:

$$Z_{kp}^{\text{eff}} = \frac{2}{\omega_p \text{Im} Y_k'(\omega_p)}.$$

- The resonance frequencies ω_p are independent of the choice of reference port k , but the eigenmodes (flux distributions) *do* depend on this choice.
- In lowest-order perturbation theory and to fourth order in the expansion, the Kerr-type interaction coefficients are:

$$\alpha_p = -12\beta_{pppp}, \quad \chi_p = -24\beta_{qqpp} \quad (\text{for } q \neq p),$$

where $\beta_{qq'pp'}$ encodes the nonlinear coupling strength between modes.

- The Lamb shift correction to the energy of mode p is:

$$\Delta_p = 6\beta_{pppp} - 12 \sum_q \beta_{qqpp}.$$

- The nonlinearity coefficients are defined as:

$$\beta_{qq'pp'} = \sum_{s=1}^N \frac{e^2}{24L_j^{(s)}} \xi_{sq} \xi_{sq'} \xi_{sp} \xi_{sp'},$$

where ξ_{sp} describes the contribution of port s to mode p :

$$\xi_{sp} = \frac{Z_{s1}(\omega_p)}{Z_{11}(\omega_p)} \sqrt{Z_{1p}^{\text{eff}}}.$$

- By the Cauchy-Schwarz inequality:

$$|\xi_{qp}| \leq 2\sqrt{\alpha_p \alpha_q},$$

indicating a bounded relationship between self- and cross-Kerr effects.

- When q and q' correspond to two different qubitlike modes, the term $\chi_{qq'}$ quantifies their total mutual interaction, which includes:

- Cavity-mediated interaction
- Direct dipole-dipole coupling

Comparison with Experiment

- The proposed theoretical method is applied to a system consisting of a single Josephson junction (JJ) coupled to a 3D cavity.
- The total admittance at the JJ port is given by:

$$Y(\omega) = j\omega C_J - \frac{j}{\omega L_J} + Y_c(\omega),$$

where:

- C_J : junction capacitance,
- L_J : Josephson inductance,
- $Y_c(\omega)$: admittance of the cavity–antenna system.
- The junction is assumed to be dissipationless, modeling a Purcell-limited qubit. Cavity losses are included via a complex admittance $Y_c(\omega)$.
- The Josephson inductance L_J is extracted from the junction's room-temperature resistance R_T using the **Ambegaokar–Baratoff relation**:

$$E_J = \frac{\hbar \Delta}{8e^2 R_T}.$$

- The only adjustable parameter in the model is C_J , which is fit to experimental data. The fitting is done by numerically solving the Hamiltonian including the ϕ^6 nonlinearity, matching the lowest three energy levels to the experimentally measured spectrum. Direct measurement of $Y_c(\omega)$ is not practical due to limited access to the antenna inside the high- Q 3D cavity.
- Instead, the cavity-antenna admittance $Y_c(\omega)$ is obtained by solving Maxwell's equations numerically using finite-element software (ANSYS HFSS).
- From the zeros of $\text{Im}Y(\omega)$ and their slopes, the system's effective Hamiltonian is built.
- The Hamiltonian is diagonalized in a truncated Hilbert space :
 - Three dressed modes are included: one qubitlike mode and two cavitylike modes.
 - Up to 10 excitations are allowed per mode.
 - Truncated Hilbert Space in Circuit QED Simulation
 - * Consider a circuit QED system with:
 - 1 qubit (nonlinear mode),
 - 2 cavity modes (linear or weakly nonlinear),
 - Each mode truncated to a maximum of N_p excitations.
 - * The total Hilbert space is the tensor product of the individual mode Hilbert spaces.
 - * After truncation, the dimension of the truncated Hilbert space is:

$$\dim(\mathcal{H}) = \prod_{p=1}^M (N_p + 1),$$

where M is the number of modes and N_p is the cutoff (maximum number of excitations) for mode p .

* Example: Suppose we truncate each of the 3 modes (1 qubit + 2 cavities) to 10 excitations:

$$\dim(\mathcal{H}) = (10+1)^3 = 11^3 = 1331.$$

* Therefore, the truncated Hilbert space contains 1331 basis states.

- The results are summarized in Table I of the paper, which compares:
 - Numerically predicted vs. measured low-energy spectrum.
 - Anharmonicities of the qubit mode.
 - Cross–Kerr shifts between modes.
- Results:
 - Energy level predictions agree with measurements at the subpercent level.
 - Anharmonicity and Kerr shift predictions match within $\sim 10\%$.

Summary: They have introduced a simple and effective method for determining the low-energy Hamiltonian of weakly nonlinear superconducting circuits, which may include both lumped and distributed elements. This approach utilizes the normal modes of the linearized classical circuit as a suitable quantum basis and relies only on the classical $N \times N$ impedance matrix for an N -qubit system. By employing a dressed-state basis, the resulting Hamiltonian parameters incorporate renormalization effects from the coupling between multilevel artificial atoms and multimode resonators, thereby significantly reducing the number of free parameters compared to Jaynes–Cummings-based models that depend on inaccessible bare quantities. This method has been validated through the calculation of the low-energy spectrum of a 3D transmon, and it holds promise for exploring the ultrastrong coupling regime in circuit QED systems.

Doubts :

1. I dont understand where this comes from. There is no galvanic (direct) short of the JJ, so from the sum rule:

$$\lim_{\omega \rightarrow 0} \frac{Z(\omega)}{j\omega} = \sum_p L_p = L_J \Rightarrow i_p \leq 1$$

2. The probability that no environmental mode is excited becomes exponentially small as coupling increases.

5 Breaking the trade-off between fast control and long lifetime of a superconducting qubit

Challenges presented:

- Recent advances in qubit design and fabrication have significantly improved coherence times. A key future challenge is the radiative decay of a qubit through its control line.
- This leads to a fundamental trade-off between:
 - Fast qubit control requires strong coupling to the control line.
 - Long qubit lifetimes require weak coupling to suppress energy loss.

What they do:

- The proposed approach introduces a second superconducting qubit strongly coupled along the control line. This qubit is called a **Josephson Quantum Filter (JQF)**.
- Key functionalities of the JQF:
 - Acts as a filter that blocks low-power microwave radiation from escaping the first qubit into the control line, thereby reducing relaxation.
 - When large magnitude strong microwave pulses are applied, the JQF becomes saturated and allows high-power signals to pass through. This behavior enables fast control without degrading coherence.
- The JQF acts as an automatic, passive decoupler between the qubit and its environment.
- Advantages of this approach: Reduction in environmental heating and mitigates inter-qubit crosstalk.

Introduction :

- Single-qubit gates are fundamental to quantum information processing.
- Implemented by applying an electromagnetic field in resonance with the qubit transition frequency to induce Rabi oscillations.
- External control requires coupling the qubit to at least one one-dimensional continuous mode.

The Trade-Off: Fast Control vs. Long life time of qubit

- Stronger coupling to the control line:
 - Enables faster gate operations.
 - Increases radiative decay into the control line.
- Weaker coupling:
 - Reduces decay and extends coherence time.
 - Requires stronger drive fields $\Rightarrow$ slower or more energy-intensive control.

- This presents a fundamental trade-off governed by the fluctuation–dissipation theorem .
 - The Fluctuation–Dissipation Theorem (FDT) is a fundamental result in statistical physics that relates:
 - * Fluctuations in a system at thermal equilibrium (e.g., thermal noise).
 - * Dissipation of energy when the system is subjected to an external perturbation.
 - The theorem implies that the way a system responds to external forces is fundamentally connected to its internal thermal fluctuations.
 - For a linear, time-invariant system in thermal equilibrium at temperature T , the FDT states:

$$S(\omega) = \frac{2k_B T}{\omega} \text{Im}[\chi(\omega)]$$

where:

- * $S(\omega)$: Power spectral density of fluctuations.
- * $\chi(\omega)$: Linear response function (susceptibility) and $\text{Im}[\chi(\omega)]$ is imaginary part of the response function, associated with dissipation.

Problem in State-of-the-Art Systems

- Superconducting qubits are leading candidates for large-scale quantum computing.
- Advances in fabrication have led to longer coherence times.
- Radiative decay into the control line is now a dominant limitation.
- Existing mitigation:
 - Use of weak coupling to the control line.
 - Compensated by applying strong microwave drive.
- Issues with this approach in large-scale systems:
 - Heating of cryogenic environment.
 - Increased crosstalk to non-targeted qubits.
 - Limited output power from control electronics.

Proposed Solution: Josephson Quantum Filter (JQF)

- A second superconducting qubit (ancillary qubit) is inserted into the control line, forming a nonlinear quantum filter.
- This filter is named the Josephson Quantum Filter (JQF).
- Key functionality:
 - Reflects single photons (low-amplitude signals) from the data qubit $\Rightarrow$ suppresses radiative decay.

- Becomes transparent to large amplitude control pulses (due to saturation) $\Rightarrow$ allows fast gate control and enables fast rabi oscillations of data qubit.

Comparison to Existing Technology :

- Contrast with the Purcell filter:
 - The Purcell filter relies on a frequency mismatch between qubit and readout resonator to realize both fast readout and long lifetime of qubit.
 - Effective for readout protection, but not suitable when control and decay signals share the same frequency.
- The JQF operates at the same frequency as the control signal and is based on nonlinearity and saturation, not frequency separation.

Model and Working Principle of the JQF:

- System Description:
 - The system consists of a data qubit and a JQF placed in a semi-infinite 1D control line.
 - The data qubit is located at the end of the line, and the JQF is placed a distance d away.
 - The JQF frequency ω_f is set to be identical to the qubit frequency ω_q .
- Physical Effects:
 - Two cooperative effects arise due to photon-mediated interaction in the waveguide:
 - * Correlated decay (When two quantum systems (like a qubit and a JQF) are coupled to the same waveguide (or field), photons emitted by one can interfere with those from the other. This creates a non-local, collective decay channel rather than each qubit decaying independently.)
 - * Energy-exchange interaction (The waveguide not only mediates decay but also allows coherent exchange of excitations (photons) between the two systems.)
- Master Equation:
 - In a rotating frame at ω_q and ω_f , the master equation is:

$$\dot{\rho} = \frac{-i}{\hbar} [H_{\text{eff}} + H_{\text{drive}}, \rho] + \sum_{i,j=q,f} \gamma_{ij}^{\text{ex}} \mathcal{D}(\sigma_i, \sigma_j) \rho$$

- σ_i is the lowering operator for qubit ($i = q$) or JQF ($i = f$).
- The correlated decay superoperator is:

$$\mathcal{D}(A, B) \rho = B \rho A^\dagger - \frac{1}{2} (A^\dagger B \rho + \rho A^\dagger B)$$

- Correlated Decay Rates:
 - Individual decay rates:

$$\gamma_{qq}^{\text{ex}} = \gamma_q^{\text{ex}}, \quad \gamma_{ff}^{\text{ex}} = \gamma_f^{\text{ex}} \cos\left(\frac{2\pi d}{\lambda_q}\right)$$

- Correlated decay rates:

$$\gamma_{qf}^{\text{ex}} = \gamma_{fq}^{\text{ex}} = \sqrt{\gamma_q^{\text{ex}} \gamma_f^{\text{ex}}} \cos\left(\frac{2\pi d}{\lambda_q}\right)$$

- Energy-Exchange Interaction:

- Effective Hamiltonian:

$$H_{\text{eff}} = \hbar J (\sigma_q^\dagger \sigma_f + \sigma_q \sigma_f^\dagger)$$

- Coupling strength:

$$J = \frac{1}{2} \sqrt{\gamma_q^{\text{ex}} \gamma_f^{\text{ex}}} \sin\left(\frac{2\pi d}{\lambda_q}\right)$$

where γ_i^{ex} (for $i = q, f$) is the respective external coupling rate, and λ_q is the qubit wavelength.

- Drive Hamiltonian:

- Drive term:

$$H_{\text{drive}} = \frac{\hbar \Omega_q}{2} \sigma_q^x + \frac{\hbar \Omega_f}{2} \cos\left(\frac{2\pi d}{\lambda_q}\right) \sigma_f^x$$

- Rabi frequencies:

$$\Omega_i = 2\sqrt{\gamma_i^{\text{ex}} \dot{n}}, \quad i = q, f$$

where $\dot{n}$ is the control photon flux.

- Working Principle of the JQF:

- Set $d = \lambda_q/2$ to:

- * Suppress hybridization ($J = 0$)

- * Maximize correlated decay ($|\gamma_{qf}^{\text{ex}}| = \sqrt{\gamma_q^{\text{ex}} \gamma_f^{\text{ex}}}$)

- Define collective modes (Individual decays of bright and dark modes) :

$$\sigma_B = N \left(-\sqrt{\gamma_q^{\text{ex}}} \sigma_q + \sqrt{\gamma_f^{\text{ex}}} \sigma_f \right), \quad \gamma_B^{\text{ex}} = \gamma_q^{\text{ex}} + \gamma_f^{\text{ex}}$$

$$\sigma_D = N \left(\sqrt{\gamma_f^{\text{ex}}} \sigma_q + \sqrt{\gamma_q^{\text{ex}}} \sigma_f \right), \quad \gamma_D^{\text{ex}} = 0$$

- When $\gamma_f^{\text{ex}} \gg \gamma_q^{\text{ex}}$, the qubit state is close to the dark state $\sigma_q \sigma_D$, and radiative decay is suppressed. N is the normalization factor.

- Approximate Master Equation in Strong Asymmetry Limit ($\gamma_f^{\text{ex}} \gg \gamma_q^{\text{ex}}$):

$$\dot{\rho} = \frac{i}{\hbar} [H_{\text{drive}}, \rho] + \mathcal{D} \left(\sqrt{\gamma_f^{\text{ex}}} \sigma_f - \sqrt{\gamma_q^{\text{ex}}} (1 + \sigma_f^z) \sigma_q \right) \rho$$

- Effective Drive Hamiltonian:

$$H_{\text{drive}} = -\frac{\hbar \Omega_f}{2} \sigma_f^x + \frac{\hbar \Omega_q}{2} (1 + \sigma_f^z) \sigma_q^x$$

- Control Behavior:
 - No control field: $\sigma_f^z = -1$ and ($H_{drive} = 0$) (JQF in ground state) $\Rightarrow$ qubit decay is completely suppressed.
 - With control field: the drive on the JQF (σ_f^x) dominates, allowing effective qubit control with Rabi frequency Ω_q .

Device Characterization :

- A superconducting transmon qubit is fabricated on a silicon substrate.
- The qubit is coupled to a coplanar waveguide control line.
- Qubit parameters:
 - Resonance frequency: $\omega_q/2\pi = 8.002$ GHz
 - Anharmonicity: $\alpha_q/2\pi = -398$ MHz
 - External coupling rate: $\gamma_q^{ex}/2\pi = 123$ kHz
- Qubit readout:
 - Via a dispersively coupled resonator.
 - Resonator frequency: $\omega_r/2\pi = 10.156$ GHz
 - External coupling rate: $\kappa^{ex}/2\pi = 2.16$ MHz
 - Dispersive frequency shift: $2\chi/2\pi = -1.87$ MHz
 - A Purcell filter is used to suppress Purcell decay via the readout line.
- An ancillary transmon qubit is used as a Josephson Quantum Filter (JQF).
 - Strongly coupled to the control line.
 - External coupling rate: $\gamma_f^{ex}/2\pi = 112$ MHz
 - Distance from the qubit: $d \approx 7.5$ mm (half the qubit wavelength).
 - Tunable resonance frequency: $\omega_f/2\pi = 6.3$ to 8.5 GHz
 - When detuned, the JQF effectively decouples from the qubit.
 - Anharmonicity: $\alpha_f/2\pi = -387$ MHz
 - Intrinsic loss rate: $\gamma_f^{in}/2\pi = 3$ MHz

Dispersive Readout:

Dispersive readout is a quantum measurement technique used in circuit quantum electrodynamics (cQED) to infer the state of a qubit by probing a coupled resonator. It relies on the fact that a qubit, when off-resonantly coupled to a resonator, can shift the resonator's frequency depending on its state—without exchanging energy with it. When a qubit is coupled to a resonator with a large detuning $\Delta = \omega_q - \omega_r$ compared to the coupling strength g , the system enters the dispersive regime:

$$|\Delta| \gg g$$

In this regime, the qubit and resonator do not exchange excitations, but the qubit imparts a state-dependent frequency shift to the resonator.

The effective Hamiltonian for the dispersive interaction is:

$$H/\hbar = \omega_r a^\dagger a + \frac{\omega_q}{2} \sigma_z + \chi a^\dagger a \sigma_z$$

where:

- ω_r : Resonator frequency
- ω_q : Qubit frequency
- $a^\dagger, a$: Photon creation and annihilation operators in the resonator
- σ_z : Pauli Z operator representing the qubit state
- χ : Dispersive shift, approximately given by $\chi \approx \frac{g^2}{\Delta}$

Mechanism

- The resonator frequency becomes dependent on the qubit state:

$$\omega_r^{\text{eff}} = \omega_r \pm \chi$$

- If the qubit is in $|0\rangle$, the resonator responds at $\omega_r - \chi$; if in $|1\rangle$, at $\omega_r + \chi$.
- A weak microwave probe at ω_r is sent into the resonator.
- The phase and amplitude of the reflected or transmitted signal depend on the qubit state.

Measurement Process

1. Prepare the qubit in an unknown state.
2. Apply a weak probe tone to the resonator.
3. Detect the output using homodyne or heterodyne detection to extract phase and amplitude information.
4. Use this information to infer the qubit's state without directly exciting it.

Measurements:

1. JQF reflection spectrum
 - Measurement Technique:
 - The JQF is characterized by measuring its reflection spectrum via the control line.
 - Both amplitude and phase of the reflection coefficient are measured as a function of frequency and probe power.
 - **Low Power Regime (−146 dBm):**

- At this small probe power, the JQF is in the *over-coupling regime*, where:

$$\gamma_f^{\text{ex}} \gg \gamma_f^{\text{in}}$$

- In this regime, the external coupling dominates over internal losses.
- Over-coupling is essential for achieving near-perfect reflection of single photons emitted by the qubit.

- **Saturation of the JQF:**

- As the probe power is increased to about -120 dBm, the JQF begins to *saturate*.
- This saturation corresponds approximately to the single-photon power level.
- Saturation leads to a reduced response of the JQF, which is crucial for selective transparency under strong driving.

- **Two-Photon Transition:**

- A second dip appears in the reflection spectrum near 7.8 GHz, which corresponds to a two-photon transition.
- This transition occurs between the ground state $|0\rangle$ and the second excited state $|2\rangle$ of the JQF.

- **High Power Regime (-100 dBm):**

- At this stronger probe power, the JQF is *completely saturated*.
- In the saturated state, the JQF no longer reflects incoming signals.
- This property allows strong qubit control fields to transmit through the JQF without attenuation.
- It demonstrates the nonlinear filtering capability of the JQF: reflecting weak signals (like qubit emission) but transmitting strong control pulses.
- The JQF acts as a power-dependent filter:
 - * Reflects low-power signals to suppress qubit relaxation.
 - * Transmits high-power signals for fast qubit control.
- This nonlinear behavior is essential for resolving the trade-off between qubit coherence and controllability.

2. Experimental Observations and Qubit–JQF Interaction

- Time-domain measurements are conducted on the qubit to study the effect of the JQF, using various JQF–qubit detunings.
- The population of the qubit in the excited state is determined from the averaged quadrature amplitude of the readout pulse.
 - The averaged amplitude is normalized by accounting for the qubit’s thermal population at each detuning.
- Qubit relaxation measurements are carried out with and without the JQF.
 - When the JQF is nearly resonant with the qubit, the qubit exhibits an exponential decay with a longer relaxation time (The relaxation time of a qubit, often denoted as T_1 , refers to the characteristic timescale over which a qubit in its excited state decays to its ground state due to energy dissipation.).

- The thermal population of the qubit increases from 2.8% to 16.2%.
- This increase is not due to thermal noise from the JQF but results from the qubit decoupling from the control line, which has a lower effective temperature than the intrinsic loss channel.
- The relaxation time T_1 and Hahn-echo coherence time T_2^E are plotted as functions of the JQF–qubit detuning.

Hahn-Echo Coherence Time T_2^E :

- T_2^E is the characteristic timescale over which a qubit maintains phase coherence in the presence of slowly varying (low-frequency) noise.
- It is measured using the *Hahn echo* technique, which is a spin-echo protocol designed to refocus dephasing caused by quasi-static noise.
- The protocol consists of three pulses:
 - (a) A $\pi/2$ pulse to place the qubit in a superposition state.
 - (b) A delay τ , during which dephasing begins.
 - (c) A π pulse to invert the qubit's state.
 - (d) Another delay τ , followed by a measurement.
- The π -pulse effectively cancels the effect of low-frequency dephasing, revealing the intrinsic coherence limit.
- The coherence decay envelope is approximately exponential:

$$\langle \sigma_x(t) \rangle \propto \exp\left(-\frac{t}{T_2^E}\right)$$

- T_2^E is related to:

$$\frac{1}{T_2^E} = \frac{1}{2T_1} + \Gamma_\phi^E$$

where:

- * T_1 : energy relaxation time.
- * Γ_ϕ^E : dephasing rate after echo correction (typically smaller than pure dephasing rate without echo).

- Both times are enhanced by approximately a factor of 4 when the JQF is near resonance with the qubit.
- The enhancement is primarily limited by intrinsic energy relaxation and thermal population of the qubit.
- The full width at half maximum (FWHM) of the enhancement spectrum is about 60 MHz.
 - This value roughly corresponds to the external coupling rate of the JQF.
 - Since 60 MHz is about 1% of the JQF frequency, a frequency-fixed JQF may be implemented using current techniques for tuning Josephson energy.
- The asymmetry in the enhancement spectrum is due to a slight deviation from the ideal JQF–qubit distance:

$$d = 0.526\lambda_q \quad \text{instead of} \quad d = 0.5\lambda_q$$

- Numerical simulations indicate that at a detuning $(\omega_f - \omega_q)/2\pi = 9\text{ MHz}$, the qubit decay times match those achievable at the ideal distance.

- Rabi oscillations of the qubit are observed under a stationary control field.
 - The photon flux is $\dot{n} = 1.5 \times 10^{10} \text{ s}^{-1}$, corresponding to input power of -101 dBm .
 - The presence of the JQF does not affect the Rabi oscillation frequency but reduces its amplitude due to thermal excitation from intrinsic qubit loss.
- The observed Rabi frequency as a function of JQF–qubit detuning remains constant.
 - This invariance is attributed to the saturation of the JQF by the strong control field.
- A trade-off in conventional qubit control is discussed:
 - The maximum Rabi frequency without a JQF is defined as:

$$\Omega_q \leq 2\sqrt{\frac{\dot{n}}{T_1}}$$

- For a fixed external coupling, the Rabi frequency is:

$$\Omega_q = 2\sqrt{\gamma_q^{\text{ex}} \dot{n}} \leq 2\sqrt{\frac{\dot{n}}{T_1}}$$

- This upper bound is only achievable if intrinsic loss, dephasing, and thermal excitations are negligible.
- With the JQF, the measured Rabi frequency exceeds the conventional upper bound.
 - This result demonstrates that the trade-off in qubit control is effectively broken by introducing the JQF.

Qubit State Manipulation and Pulses on the Bloch Sphere :

Qubit states can be visualized as points on the Bloch sphere:

- $|0\rangle$ (ground state): North pole
- $|1\rangle$ (excited state): South pole
- Superpositions like $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$: Equator

A microwave pulse causes the qubit's state vector to rotate around the Bloch sphere, usually about the X or Y axis, depending on the pulse phase.

What is a π Pulse? A π -pulse rotates the qubit state vector by 180° (radians: π) around an axis (e.g., the X -axis).

$$|0\rangle \xrightarrow{\pi \text{ pulse}} |1\rangle, \quad |1\rangle \xrightarrow{\pi \text{ pulse}} |0\rangle$$

What is a $\pi/2$ Pulse? A $\pi/2$ -pulse rotates the qubit state vector by 90° (radians: $\frac{\pi}{2}$).

$$|0\rangle \xrightarrow{\pi/2 \text{ pulse}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$\pi/2$ -pulses are commonly used in Ramsey and spin echo experiments to put the qubit into and out of superpositions.

How is the Pulse Sent?

In superconducting qubit systems, π and $\pi/2$ pulses are implemented using resonant microwave signals:

- (a) *Frequency Matching*: The qubit has a transition frequency ω_q (e.g., 5 GHz). A microwave tone at frequency ω_q is applied via a control line.
- (b) *Envelope Shaping*: The pulse is modulated with a Gaussian or square envelope. The area under the pulse (amplitude $\times$ duration) determines the rotation angle:

$$\theta = \Omega t$$

where:

- θ = rotation angle (π or $\pi/2$)
 - Ω = Rabi frequency (qubit rotation rate)
 - t = pulse duration
- (c) *Phase Control*: The phase of the pulse determines the rotation axis:
- $0^\circ \rightarrow$ X-axis rotation
 - $90^\circ \rightarrow$ Y-axis rotation
- (d) *Implementation*:
- A digital-to-analog converter (DAC) generates the pulse.
 - The pulse is sent through a transmission line to the qubit.
 - The pulse duration and amplitude are calibrated such that:

$$t_\pi = \frac{\pi}{\Omega}, \quad t_{\pi/2} = \frac{\pi}{2\Omega}$$

If a qubit has a Rabi frequency of 10 MHz:

$$t_\pi = \frac{\pi}{\Omega} = \frac{\pi}{2\pi \cdot 10^7} = \frac{1}{2 \cdot 10^7} = 50 \text{ ns}$$

$$t_{\pi/2} = \frac{t_\pi}{2} = 25 \text{ ns}$$

So:

- A 25 ns pulse $\rightarrow \pi/2$ rotation (puts into superposition)
- A 50 ns pulse $\rightarrow \pi$ rotation (flips state)

Rabi Decay Time:

Rabi oscillations occur when a qubit is driven by a resonant microwave field. The qubit undergoes coherent oscillations between the ground and excited states with a frequency called the *Rabi frequency* Ω . However, due to decoherence mechanisms (such as energy relaxation and dephasing), these oscillations decay over time.

The *Rabi decay time*, often denoted as T_R , is the characteristic time over which the amplitude of the Rabi oscillations diminishes exponentially:

$$P_1(t) = A \sin^2\left(\frac{\Omega t}{2}\right) \cdot e^{-t/T_R} + C$$

where:

- $P_1(t)$: Probability of measuring the qubit in the excited state at time t
- A : Amplitude of oscillation
- Ω : Rabi frequency
- T_R : Rabi decay time
- C : Offset due to state preparation and measurement errors

Qubit Gate Error and Gate Fidelity: In quantum computing, accurate qubit control is essential. Gate operations (such as X , Y , Z , H , and $CNOT$) must be implemented with high precision. Two key metrics to quantify the quality of gate operations are:

Gate fidelity measures how close the actual operation $\mathcal{E}$ is to the ideal unitary gate U .

For a pure state $|\psi\rangle$, the *state fidelity* is:

$$F(|\psi\rangle, \rho) = \langle \psi | \rho | \psi \rangle$$

For a quantum process, the *average gate fidelity* over all input states is:

$$F_{\text{avg}}(U, \mathcal{E}) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle\langle\psi|) U | \psi \rangle$$

Alternatively, using the *process fidelity*:

$$F_{\text{avg}} = \frac{dF_\chi + 1}{d + 1}$$

where:

- F_χ is the fidelity between the ideal and actual process matrices
- d is the Hilbert space dimension (e.g., $d = 2$ for a single qubit)

Gate error is defined as:

$$\epsilon = 1 - F_{\text{avg}}$$

It quantifies the probability that a gate deviates from its ideal behavior. Lower error rates indicate more reliable gate performance. Randomized Benchmarking (RB) measures average fidelity across many sequences, robust to state preparation and measurement (SPAM) errors.

3. Effect of the JQF on Qubit Control Performance

- To assess whether the JQF negatively impacts qubit control, both the Rabi frequency and Rabi decay time of the qubit are measured as functions of control amplitude.
- In the presence of the JQF:
 - When the JQF is not fully saturated (i.e., $\Omega_f/\gamma_f^{\text{ex}} \ll 1$), the Rabi frequency is reduced.
 - The Rabi decay time is also shortened compared to the case without the JQF.
- In the high-power regime $\Omega_f/\gamma_f^{\text{ex}} \gg 1$:

- The Rabi frequency and decay time become nearly identical with and without the JQF.
- Numerical simulations are conducted by modeling the JQF as a two-level system (instead of a transmon) with the same parameters.
- In the two-level JQF case:
 - The Rabi decay time remains shorter than that without the JQF, even when the JQF is nearly fully saturated.
 - This occurs because a two-level JQF, even when saturated, remains coupled to the qubit via the correlated decay pathway.
 - This introduces additional decay channels for the qubit through the JQF.
- In contrast, the transmon JQF behaves differently:
 - When excited to higher energy levels, it becomes decoupled from the qubit due to its anharmonic energy level structure.
 - The correlated decay is effectively lost, and thus the JQF no longer impacts the qubit's Rabi oscillations.
- Numerical simulations show that the minimum Rabi cycle error is achieved when:

$$|\alpha_f| \approx \gamma_f^{\text{ex}}$$

where α_f is the anharmonicity of the transmon JQF.

- This condition aligns with experimental observations and supports optimal operation of the JQF-enhanced architecture.
- When sequential control pulses are applied:
 - The JQF is expected to be saturated during each gate operation.
 - As a result, it has little impact on overall gate fidelity.
- Randomized benchmarking is used to measure average gate error of Clifford gates on the qubit.
- Experimental results:
 - Gate errors are shown as a function of control pulse duration with and without the JQF.
 - In both cases, the average gate error approaches the coherence limit.
- A small increase in gate error with the JQF is observed:
 - This is attributed to incomplete saturation of the JQF at the start and end of each control pulse.
- The coherence limit of the qubit gate error is primarily determined by its external coupling rate.

Conclusion :

- The trade-off between fast control and long qubit lifetime was successfully resolved by implementing a Josephson Quantum Filter (JQF) on the qubit control line.
- Experimental findings confirmed that:
 - The JQF effectively suppresses radiative decay of the qubit.
 - The Rabi frequency and gate fidelity remain largely unaffected compared to the system without the JQF.

- With the JQF, it becomes feasible to:
 - Enhance the qubit–control-line coupling.
 - Reduce the required Rabi drive power by three orders of magnitude, compared to a weakly coupled system achieving a 1-ms qubit lifetime without a JQF.
- The JQF contributes to scalable quantum computing by:
 - Reducing environmental heating.
 - Suppressing crosstalk between qubits.
- More generally, the experiment shows that:
 - A nonlinear element (JQF) functions as a power-dependent variable boundary condition for microwave modes.
 - This principle is applicable to other parametric control tasks, such as:
 - * Two-qubit gates.
 - * Single-photon generation.
 - * Active cooling of quantum systems.

Doubt :

As shown in Fig. 3d, the observed Rabi frequency with the JQF exceeds this upper bound (indicated by the shadowed area in Fig. 3d), which demonstrates that we break the trade-off in qubit control. I don't understand the correspondence between the two and which shadow area is being referred to?

6 Fully directional quantum-limited phase-preserving amplifier

Challenge Presented :

- Photon loss in circulators and other components reduces overall quantum efficiency, even for near-quantum-limited amplifiers.
- Ferrite-based circulators require strong magnetic fields, which are incompatible with superconducting circuits. Their bulky size prevents direct on-chip integration.
- Many existing amplifiers allow noise from the output to leak back into the input, degrading performance.
- Key challenge is to achieve directional, quantum-limited amplification without ferrite circulators.
- Thermalization causes the output port, connected to other components, to emit thermal photons back into the amplifier, introducing unwanted noise and backaction.

What they do :

- The authors propose a new amplifier design based on a superconducting Josephson device. The circuit has four ports (physical connections for input/output signals) and four internal modes (distinct resonant frequencies or oscillations inside the system).
- Fully directional means the amplifier passes signals only in one direction (input $\rightarrow$ output) while suppressing reverse propagation (reverse isolation being greater than the forward gain). Example: Forward gain = +20 dB, reverse isolation = -30 dB $\Rightarrow$ reverse suppression stronger than forward amplification.
- Quantum-limited means it adds the minimum noise allowed by quantum mechanics during amplification.
- Phase-preserving means both quadratures of the signal (X and Y) are amplified equally, preserving phase information. Whereas a phase-sensitive amplifier (PSA) amplifies only one specific quadrature of an input signal while deamplifying the orthogonal quadrature.
- The amplifier uses six different parametric interactions (frequency-mixing processes driven by pump tones). These interactions couple every mode to every other mode in a carefully engineered way. Interference between these processes is tuned so that amplification happens only in the forward direction while the reverse path is canceled (Reverse isolation = suppression of signals traveling from output back to input).
- Matched ports means the impedance of the amplifier matches the connected circuits (usually 50 Ω), avoiding unwanted reflections. Impedance-matched ports allow direct connection to other superconducting quantum circuits without extra matching components.
- Directionality prevents signals from a mismatched output from traveling back into the amplifier. This ensures stable operation even if the load at the output is not perfectly matched.
- Minimal backaction means only vacuum noise goes back to the signal source.

Minimal Scattering Matrix of fully directional amplifier:

- They describe a linear amplifier in the language of input-output theory using its scattering matrix representation.
- For an amplifier with N ports:
 - Each port has an field operator a_n^{in} and its Hermitian conjugate $a_n^{\text{in}\dagger}$.
 - Each port also has an output field operator a_n^{out} and $a_n^{\text{out}\dagger}$.
 - These operators represent the quantum fields (or modes) at that port, including both the signal and noise components.
- The input and output fields are related by the $2N \times 2N$ scattering matrix $\tilde{S}$:

$$\begin{bmatrix} a_1^{\text{out}} \\ a_1^{\text{out}\dagger} \\ \vdots \\ a_N^{\text{out}} \\ a_N^{\text{out}\dagger} \end{bmatrix} = \tilde{S} \begin{bmatrix} a_1^{\text{in}} \\ a_1^{\text{in}\dagger} \\ \vdots \\ a_N^{\text{in}} \\ a_N^{\text{in}\dagger} \end{bmatrix}$$

This matrix fully captures how inputs are transformed into outputs.

- The reason for explicitly separating a and $a^\dagger$ in the vector because a general amplifier does not necessarily amplify the two quadratures equally.

The four requirements for fully directional quantum limited phase preserving amplification are:

1. The scattering matrix $\tilde{S}$ can always be transformed into a block-diagonal form using a proper mode basis (In the original port basis, the scattering matrix $\tilde{S}$ might be complicated, with couplings between a and $a^\dagger$ all mixed together. By applying a linear transformation L , we reorganize the variables into a *proper* mode basis where the matrix becomes simpler. This procedure is analogous to diagonalizing a matrix to reveal its structure more clearly except here, the goal is not a diagonal form, but a block-diagonal form):

$$L\tilde{S}L^{-1} = \begin{pmatrix} S & 0 \\ 0 & S^* \end{pmatrix}$$

- L is the linear transformation from the original mode basis (Eq. (1)) to the proper basis.
- S is an $N \times N$ matrix often referred to as the scattering matrix, even though it can mix a_n and $a_m^\dagger$.
- The conjugate relationship between the two submatrices (S and S^*) ensures phase-preserving amplification.
- In the graph-based notation, this property ensures that the full scattering graph can be decomposed into two subgraphs.

A *block-diagonal matrix* is a matrix where the only nonzero entries are inside square submatrices (blocks) along the diagonal, and all off-block-diagonal entries are zero.

$$M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$$

Here:

- A and B are themselves matrices (blocks).
- The top-right and bottom-left sections are zero.

In our case:

$$L\tilde{S}L^{-1} = \begin{bmatrix} S & 0 \\ 0 & S^* \end{bmatrix}$$

means the transformed scattering matrix has two independent submatrices S and S^* that do not mix.

- In the *original port basis*, a and $a^\dagger$ are all mixed together in $\tilde{S}$, making the structure hard to see.
- By choosing the right mode basis (through L), we can separate the “normal” and “conjugate” parts into two independent chunks.

Why S and S^* :

- For *phase-preserving amplifiers*, physics demands that the two blocks are complex conjugates.
- This guarantees that both quadratures of the signal are amplified equally, preserving the quantum information.

2. The scattering matrix $\tilde{S}$ must be *symplectic*, meaning:

$$\tilde{S}^T \cdot J \cdot \tilde{S} = J$$

where J is the symplectic matrix:

$$J = \begin{bmatrix} 0 & 1 & & & \\ -1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{bmatrix}.$$

This condition means:

- The device preserves the information content of the input signal.
- Equivalently, it preserves the commutation relations of the input field operators.
 - In quantum optics, the *information content* of a signal is encoded in its quadratures x and p , or equivalently in the annihilation/creation operators a and $a^\dagger$.
 - These quadratures obey the canonical commutation relation:

$$[x, p] = i \quad (\hbar = 1),$$

which sets the quantum uncertainty limit.

- Preserving the commutation relation is essential:
 - * If $[x, p] < i$, phase-space area is reduced $\Rightarrow$ irreversible loss of quantum information.
 - * If $[x, p] > i$, extra unphysical correlations are introduced $\Rightarrow$ violates realizability in quantum mechanics.

- Physically, a symplectic transformation can rotate, mix, or squeeze modes, but must preserve phase-space area, meaning the fundamental quantum noise constraints and information content remain unchanged.

3. The subscattering matrix between the input port (port 1) and output port (port 2) must be:

$$[S]_{1,2} = \begin{pmatrix} 0 & 0 \\ \sqrt{G}e^{i\phi} & 0 \end{pmatrix}$$

where:

- $S_{21} = \sqrt{G}e^{i\phi}$ is the gain from input to output.
- This condition ensures that:
 - Both input and output ports are *matched*.
 - Reverse isolation is infinite ($1/|S_{12}| \rightarrow \infty$).
 - (a) Meaning of the entries: The element $S_{21} = \sqrt{G}e^{i\phi}$ means that the amplitude of the signal going from port 1 (input) to port 2 (output) is $\sqrt{G}$, with a phase shift ϕ . All other entries in the given submatrix are zero, meaning no direct coupling in those directions.
 - (b) Why the ports are matched: Matching means no reflection at a port. Mathematically, $S_{11} = 0$ means no signal from the input reflects back to the input, and $S_{22} = 0$ means no signal from the output reflects back to the output. Both conditions are satisfied here.
 - (c) Why the reverse isolation is infinite: Reverse isolation measures how much signal injected into the output port returns to the input port, quantified by $|S_{12}|$. If $S_{12} = 0$, then $\frac{1}{|S_{12}|} \rightarrow \infty$, meaning no signal flows backward from output to input.
 - (d) Summary: The matrix specifies: (i) $S_{21} = \sqrt{G}e^{i\phi}$ — strong forward gain with phase control; (ii) $S_{12} = 0$ — infinite reverse isolation; (iii) $S_{11} = S_{22} = 0$ — perfect impedance matching.

4. The coupling between auxiliary ports and the main ports (input/output) must satisfy:

$$\sum_{p=3}^N |S_{1p}|^2 = 1, \quad \sum_{p=3}^N |S_{2p}|^2 \approx G \quad \text{for } G \gg 1$$

This ensures backaction and added noise are quantum-limited when only vacuum noise enters the output and auxiliary ports.

- In summary:
 - Solving the above four set of conditions one can obtain equations for N^2 complex elements of S (**yet to understand derivation on this from supplementary section**).
 - * Solutions exist only for $N \geq 4$.
 - * At least four ports are required: input, output, and two ancillary ports (Ancillary means supporting or secondary).
 - S is the scattering matrix. This is the complete $2N \times 2N$ symplectic matrix that describes how both annihilation operators a and creation operators $a^\dagger$ are transformed by the device.

- $\tilde{S}$ is the full scattering matrix. This is just the upper-left $N \times N$ block of $\tilde{S}$ in the proper mode basis after block-diagonalization.
- For a four-port amplifier system, there are two distinct types of 4×4 scattering matrices that satisfy the desired properties for a fully directional amplifier.
 - These are called the *cis amplifier* (C amp) and the *trans amplifier* (T amp).
 - The names “cis” and “trans” come from the topology of their scattering graphs.
- The minimal form (Minimal form means the simplest representation that still captures all essential properties without unnecessary parameters) of the scattering matrices for the two cases are:

- C amp:

$$S_C = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \sqrt{G} & 0 & 0 & \sqrt{G+1} \\ \sqrt{G+1} & 0 & 0 & \sqrt{G} \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- T amp:

$$S_T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \sqrt{G+1} & 0 & 0 & \sqrt{G} \\ 0 & 1 & 0 & 0 \\ \sqrt{G} & 0 & 0 & \sqrt{G+1} \end{pmatrix}$$

where $G \geq 0$ is the power gain.

- The mode bases for the input and output are different for the two amplifier types:
 - For the C amp:

$$A_C^{\text{in,out}} = \begin{pmatrix} a_1 \\ a_2^\dagger \\ a_3 \\ a_4^\dagger \end{pmatrix}_{\text{in,out}}$$

- For the T amp:

$$A_T^{\text{in,out}} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4^\dagger \end{pmatrix}_{\text{in,out}}$$

- These matrices have been made positive and real by applying an appropriate phase shift to the incoming and outgoing wave modes (The original scattering matrices are generally complex-valued because waves at each port carry complex amplitudes with both magnitude and phase. Phase shifts arise naturally from physical propagation delays and device properties. By applying a change of phase to the incoming and outgoing waves (i.e., redefining the reference phase of each mode), the matrices can be transformed so that their elements become real and positive. This is mathematically equivalent to multiplying the scattering matrix by diagonal unitary matrices of phase factors (rotations in complex plane), which does not change the physical behavior but simplifies the representation).
 - The specific forms above are for the *special case* of a resonant input signal frequency.

- Key similarities:
 - In the high-gain limit ($G \gg 1$), both amplifier types provide the *same* scattering relation between input and output ports.
- Key differences:
 1. Matching:
 - C amp: All ports are matched (no reflection gain).
 - T amp: One ancillary port exhibits reflection gain.
 2. Coupling of frequency components:
 - C amp: Couples *conjugate* frequency components between input and output (annihilation operator in one port couples to creation operator in another).
 - T amp: Couples *same* frequency components between input and output (annihilation couples to annihilation).
- Role of the two auxiliary ports (ports 3 and 4):
 - When terminated with matched cold loads:
 1. They act as “dumps” for signals entering from the output port, preventing signals from returning to the input port (providing isolation).
 2. Their mode-conjugation ensures that vacuum noise is injected into the input port, as required by the minimum quantum backaction principle.

Mode Coupling of the 4PFDA:

- After determining the minimal scattering matrices for the amplifiers, the next step is to understand how these matrices arise from a coupled-mode system.
- In general, such systems can have more modes than ports:
 - Increasing the number of modes can help increase the amplifier’s bandwidth.
 - Here, the focus is on the minimal case where the number of modes equals the number of ports, i.e., an N -port N -mode system.
- The scattering matrix S for this coupled-mode system can be expressed as:

$$S = (\Sigma + M)^{-1}(\Sigma - M),$$

where

$$\Sigma = \begin{pmatrix} \kappa_1/2 & 0 & \cdots & 0 \\ 0 & \kappa_2/2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \kappa_N/2 \end{pmatrix}$$

represents the amplitude damping rates of the modes due to their coupling to the environment.

- The matrix M describes couplings between modes (M_{mn}) and detunings:

$$M = \begin{pmatrix} i\Delta_1 & M_{12} & \cdots & M_{1N} \\ M_{21} & i\Delta_2 & \cdots & M_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ M_{N1} & M_{N2} & \cdots & i\Delta_N \end{pmatrix},$$

where $\Delta_n = \omega_{\text{in},n} - \omega_n$ is the detuning of the input drive from mode n .

- Equation for the scattering matrix S can be interpreted as a generalized reflection coefficient between two systems of different impedances.
- Given the scattering matrix S , the coupling matrix M can be extracted using:

$$M = (I - S)(I + S)^{-1},$$

where I is the identity matrix.

- Using the minimal scattering matrices S_C and S_T , the coupling Hamiltonians of the C amp and T amp are found as:

$$H_C/\hbar = \frac{1}{2} (g_{12}a_1a_2 + g_{13}a_1a_3^\dagger + g_{14}a_1a_4 + g_{23}a_2a_3 + g_{24}a_2a_4^\dagger + g_{34}a_3a_4 + \text{h.c.}),$$

$$H_T/\hbar = \frac{1}{2} (g_{12}a_1a_2^\dagger + g_{13}a_1a_3^\dagger + g_{14}a_1a_4 + g_{23}a_2a_3^\dagger + g_{24}a_2a_4 + g_{34}a_3a_4^\dagger + \text{h.c.}),$$

where a_n and $a_n^\dagger$ are annihilation and creation operators for mode n .

- The coupling rates g_{mn} are given by:

$$g_{mn} = \pm i\sqrt{\kappa_m \kappa_n} \frac{\sqrt{G-1}}{\sqrt{G+1}},$$

for terms involving two annihilation operators (two-mode squeezing), and

$$g_{mn} = \pm i\sqrt{\kappa_m \kappa_n},$$

for terms involving one annihilation and one creation operator (frequency conversion).

- The sign (phase) of each coupling term is uniquely determined by the corresponding scattering matrix.
- Physical interpretation of the couplings:
 - Two-mode squeezing terms lead to phase-preserving amplification with photon-number gain G in reflection and gain $G - 1$ in transmission.
 - Frequency conversion terms lead to perfect photon conversion between modes, denoted as conversion efficiency $C = 1$.

- More generally, the conversion efficiency C is related to the coupling rate by an additional prefactor $\sqrt{C}(1 + (1 - C)^{-1/2})$. Specifically, the effective coupling rate becomes:

$$g_{mn}^{(\text{effective})} = g_{mn} \times \sqrt{\frac{C}{1 - C}}.$$

This factor $\sqrt{\frac{C}{1 - C}}$ arises from the underlying physics of coupled-mode theory and ensures that the scattering parameters correctly describe a conversion process with efficiency C .

- * When $C \rightarrow 1$, the prefactor $\sqrt{\frac{C}{1 - C}} \rightarrow \infty$, representing very strong coupling to achieve near-perfect conversion.
- * When $C \rightarrow 0$, the prefactor approaches zero, indicating very weak coupling and negligible conversion.
- The coupling Hamiltonians are graphically depicted
 - The total phases of the three couplings associated with each mode are either $\pi/2$ or $-\pi/2$.
 - Each closed loop of three coupled modes forms a three-port directional amplifier or circulator, known from prior works.
 - The coupling graphs of C amp and T amp correspond to two different ways to connect a three-port directional amplifier to a three-port circulator.
- Previous works:
 - One cited reference proposed a four-port four-mode system with five couplings for quantum-limited directional phase-preserving amplification.
 - However, in that system the input port exhibits unity reflection, which is undesirable as it can cause backaction on the signal source.
 - The present four-port four-mode system with six properly arranged couplings represents the minimal construction of a fully directional quantum-limited phase-preserving amplifier without such reflection.
 - For fully directional phase-sensitive amplification, a three-port three-mode system with five couplings suffices.

Effects of Imperfect Conversion in 4PFDA:

- Until now, amplifier performance has been discussed under ideal conditions:
 - Perfect frequency conversion ($C = 1$)
 - Resonant input signal (no frequency detuning)
- This section considers more practical operating conditions:
 - Imperfect frequency conversion ($C < 1$)
 - Detuned input signals (frequency different from resonance)

- Under these realistic conditions, both cis amplifier (C amp) and trans amplifier (T amp) show superior performance compared to existing directional amplifiers that rely on interference between multiple parametric processes.
- The effect of imperfect frequency conversion ($C < 1$) on amplifier performance is studied:
 - Imperfect conversion generally causes reflections at input and output ports.
 - They show scattering matrix elements between input and output ports of C amp and T amp as functions of phase-preserving gain G .
 - Two frequency conversion values are considered: $C = 0.99$ and $C = 0.999$, both achievable in experiments.
- For the C amp:
 - Reflection coefficients S_{11} and S_{22} and reverse transmission S_{12} increase as gain G between paired modes increases.
 - These quantities eventually saturate for high gain values ($G > 40$ dB).
 - Better frequency conversion leads to better forward gain S_{21} and improved isolation in the reverse direction S_{12} .
 - At conversion $C = 0.99$, the amplifier achieves about -15 dB matching on both input and output ports while providing 20 dB forward gain.
 - Achieving similar input matching and gain with a single-parametric reflection amplifier typically requires at least two circulators before the amplifier.
- For the T amp:
 - Both reflections and forward gain increase rapidly once the gain coupling between paired modes exceeds $\frac{4}{1-C}$.
 - Despite this, impedance matching at both input and output ports remains better than -20 dB while forward gain approaches 20 dB.
 - This performance can be achieved with phase-preserving gain around 17 dB between coupled modes for both considered frequency conversion values.
 - Reverse isolation (suppression of S_{12}) is largely insensitive to the strength of phase-preserving gain.
- Key points:
 - Both amplifiers tolerate imperfect frequency conversion well, though C amp shows increased reflections and reverse transmission at high gain.
 - T amp maintains excellent impedance matching and isolation over a broad range of gain values.
 - Experimentally feasible frequency conversion efficiencies ($C = 0.99$ or 0.999) enable high forward gain and good matching without additional circulators.

Frequency dependence of 4PFDA performance:

- The amplifier performance is analyzed for *detuned input signals*, i.e., signals whose frequency deviates from the resonant frequency.

- The scattering matrix for detuned input signals is derived from the amplifier Hamiltonian.
 - These scattering matrices become *complex-valued functions* depending on the amount of signal detuning.
 - Detailed expressions and derivations are provided in the Supplemental Material [26].
- They present four key scattering matrix elements between input and output ports for two frequency conversion strengths:

$$C = 1 \quad \text{and} \quad C = 0.99,$$

while maintaining a fixed forward gain of

$$|S_{21}|^2 = 20 \text{ dB}$$

at the resonant input frequency.

- It is assumed, without loss of generality, that all modes share the same linewidth κ , simplifying comparisons between the two amplifier types.
- A major difference between the cis amplifier (C amp) and trans amplifier (T amp) lies in the *frequency dependence of their impedance matching*:
 - For the C amp, impedance matching *rapidly deteriorates* as the input signal detunes away from resonance.
 - For the T amp, good impedance matching is maintained over a *much wider range* of detuning.
- The robustness of impedance matching in the T amp arises because:
 - Two out of the three parametric processes directly involving the input and output modes correspond to *frequency conversions* (these processes take an input signal at frequency ω_1 and convert it to a signal at frequency ω_2 . The important feature: no net amplification occurs in the same mode — the energy is transferred between modes at different frequencies).
 - Frequency conversion processes inherently *reduce reflections*, enhancing matching across frequencies.
- In contrast, the C amp has two of its three parametric processes as *amplifications*, which tend to:
 - Produce *amplified reflections*, causing impedance mismatch at detuned frequencies.
- When examining the forward gain and reverse isolation as functions of detuning:
 - Both amplifier types exhibit *similar frequency dependence* in forward gain and isolation.
 - However, the C amp achieves *better reverse isolation* near the resonant frequency.
- Overall, the T amp is considered *easier to operate* because:
 - It requires *less gain and frequency conversion* to achieve the same forward gain as the C amp.
 - Its impedance matching is more *robust over frequency detuning*, making it more tolerant to input frequency variations.

Bandwidth of 4FPDA:

- A useful metric for characterizing the frequency-dependent performance of an amplifier is its *bandwidth*.
 - Bandwidth is typically defined as the frequency range over which the amplifier's gain drops by 3 dB from its desired value.
- For a resonator-based single-pump amplifier (such as a two-port, two-mode phase-preserving amplifier), the bandwidth B is related to the geometric mean of the linewidths of its two modes, denoted by $\bar{\kappa}$, and the gain G by:

$$\bar{\kappa} \approx B\sqrt{G},$$

which expresses the well-known *fixed gain-bandwidth product*.

- For amplifiers involving multiple parametric processes, such as the cis amplifier (C amp) and trans amplifier (T amp), the bandwidth definition is more complex.
 - Multiple scattering parameters affect their performance.
 - Therefore, bandwidth is defined as the frequency range over which *all* relevant scattering parameters maintain desired performance.
 - In these amplifiers, the four scattering parameters between the input and output ports are all relevant.
- The bandwidth of the C amp and T amp is thus defined as the frequency range where all the following conditions are met:

$$|S_{11}|^2, |S_{22}|^2 \leq 0.01, \quad |S_{12}|^2 \leq \frac{1}{G}, \quad |S_{21}|^2 \geq \frac{G}{2},$$

where G is the forward gain at the resonant input frequency.

- Explanation of these conditions:
 - The input matching condition $|S_{11}|^2, |S_{22}|^2 \leq 0.01$ ensures less than 1% reflection, comparable to commercially available circulators used in circuit-QED experiments.
 - The reverse isolation condition $|S_{12}|^2 \leq \frac{1}{G}$ prevents the formation of positive feedback loops caused by reflections from downstream components and the signal source.
 - * This guards against signals being reflected from the output back to the input, which could destabilize the amplifier.
 - * The definition assumes the worst-case scenario of full reflection from downstream components to provide a lower bound for reverse isolation bandwidth.
 - * Reverse isolation also protects the signal source from reflected noise originating from downstream components.
 - * The required level of reverse isolation can vary depending on how susceptible the signal source is to such disturbances, making it application dependent.
- Bandwidth is plotted as a function of forward gain for the different conditions.
 - For the C amp, the bandwidth limited by the impedance-matching condition is about 10 times narrower than bandwidths set by forward gain and reverse isolation.
 - * Hence, impedance matching defines the overall amplifier bandwidth for C amp.

- For the T amp, the 3-dB bandwidth of the forward gain sets the amplifier bandwidth.
- Both amplifiers exhibit 3-dB bandwidths of forward gain that are approximately twice as large as that of a resonator-based single-parametric amplifier (e.g., Josephson Parametric Amplifiers, JPAs).
 - The scaling of bandwidth with gain is similar across these amplifier types.
- Overall, the T amp is considered more favorable for most applications due to:
 - Wider impedance matching bandwidth.
 - Larger effective operational bandwidth at comparable gain.

Conclusion:

- The work presented two fully directional, quantum-limited, phase-preserving parametric amplifiers with matched input and output ports and perfect reverse isolation. These amplifiers are implemented in four-port, four-mode systems using carefully arranged two-mode squeezing and frequency-conversion couplings.
- T amplifier: Has frequency-conversion coupling between its input and all other modes. Offers superior operating bandwidth compared to single-pump parametric amplifiers and three-port directional amplifiers based on multiparametric couplings.
- C amplifier: Has all ports matched. Is robust against impedance mismatch on auxiliary ports under practical operating conditions.
- While the design suggests six couplings are needed, the number of active parametric couplings can be reduced by exploiting frequency degeneracy among the four modes. This ensures passive frequency-conversion couplings always exist.

Doubts :

1. Is there a way to determine a relation between Gain and Bandwidth for multiple scattering parameters.
2. I don't understand the difference between the effects of frequency conversion for $C=0.99$ and $C=0.999$ on amplifier performance.

7 Faithful conversion of propagating quantum information to mechanical motion

Challenge presented : Mechanical Oscillators combined with superconducting quantum circuits haven't been used to manipulate qubit states.

1. What they do:

- They show how to convert superpositions of zero and one photons into vibrations of a tiny mechanical resonator. The conversion works with higher accuracy than what's possible using only classical physics.
- This capability is key for using mechanical resonators as links between microwave and optical quantum systems or as quantum memory units. It also helps explore whether quantum mechanics still works for very massive systems.

2. Background

- Quantum communication networks using superconducting qubits and modular quantum computing architectures require the ability to:
 - (a) Store propagating microwave fields.
 - (b) Amplify these fields.
 - (c) Shift their frequency.
- A single electromechanical device can provide all of these functionalities by rapidly varying the parametric coupling between:
 - (a) Mechanical motion.
 - (b) Microwave fields.

3. State Capture via Parametric Coupling

- Sudden switching off the interaction between a microwave field and mechanical motion enables:
 - (a) Conversion of the state of a propagating field into a mechanical resonator's motional state.
 - (b) Trapping (storage) of this state within the mechanical resonator.
- This capture process is essential for general quantum information processing.

4. Gaussian vs Non-Gaussian States

- In quantum information, states with non-Gaussian statistics are crucial, for example:
 - (a) Qubits encoded as superpositions of zero and one photons.
- Systems restricted to Gaussian states can be simulated efficiently on a classical computer.
- In most electromechanical devices, the governing equations are linear, meaning:
 - (a) Gaussian input states remain Gaussian during evolution.
 - (b) No intrinsic generation of non-Gaussian states occurs.

5. Accessing Non-Gaussian Mechanical States

- To achieve non-Gaussian mechanical states with propagating microwave fields, one requires:

- (a) A non-Gaussian microwave source, or
- (b) A nonlinear detector, for example, a single-photon counter.
- Without relying on propagating fields, non-Gaussian mechanical states have been achieved by:
 - (a) Resonant coupling of a mechanical system to a qubit.
- More recent proposals involve parametric coupling via an intermediate cavity bus:
 - (a) This may allow access to non-Gaussian states.
 - (b) However, predicted transfer fidelity is low due to intrinsic losses in the cavity.
- Fabricating such hybrid devices typically reduces the coherence of:
 - (a) The mechanical resonator.
 - (b) The qubit.
 - (c) The cavity.

6. Contribution of This Work

- The authors demonstrate conversion of non-Gaussian states from propagating microwave fields into the motion of a micrometre-scale mechanical resonator.
- They employ an electromechanical device to:
 - (a) Capture single photons from a superconducting qubit.
 - (b) Store these photons in the mechanical resonator.
 - (c) Amplify the stored states.
- The density matrix of the mechanical resonator is reconstructed using:
 - (a) Quantum state tomography.

7. Storage of Quantum States

- The experiment demonstrates that a quantum state can be stored for a timescale exceeding $100 \mu\text{s}$.
- This represents an improvement of more than four orders of magnitude compared to previous work that stored a non-Gaussian state in an electromechanical device.

8. Characterizing the Capture Process

- To study how the capture process affects arbitrary propagating qubit states, the experiment captures superpositions of zero and one photons.
- The preservation of quantum information is quantified by the average fidelity:

$$F_{\text{avg}} = 0.83^{+0.03}_{-0.06}$$

where the bounds indicate the 90% confidence interval.

- This fidelity exceeds the maximum achievable using only classical resources, showing that the electromechanical device is suitable for quantum information transduction.

9. Device Design:

- The device consists of a tunable inductor–capacitor (LC) circuit coupled to a mechanical resonator.
- The coupling and tunability come from the capacitor’s upper plate, which is a 100 nm thick suspended and tensioned aluminium membrane free to vibrate.
- The fundamental vibrational mode of the membrane resembles a drumhead and has a mechanical resonance frequency:

$$\frac{\omega_m}{2\pi} \approx 9.3 \text{ MHz.}$$

- If displaced by the resonator’s zero-point motion of 6.4 fm, the LC circuit’s resonance frequency shifts by:

$$\frac{g_0}{2\pi} \approx 280 \text{ Hz.}$$

- The LC circuit also couples inductively to propagating microwave fields in a nearby transmission line at a rate:

$$\frac{\kappa_{LC}}{2\pi} \approx 3 \text{ MHz.}$$

- The LC circuit is tuned into resonance with a narrow-band, fixed-frequency photon source using a third electrode.
- A DC voltage $V_{d.c.}$ applied between this electrode and the membrane controls the static separation between the membrane and the microwave electrode, thereby adjusting the circuit’s resonance frequency.

10. Connection to the Photon Source

- The electromechanical device is connected to an on-demand single-photon source via a microwave network.
- This system consists of a transmon qubit with transition frequency:

$$\frac{\omega_q}{2\pi} = 5.652 \text{ GHz}$$

embedded in a microwave cavity.

- When the qubit is in the ground state $|g\rangle$, the cavity resonance frequency is:

$$\frac{\omega_c^g}{2\pi} = 7.290 \text{ GHz.}$$

- When the qubit is in the excited state $|e\rangle$, the cavity resonance frequency becomes:

$$\frac{\omega_c^e}{2\pi} = \frac{\omega_c^g - 2\chi}{2\pi} = 7.283 \text{ GHz,}$$

where χ is the dispersive shift.

11. Single-Photon Generation

- A control pulse is used to drive the transition:

$$|g\rangle|0\rangle \rightarrow |e\rangle|1\rangle,$$

where $|0\rangle$ and $|1\rangle$ correspond to zero and one cavity photons.

- The cavity state evolves into a propagating field through the transmission line.
- The emitted photon has:
 - (a) Center frequency ω_c^e .
 - (b) Narrow bandwidth $\kappa_c/2\pi = 60$ kHz, matched to the electromechanical device's requirements.

12. Parametric Coupling Mechanism

- The propagating microwave field is parametrically coupled to the membrane's motion by applying pump tones to the LC circuit.
- To capture the state of the propagating field, a pump is applied that is detuned below the LC resonance by:

$$\Delta_r = -\omega_m.$$

- This red-detuned pump creates an interaction that allows the exchange of quantum states between:
 - (a) The input microwave field.
 - (b) The mechanical resonator.
- For optimal capture efficiency, the coupling rate:

$$\Gamma_r(t) = \frac{4g_0^2 n_r(t)}{\kappa_{LC}}$$

must be modulated according to the temporal envelope of the input field.

- Here, $n_r(t)$ is the number of photons induced in the LC circuit by the red-detuned pump.

13. Amplification via Blue-Detuned Pump

- If instead a blue-detuned pump is applied with:

$$\Delta_b = +\omega_m,$$

a two-mode squeezer interaction is generated.

- This interaction amplifies both:
 - (a) The motion of the mechanical resonator.
 - (b) The incident microwave field.
- During amplification, the LC circuit emits a propagating field with a temporal envelope that increases exponentially at a rate:

$$\Gamma_b/2,$$

where:

$$\Gamma_b(t) = \frac{4g_0^2 n_b(t)}{\kappa_{LC}}$$

and $n_b(t)$ is the average photon number induced in the LC circuit by the blue-detuned pump.

- The state of the emitted field depends jointly on:

- (a) The pre-amplification state of the resonator.
- (b) The pre-amplification state of the incident microwave field.

14. Experimental Protocols

- Two protocols are used to characterize the capture process:
 - (a) Calibration protocol.
 - (b) Capture protocol.
- The goal is to compare:
 - (a) The state at the input of the electromechanical device.
 - (b) The state stored in the resonator after capture.
- These protocols are first tested using coherent microwave signals whose frequency and band-width match those produced by the circuit quantum electrodynamics (cQED) system.

15. Calibration Protocol

- In the calibration protocol, the input field is amplified directly and then measured.
- This is implemented by applying the blue-detuned pump at the same time as the input field.
- In this case, the electromechanical device functions as a linear phase-preserving amplifier:
 - (a) The amplifier's input is the incident microwave field.
 - (b) The amplifier's output is the reflected microwave field.
- The incident and reflected pulsed fields have different temporal envelopes, but can be related via an appropriate filter.
- The energy gain is given by:

$$\cosh^2 \frac{r}{2},$$

where:

$$r = \Gamma_b \tau_b$$

and τ_b is the pump duration.

- In this interpretation, the fluctuations in the resonator's motion act as the source of the amplifier's added noise.
- The amplifier reaches the quantum limit when the resonator is in its ground state.

16. Capture Protocol

- In the capture protocol, the red-detuned pump is first applied coincident with the input field.
- This captures the field's state into the resonator.
- After capture, the blue-detuned pump is applied to amplify the resonator's stored state.
- Here, the amplifier's input is regarded as the mechanical resonator state rather than the incident field.
- The output remains the reflected field, but now the added noise originates from the vacuum fluctuations of the incident field.

- In this interpretation, the device operates as a linear phase-conjugating amplifier with an energy gain:

$$\sinh^2 \frac{r}{2}.$$

17. State Tomography

- Operating the device as a low-noise amplifier allows quantum state tomography to be performed on:
 - (a) The input microwave field.
 - (b) The mechanical resonator's motion.
- In each run of either protocol:
 - (a) A voltage signal $V(t)$ is recorded at the detector during amplification.
 - (b) From each $V(t)$ record, a pair of quadrature amplitudes (X, Y) is extracted for the relevant state.
- Repeating the measurement yields a collection of quadrature data points.
- These are processed via a maximum likelihood state tomography method to reconstruct a density matrix ρ .
- The density matrix of the input microwave field is denoted ρ_e , and that of the mechanical resonator is denoted ρ_m .

They test the conversion of non-Gaussian states (single photons) into mechanical motion using an electromechanical device.

18. Calibration (Vacuum Injection):

- Run the calibration and capture protocols without generating single photons.
- Inject a vacuum state to determine the gain of the detector.
- This gain is used to scale the measured quadratures X and Y into units of $\sqrt{\text{quanta}}$.

19. Initial State Preparation:

- Cool the mechanical resonator close to its quantum ground state.
- Achieved occupancy ≈ 0.1 quanta before each protocol.

20. State Tomography Results:

- For both input microwave and mechanical resonator states, reconstructed density matrices show significant elements only along the diagonal (Fock basis).
- Single-photon detection probability:

$$\rho_{e,11} = 0.33^{+0.02}_{-0.01}$$

- Single-phonon probability occupying the mechanical mode after capture:

$$\rho_{m,11} = 0.26^{+0.01}_{-0.02}$$

21. Distinguishing from Thermal/Coherent States:

- Calculate second-order coherence:

$$g_m^{(2)} = 0.89^{+0.05}_{-0.17}$$
- For thermal/coherent states, $g_m^{(2)} \geq 1$.
- The measured $g_m^{(2)} < 1$ confirms non-classicality of the captured state.

22. Storage Time Measurement:

- Vary storage time τ_s to test retention of non-Gaussian states.
- Use master equation modeling with τ_m (characteristic storage time) as the only free parameter.
- Extracted storage time:

$$\tau_m = 137 \pm 6 \mu s$$
- This is ~ 10 times longer than capture time.

23. Testing Arbitrary Qubit State Capture:

- Map $\mathcal{E}$ describes transformation from incident state to captured state.
- Quality measured by average fidelity (measures how indistinguishable the output of the process is from the input, averaged over all pure input states):

$$F_{\text{avg}} = \int d\Psi \langle \Psi | \mathcal{E}(|\Psi\rangle\langle\Psi|) | \Psi \rangle$$

- It suffices to test single-photon state and superpositions of $|0\rangle$ and $|1\rangle$.

24. Creating Superpositions:

- Prepare transmon qubit in:

$$\frac{1}{\sqrt{2}} (|g\rangle + e^{i\phi} |e\rangle)$$
- Transfer qubit state to cavity via $|g, 0\rangle \rightarrow |e, 1\rangle$ transition.
- Let cavity state evolve into a propagating microwave field.
- Apply capture protocol to map field state to mechanical motion.

25. Fidelity Results:

- Reconstruct both ρ_e (input) and ρ_m (mechanical) for this set of states.
- Obtained:

$$F_{\text{avg}} = 0.83^{+0.03}_{-0.06}$$
- This exceeds the classical limit $F_{\text{avg}} \leq 2/3$.

26. Implications:

- Demonstrates ability to transfer quantum states (including non-Gaussian) to a mechanical resonator.

- Enables potential for:
 - Quantum transduction between microwave and optical domains.
 - Mechanical quantum memories for feed-forward operations in quantum computing (teleportation, error correction).
 - Long-lived storage (minute-scale) possible with reduced mechanical dissipation via micro-fabrication advances.

Gaussian states:

In continuous-variable quantum systems (such as modes of the electromagnetic field or mechanical oscillators), we can describe a quantum state in phase space using the quadrature operators:

$$\hat{x} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = \frac{1}{\sqrt{2}i}(\hat{a} - \hat{a}^\dagger)$$

These are analogous to position and momentum.

A complete description of the state can be given by its Wigner function $W(x, p)$.

A Gaussian state is one whose Wigner function is a multivariate Gaussian distribution:

$$W(\mathbf{X}) = \frac{\exp\left[-\frac{1}{2}(\mathbf{X} - \bar{\mathbf{X}})^T \mathbf{\Sigma}^{-1}(\mathbf{X} - \bar{\mathbf{X}})\right]}{(2\pi)^n \sqrt{\det \mathbf{\Sigma}}}$$

where:

- $\mathbf{X} = (x_1, p_1, x_2, p_2, \dots)$ is the vector of quadratures.
- $\bar{\mathbf{X}}$ is the vector of mean values of the quadratures (first moments).
- $\mathbf{\Sigma}$ is the covariance matrix (second moments).

Because this is a Gaussian function in x and p , the state is fully determined by:

1. First moments: average displacement in phase space (where the Gaussian is centered).
 2. Second moments: variances and covariances that define the shape and orientation of the Gaussian profile.
- Vacuum state $|0\rangle$: circular Gaussian centered at the origin.
 - Coherent state $|\alpha\rangle$: same shape as the vacuum, but displaced in phase space.
 - Squeezed state: ellipse in phase space, noise reduced in one quadrature and increased in the other.
 - Thermal state: circular Gaussian with larger radius due to thermal noise.
 - Mathematically simple: only need means and variances to fully describe.
 - Under linear transformations (beam splitters, displacements, phase shifts, squeezing), Gaussian states remain Gaussian.
 - If an entire quantum process is Gaussian, it is classically simulatable — no quantum computational speedup.

Non-Gaussian states in detail:

A non-Gaussian state is any state whose Wigner function is not Gaussian in shape. This can mean:

- The function has oscillations or ripples.
 - It may take on negative values (a purely quantum effect).
 - Higher-order correlations beyond the second moments are needed to describe it.
1. Fock states (number states): The $n = 1$ state (single photon) has a Wigner function with a central dip that goes negative at the origin. These states have a fixed photon number and cannot be represented as a Gaussian distribution.
 2. Schrödinger cat states (superpositions of coherent states): Interference fringes in their Wigner function make them strongly non-Gaussian.
 3. Photon-added or photon-subtracted states: Taking a Gaussian state and adding or subtracting a photon breaks Gaussianity.
 4. Superpositions such as

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

have phase-space features that no Gaussian can reproduce.

Doubt: From each $V(t)$ record, a pair of quadrature amplitudes (X, Y) is extracted for the relevant state. How is it done experimentally? (I would want to know the detailed process)

Is there a mathematical framework to derive/express on how does red detuned pump lead to exchange interaction and blue mode leads to two mode squeezer interaction.

8 Magnon-photon strong coupling for tunable microwave resonators

- Generic theoretical framework means the authors developed a general model that is not tied to a specific device geometry or material, capturing the essential physics so it can be broadly applied.
- Nonreciprocal microwave circulation refers to routing microwave signals in a direction-dependent way (e.g., port $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 1$), preventing reverse flow. This is important for isolating quantum devices and controlling signal flow.
- Multimode cavity magnonic system describes a platform with a microwave cavity (supporting multiple electromagnetic modes) coupled to magnon modes (collective spin-wave excitations in a magnetic material). Multiple modes increase design flexibility.
- Assess the optimal performance means the model is used to predict achievable metrics like isolation, insertion loss, and bandwidth, accounting for realistic losses and couplings.
- Isolation > 56 dB: suppression of reverse transmission, where 56 dB means reverse power is reduced by a factor $\sim 4 \times 10^{-6}$.
- Insertion loss < 0.05 dB: negligible loss in forward direction, meaning $|S_{21}| \approx 1$.
- Flexible bandwidth control means the device's operating bandwidth can be tuned (e.g., between wide-band and narrowband) by adjusting parameters.
- High-quality-factor superconducting cavity based magnonic platforms use superconducting cavities with extremely low loss, enabling high isolation and low insertion loss.
- Spin density is the number of magnetic spins per unit volume in the magnetic material, which determines collective coupling strength g to the microwave field.
- High-spin-density material (e.g., YIG) $\Rightarrow$ strong collective coupling $g \propto \sqrt{N}$, allowing operation in the strong-coupling regime where mode splitting increases usable bandwidth for circulation.
- Low-spin-density material (e.g., V[TCNE]) $\Rightarrow$ smaller g , favoring narrow-band circulator design, which can be advantageous for controlled frequency response in noise-sensitive measurements.
- Highly integratable means the material/device can be easily fabricated and integrated on-chip, which is important for superconducting quantum circuits.
- Narrow-band, low-loss, high-isolation circulators are useful for quantum microwave measurements, where isolation protects the qubit and improves measurement fidelity.
- The same theoretical framework applies to other coupled systems (e.g., optomechanical, electromechanical, photonic), enabling design of tunable nonreciprocal devices in both classical and quantum domains.
- Nonreciprocal microwave devices are essential in classical and quantum information processing to protect sensitive measurements from reflected signals. Traditional nonreciprocity uses magnetic materials, but recent nonmagnetic approaches often have limitations like strict phase matching and complexity. Due to the need for low-loss, tunable, and compact circulators, cavity magnonic systems, which enable strong magnon-photon interactions, have gained interest. While most prior works focus on single-mode reciprocal coupling, recent studies explore nonreciprocal coupling in two-port

systems. Research into three-port nonreciprocal magnonic devices is motivated by applications in sensitive cryogenic microwave measurements.

What they do:

- Presented a generic theoretical model for nonreciprocal multimode cavity magnonic systems.
- Achieved nonreciprocal microwave circulation with high isolation, low insertion loss, and flexible control by exploiting:
 - Selective coupling between magnon and microwave modes with different chiralities. For microwave or magnon modes, chirality often means whether the mode circulates clockwise or counterclockwise, or equivalently, whether it has left-handed or right-handed circular polarization. Coupling magnon modes to microwave modes of different chiralities means selectively interacting with modes that have these distinct directional or polarization characteristics.
 - Interference effects between different paths in a three-port system.
- Proposed a practical device design using a high-quality-factor superconducting ring resonator coupled to a high-Q magnon mode in a low-Gilbert-damping magnetic medium under a bias magnetic field. Gilbert damping is a parameter that describes how quickly the magnetization in a magnetic material loses energy (or “damps”) and relaxes back to its equilibrium state after being disturbed. Mathematically, it appears as a damping term (often denoted by α) in the Landau-Lifshitz-Gilbert (LLG) equation, which governs the dynamics of the magnetization vector in ferromagnets. A low-Gilbert-damping material has a small α , meaning it loses energy very slowly. This results in:
 - Long-lived spin waves or magnons (collective excitations of the magnetic spins).
 - Narrow resonance linewidths in magnetic resonance experiments.
 - Better coherence and stronger magnon-photon coupling since the magnons do not decay quickly.
- Discussed two material platforms:
 - Yttrium iron garnet (YIG): high-spin-density, enabling strong-coupling regime and broader circulation bandwidth.
 - Vanadium tetracyanoethylene (V[TCNE]₂): low-spin-density, highly integratable, suited for narrow-bandwidth operation.
- Commercial circulators typically aim to work over a very wide frequency range (an octave bandwidth or more), so they can handle signals across many frequencies. An octave bandwidth refers to a frequency range where the highest frequency is twice the lowest frequency. This work focuses instead on cavity-enhanced circulation, which uses resonant cavities to boost the circulator’s performance. Because of this, it sacrifices wide bandwidth — meaning the device works well only in a narrower frequency range. But this trade-off results in much higher isolation (better suppression of signals going the wrong way) and much lower insertion loss (less signal power lost passing through). These traits — high isolation and low loss — are critical for delicate single-photon-level quantum measurements, where even tiny losses or back reflections can ruin the measurement.

Theoretical model (Coupled mode theory):

- The circulator has a three-port superconducting ring resonator, which supports two microwave modes that rotate in opposite directions (counter-rotating) and have the same frequency (degenerate).
- These two degenerate modes circulate within the ring resonator in clockwise and counterclockwise directions, enabling interactions with external signals at three distinct ports.
- The ring resonator is placed in alignment with a ferrimagnetic disk that has similar physical dimensions to ensure maximum spatial overlap between the resonator's microwave fields and the magnetic material.
- A static magnetic field is applied perpendicular (out-of-plane) to the disk, which magnetizes the ferrimagnetic material and sets its operating point.
- This ferrimagnetic disk supports a uniform magnon mode — a collective excitation of the magnetic spins in the material that oscillates coherently.
- The resonant frequency of this magnon mode, denoted by ω_m , depends linearly on the magnitude of the applied external magnetic field $|B_0|$.
- Mathematically, this relationship is expressed as $\omega_m \approx \gamma|B_0|$, where γ is the gyromagnetic ratio, a constant characterizing the coupling between magnetic moment and magnetic field. The ratio is

$$\gamma = 2.8 \text{ MHz/Oe}$$

- The system Hamiltonian is given by

$$\frac{H}{\hbar} = \omega_{ccw} a_{ccw}^\dagger a_{ccw} + \omega_{cw} a_{cw}^\dagger a_{cw} + \omega_m m^\dagger m + \frac{H_{\text{int}}}{\hbar}.$$

Here, $\hbar$ is the reduced Planck constant, and the Hamiltonian is expressed in units of angular frequency.

- The operators a_{ccw} and $a_{ccw}^\dagger$ are the annihilation and creation operators, respectively, for the counterclockwise (ccw) microwave mode. Similarly, a_{cw} and $a_{cw}^\dagger$ are the annihilation and creation operators for the clockwise (cw) microwave mode. The operators m and $m^\dagger$ correspond to the annihilation and creation operators for the magnon mode.
- The terms ω_{ccw} , ω_{cw} , and ω_m represent the resonant angular frequencies of the ccw microwave mode, cw microwave mode, and magnon mode, respectively.
- Since the ccw and cw microwave modes are orthogonal (meaning they represent independent modes that do not overlap), their mutual coupling is negligible and is not considered. Instead, each microwave mode couples only to the magnon mode.
- The interaction Hamiltonian describing this coupling is given by

$$\frac{H_{\text{int}}}{\hbar} = -g_{ccw}(a_{ccw} + a_{ccw}^\dagger)(m + m^\dagger) - g_{cw}(a_{cw} + a_{cw}^\dagger)(m + m^\dagger).$$

- Here, g_{ccw} and g_{cw} are the coupling strengths between the magnon mode and the counterclockwise and clockwise microwave modes, respectively. These parameters quantify how strongly the magnon mode interacts with each microwave mode.

- The terms $(a_{ccw} + a_{ccw}^\dagger)$ and $(a_{cw} + a_{cw}^\dagger)$ correspond to the microwave mode's position quadratures, while $(m + m^\dagger)$ is the magnon mode's position quadrature. The product terms represent the energy exchange interactions between magnons and photons in the microwave modes.
- The negative sign in front of the interaction terms indicates the nature of the coupling, which can be derived from the system's underlying physics (e.g., dipole interaction).
- The system under study consists of three modes: two microwave modes (counterclockwise (ccw) and clockwise (cw)) and one magnon mode. The vector of annihilation operators is denoted as

$$\mathbf{a} = \begin{pmatrix} a_{ccw} \\ a_{cw} \\ m \end{pmatrix}.$$

- The input fields entering the system at three ports are represented by the vector

$$\mathbf{s}_{in} = \begin{pmatrix} s_{in1} \\ s_{in2} \\ s_{in3} \end{pmatrix},$$

and the output fields leaving the system are similarly given by

$$\mathbf{s}_{out} = \begin{pmatrix} s_{out1} \\ s_{out2} \\ s_{out3} \end{pmatrix}.$$

- The Heisenberg-Langevin equation under the rotating wave approximation (RWA) governs the dynamics of the internal modes:

$$\dot{\mathbf{a}} = \mathbf{M}_1 \mathbf{a} + \mathbf{K} \mathbf{s}_{in},$$

where:

- $\mathbf{M}_1$ is a 3×3 matrix describing the internal system dynamics including resonance frequencies and dissipation.
- $\mathbf{K}$ is a 3×3 matrix describing the coupling from the input fields to the internal modes.
- The input-output relation connects the output fields to the inputs and internal modes as:

$$\mathbf{s}_{out} = \mathbf{C} \mathbf{s}_{in} + \mathbf{M}_2 \mathbf{a},$$

where:

- $\mathbf{C}$ is a 3×3 matrix describing the direct scattering (or direct transmission) between ports without involving the cavity modes.
- $\mathbf{M}_2$ is a 3×3 matrix describing how the internal modes contribute to the output fields.
- The matrix $\mathbf{M}_1$ is explicitly given by

$$\mathbf{M}_1 = \begin{pmatrix} -i\omega_{ccw} - \frac{\kappa_{ccw}}{2} & 0 & ig_{ccw} \\ 0 & -i\omega_{cw} - \frac{\kappa_{cw}}{2} & ig_{cw} \\ ig_{ccw} & ig_{cw} & -i\omega_m - \frac{\kappa_m}{2} \end{pmatrix},$$

where:

- κ_{ccw} , κ_{cw} , κ_m are the total dissipation rates (losses) for each mode.
- The coupling matrices $\mathbf{K}$ and $\mathbf{M}_2$ describe how the three outgoing waves couple to the three incoming waves with two resonant modes. They satisfy the energy conservation relation

$$\mathbf{K}\mathbf{K}^\dagger = \Gamma_e,$$

where Γ_e ensures no energy is lost in the coupling process.

- In general, the matrix $\mathbf{K}$ can be written as

$$\mathbf{K} = -\mathbf{M}_2^\dagger \mathbf{C} = \begin{pmatrix} \sqrt{\kappa_{ccw,e1}} & \sqrt{\kappa_{ccw,e2}}e^{i\alpha} & \sqrt{\kappa_{ccw,e3}}e^{i\eta} \\ \sqrt{\kappa_{cw,e1}}e^{i\beta_1} & \sqrt{\kappa_{cw,e2}}e^{i\beta_2} & \sqrt{\kappa_{cw,e3}}e^{i\beta_3} \\ 0 & 0 & 0 \end{pmatrix},$$

where:

- $\kappa_{ccw,e1(2,3)}$ and $\kappa_{cw,e1(2,3)}$ are the external coupling rates of the ccw and cw modes to ports 1, 2, and 3.
- α and η are relative phases for excitation ports 2 and 3 relative to port 1 for the ccw mode.
- β_1 is the phase difference between cw and ccw modes at port 1.
- $(\beta_2 - \beta_1)$ and $(\beta_3 - \beta_1)$ are the relative phase differences of the cw mode between ports 2 and 3 relative to port 1.
- Matrix $\mathbf{C}$ relates incoming and outgoing fields directly without coupling to internal modes. It must satisfy unitarity due to energy conservation:

$$\mathbf{C}^\dagger \mathbf{C} = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix.

- The specific form of $\mathbf{C}$ depends on the physical implementation of the ports. For example:
 - For open-ended (capacitive) coupling, $\mathbf{C} = \mathbf{I}$ (identity matrix).
 - For short-ended (inductive) coupling, $\mathbf{C} = -\mathbf{I}$.
- In their work, they assume inductive coupling with negligible dissipative cross-coupling between ports, $\mathbf{C}$ is chosen as

$$\mathbf{C} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

- Using the above matrices, the equation of motion can be solved in the frequency domain (ω) to obtain the scattering matrix $\mathbf{S}[\omega]$, which relates output to input fields in frequency space:

$$\mathbf{s}_{out}[\omega] = \mathbf{S}[\omega]\mathbf{s}_{in}[\omega].$$

- The scattering matrix is given by

$$\mathbf{S}[\omega] = \mathbf{C} + \mathbf{M}_2(-i\omega\mathbf{I} - \mathbf{M}_1)^{-1}\mathbf{K},$$

where:

- $\mathbf{I}$ is the 3×3 identity matrix.
- The term $(-i\omega\mathbf{I} - \mathbf{M}_1)^{-1}$ corresponds to the Green's function or system response in frequency domain.
- The scattering matrix $\mathbf{S}[\omega]$ completely describes how input signals are transmitted, reflected, or converted at the three ports considering the internal cavity and magnon dynamics.

Derivation : The time-domain equation of motion (Heisenberg-Langevin equation):

$$\dot{\mathbf{a}} = \mathbf{M}_1 \mathbf{a} + \mathbf{K} \mathbf{s}_{\text{in}},$$

and the input-output relation:

$$\mathbf{s}_{\text{out}} = \mathbf{C} \mathbf{s}_{\text{in}} + \mathbf{M}_2 \mathbf{a}.$$

where $\mathbf{a}$ is the vector of mode amplitudes, $\mathbf{s}_{\text{in}}, \mathbf{s}_{\text{out}}$ are the vectors of input and output fields, and $\mathbf{M}_1, \mathbf{M}_2, \mathbf{K}, \mathbf{C}$ are system matrices, and I is the identity matrix.

Step 1: Take the Fourier transform (frequency domain)

Using the convention $\frac{d}{dt} \rightarrow -i\omega$:

$$-i\omega \mathbf{a}(\omega) = \mathbf{M}_1 \mathbf{a}(\omega) + \mathbf{K} \mathbf{s}_{\text{in}}(\omega)$$

Rearranged:

$$(-i\omega I - \mathbf{M}_1) \mathbf{a}(\omega) = \mathbf{K} \mathbf{s}_{\text{in}}(\omega).$$

Step 2: Solve for $\mathbf{a}(\omega)$

$$\mathbf{a}(\omega) = (-i\omega I - \mathbf{M}_1)^{-1} \mathbf{K} \mathbf{s}_{\text{in}}(\omega).$$

Step 3: Substitute $\mathbf{a}(\omega)$ into the output relation

$$\mathbf{s}_{\text{out}}(\omega) = \mathbf{C} \mathbf{s}_{\text{in}}(\omega) + \mathbf{M}_2 \mathbf{a}(\omega).$$

Substituting $\mathbf{a}(\omega)$:

$$\mathbf{s}_{\text{out}}(\omega) = \mathbf{C} \mathbf{s}_{\text{in}}(\omega) + \mathbf{M}_2 (-i\omega I - \mathbf{M}_1)^{-1} \mathbf{K} \mathbf{s}_{\text{in}}(\omega).$$

Step 4: Write in terms of the scattering matrix $\mathbf{S}[\omega]$

The scattering matrix $\mathbf{S}[\omega]$ relates inputs to outputs as:

$$\mathbf{s}_{\text{out}}(\omega) = \mathbf{S}[\omega] \mathbf{s}_{\text{in}}(\omega).$$

From the above expression:

$$\boxed{\mathbf{S}[\omega] = \mathbf{C} + \mathbf{M}_2 (-i\omega I - \mathbf{M}_1)^{-1} \mathbf{K}.$$

- The matrix $\mathbf{C}$ describes the direct scattering between input and output ports *without* involving the cavity or internal modes. It depends on how the ports physically connect to the system.
- Physically, each port can be thought of as a transmission line terminated either by:
 - An open circuit (open-ended), which acts like a capacitor at the end of the line.

- A short circuit (short-ended), which acts like an inductor at the end of the line.
- The reflection coefficient r for a wave incident on the end of a transmission line is:

$$r = \frac{Z_L - Z_0}{Z_L + Z_0},$$

where

- Z_L is the load impedance (termination).
- Z_0 is the characteristic impedance of the line.
- For an open circuit:

$$Z_L \rightarrow \infty \implies r = \frac{\infty - Z_0}{\infty + Z_0} = +1.$$

This means the wave reflects with no phase change (reflection coefficient $+1$).

- For a short circuit:

$$Z_L = 0 \implies r = \frac{0 - Z_0}{0 + Z_0} = -1.$$

This means the wave reflects with a π phase shift (reflection coefficient -1).

- Since the direct scattering matrix $\mathbf{C}$ represents how an incoming wave is directly reflected or transmitted at the ports without interacting with the cavity, these reflection coefficients translate into:

$$\mathbf{C} = \mathbf{I} \quad \text{for open-ended (capacitive) coupling,}$$

and

$$\mathbf{C} = -\mathbf{I} \quad \text{for short-ended (inductive) coupling.}$$

- Intuitively:
 - For open-ended ports, the wave reflects back with the same sign, so $\mathbf{C} = \mathbf{I}$.
 - For short-ended ports, the wave reflects back with a π phase shift (sign flip), so $\mathbf{C} = -\mathbf{I}$.
- This phase difference critically affects interference and coupling between ports and internal modes, and thus plays a key role in the device's behavior.

Theoretical model (Circulation under threefold symmetry):

- The system exhibits threefold rotational symmetry (Threefold geometrical rotational symmetry means the system looks exactly the same after rotating it by 120 degrees around a central axis).
- Microwave modes a_{ccw} and a_{cw} are degenerate (have the same frequency).
- Each mode couples externally to three waveguides with equal rates $\kappa_e/3$.
- Total dissipation rate for each mode is $\kappa_{\text{ccw}} = \kappa_{\text{cw}} = \kappa_e + \kappa_i$.
- Here, κ_i is the intrinsic microwave dissipation, and κ_e is the total external coupling rate.

- The relative excitation phase difference between port 2 (3) and port 1 is given by

$$\alpha = -\eta = -\frac{2n\pi}{3},$$

where n is the mode number integer.

- For the two degenerate fundamental circulation microwave modes ($n = 1$), the conditions are:

$$\omega_{\text{ccw}} = \omega_{\text{cw}}, \quad \alpha = -\eta = -\frac{2\pi}{3}.$$

- This phase difference arises from the threefold rotational symmetry of the system, because the fundamental mode corresponds to an azimuthal mode number of 1 with a 2π phase shift along the ring circumference.
- Under threefold rotational symmetry, the relative phase difference between any two ports is $\pm 2\pi/3$, depending on the propagation direction of the mode.
- The two modes (counterclockwise and clockwise) are orthogonal, and the relative excitation phase β_1 between a_{ccw} and a_{cw} does not affect the scattering matrix. For simplicity, β_1 can be set to zero.
- *Chirality of microwave modes:* The two circular microwave modes in the system have opposite chiralities: one is counterclockwise (ccw) and the other is clockwise (cw).
- *Magnon mode chirality:* The magnon mode, which is a collective spin excitation in the magnetic material, also has a well-defined chirality that matches one of the microwave modes.
- *Selective coupling rule:* The magnon mode interacts only with the microwave mode that shares the same chirality. This means:

$$g_{\text{ccw}} = g \quad (\text{significant coupling})$$

$$g_{\text{cw}} \approx 0 \quad (\text{negligible coupling})$$

- *Physical origin:* This selective coupling arises because the magnon mode's spin precession and the microwave mode's electromagnetic field rotation must match in handedness for efficient interaction.
- *Implication:* Only the ccw microwave mode effectively exchanges energy with the magnon mode, while the cw mode remains essentially uncoupled, simplifying the system dynamics.
- The two degenerate microwave modes can be represented in different eigenmode bases by applying a unitary rotation matrix:

$$U(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- After this basis rotation, the interaction matrix M_1 transforms accordingly, resulting in new coupling strengths:

$$g_{\text{ccw}} = g \sin \theta, \quad g_{\text{cw}} = -g \cos \theta$$

- This means the coupling to the magnon mode is now distributed between the two rotated microwave modes, depending on the angle θ .

- Importantly, this unitary rotation of the mode basis does not affect the physical observables or the fundamental behavior of the system.
- Therefore, the choice of basis is a mathematical convenience, and the physical results remain invariant under such transformations. For threefold geometrical symmetry the coupling matrix is

$$K = \sqrt{\frac{\kappa_e}{3}} \begin{pmatrix} 1 & e^{-i\frac{2\pi}{3}} & e^{i\frac{2\pi}{3}} \\ 1 & e^{i\frac{2\pi}{3}} & e^{-i\frac{2\pi}{3}} \\ 0 & 0 & 0 \end{pmatrix}$$

The last row is zero because the magnon mode does not directly couple externally to the ports. However, in the given matrix, the first two elements are both 1 because they represent the amplitude of coupling, not the phase difference (for the ccw mode). The phase difference appears only starting from the second and third columns for port 2 and 3, as complex exponentials.

- The scattering matrix S of the system exhibits the following symmetry relations due to the threefold rotational symmetry of the device:

$$S_{11} = S_{22} = S_{33} = -1 + \frac{\kappa_e}{3 \left[\frac{\kappa_{cw}}{2} - i(\omega - \omega_{cw}) \right]} + \frac{\kappa_e \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right]}{3 \left\{ g^2 + \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right] \left[\frac{\kappa_{ccw}}{2} - i(\omega - \omega_{ccw}) \right] \right\}}, \quad (2)$$

$$S_{12} = S_{23} = S_{31} = e^{i2\pi/3} \frac{\kappa_e}{3 \left[\frac{\kappa_{cw}}{2} - i(\omega - \omega_{cw}) \right]} + e^{-i2\pi/3} \frac{\kappa_e \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right]}{3 \left\{ g^2 + \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right] \left[\frac{\kappa_{ccw}}{2} - i(\omega - \omega_{ccw}) \right] \right\}}, \quad (3)$$

$$S_{21} = S_{13} = S_{32} = e^{-i2\pi/3} \frac{\kappa_e}{3 \left[\frac{\kappa_{cw}}{2} - i(\omega - \omega_{cw}) \right]} + e^{i2\pi/3} \frac{\kappa_e \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right]}{3 \left\{ g^2 + \left[\frac{\kappa_m}{2} - i(\omega - \omega_m) \right] \left[\frac{\kappa_{ccw}}{2} - i(\omega - \omega_{ccw}) \right] \right\}}$$

$$S_{11} = S_{22} = S_{33}, \quad S_{12} = S_{23} = S_{31}, \quad S_{21} = S_{13} = S_{32}.$$

These equalities imply that the reflection and transmission coefficients are identical when cyclically permuting the ports, consistent with the rotational symmetry.

- If the threefold symmetry is broken, for example, by designing the three waveguides with different physical dimensions, the external coupling rates κ_e for the microwave modes to the three ports will no longer be equal.
- Consequently, the reflection coefficients will generally differ:

$$S_{11} \neq S_{22} \neq S_{33}.$$

Similarly, other transmission parameters between ports will also lose their degeneracy.

- Breaking this symmetry degrades the performance of the circulator, as the system will only be optimized for a specific connection configuration and magnetic bias condition.

- Therefore, maintaining the threefold rotational symmetry is crucial to ensure uniform and optimal circulation performance across all ports.
- Although the microwave counterclockwise (ccw) and clockwise (cw) modes are degenerate in frequency due to the device's structural symmetry, the coupling with the magnon mode introduces non-reciprocity in the scattering matrix.
- The selective coupling rule states that the magnon mode only couples strongly with one of the microwave modes (typically the ccw mode), breaking the symmetry in interaction and leading to nonreciprocal behavior.
- When there is no magnon-photon coupling ($g = 0$), the scattering parameters S_{12} and S_{21} are reciprocal in amplitude, meaning:

$$|S_{12}| = |S_{21}|.$$

There is no amplitude difference or isolation between forward and backward transmission.

- When magnon-photon coupling is turned on ($g \neq 0$), interference effects between the coupled magnon and microwave modes create nonreciprocal transmission characteristics, resulting in signal circulation.
- The isolation ratio between S_{12} and S_{21} , defined as the difference in transmitted amplitudes in opposite directions, depends on:
 - The coupling strength g between the magnon and microwave mode.
 - The detuning between the microwave mode resonance frequency $\omega_{ccw/cw}$ and the magnon resonance frequency ω_m , i.e., $\omega_{ccw/cw} - \omega_m$.
- By tuning g and the detuning, one can control and optimize the degree of nonreciprocity and isolation in the circulator.
- *Frequency degeneracy of ccw and cw modes:*
 - In planar microwave ring resonators (e.g., microstrip rings), the resonant frequencies of the counterclockwise (ccw) and clockwise (cw) modes with the same mode number are highly degenerate.
 - This degeneracy holds unless additional geometric perturbations are introduced.
 - Therefore, under ideal conditions with threefold geometrical symmetry, we can treat $\omega_{ccw/cw}$ as degenerate.
- *External coupling rates to waveguides:*
 - The external coupling rates $\kappa_{ccw,e1(2,3)}$ and $\kappa_{cw,e1(2,3)}$ between the ccw/cw modes and the three waveguides depend primarily on the impedance matching between the ring and waveguides.
 - These impedance values depend on the geometrical dimensions and the dielectric substrate material.
 - Precise lithographical patterning allows fabrication of waveguides on the same chip with nearly identical geometry, achieving threefold symmetry.
 - As a result, the degeneracy of external coupling rates can be well controlled and accurately modeled at microwave frequencies.

- *Dynamic tuning of external coupling:*
 - Some works demonstrate dynamic tuning of external coupling rates during experiments using superconducting quantum interference devices (SQUIDs).
- *Microwave intrinsic dissipation κ_i of the superconducting ring:*
 - The intrinsic dissipation rate κ_i is minimal at cryogenic temperatures, typically on the order of a few megahertz.
 - Due to system symmetry, intrinsic dissipation is effectively evenly distributed among the three ports.

Material Platforms (YIG-based circulator):

- Yttrium iron garnet (YIG) is widely used as a magnon medium due to excellent magnetic properties such as long spin lifetime and wide tunability.
- YIG has a large saturation magnetization ($4\pi M_s = 1750$ G), about 20 times higher than that of V[TCNE]₂, making it a high-spin-density material.
- Coherent coupling between magnons and microwave photons has been demonstrated in YIG thin films and spheres, with coupling strengths ranging from several MHz to a few GHz.
- Coupling strength depends on engineering mode overlap, mode volume, and resonant frequencies.
- Both YIG thin films and bulk YIG are promising for high-performance circulators due to low Gilbert damping factors.
- For YIG thin films magnetized out-of-plane, the bias field must overcome a large demagnetizing field (~ 1750 Oe) for full saturation and effective magnon resonance excitation ($\omega_m = \gamma|B_0 - H_d|$).
- Bulk YIG has a much smaller demagnetizing field (tens of oersted). Bulk YIG helps for easier saturation with lower magnetic fields and less interference with superconducting devices.
- Using YIG thin films may introduce additional flux into superconducting microwave resonators compared to bulk YIG. Thin film helps with strong magnon-photon coupling and integration into compact or chip-scale systems, but their use is limited only if one can manage higher bias fields and flux considerations.
- Their work demonstrates the use of a bulk YIG disk as the magnon medium to achieve microwave circulation under the strong magnon-photon coupling regime, where the coupling strength satisfies

$$g > \kappa_{ccw}, \kappa_{cw}, \kappa_m,$$

and the superconducting microwave ring resonator is overcoupled such that

$$\kappa_e > \kappa_i.$$

- In this regime, the magnon resonance can be detuned from the microwave resonance, entering the dispersive coupling regime to minimize loss caused by ferrimagnetic damping (Loss of energy from the precessing magnetization in ferrimagnetic materials, leading to the decay of spin excitations (magnons) over time).

- The tunability of the magnon resonance provides additional functionalities, including:
 - Adjustable isolation ratios
 - Isolation ratio measures how well a circulator or isolator suppresses signal transmission in the undesired direction compared to the desired one. It is often expressed in decibels (dB) as:

$$\text{Isolation Ratio (dB)} = 20 \log_{10} \left(\frac{|S_{\text{forward}}|}{|S_{\text{reverse}}|} \right)$$

where $|S_{\text{forward}}|$ and $|S_{\text{reverse}}|$ are the transmission amplitudes in the forward and reverse directions, respectively.

- Switchable signal propagation directions (Switchable signal propagation means the ability to dynamically change the direction in which the signal travels through a circulator or similar device).
- To withstand the relatively strong bias magnetic field required, superconductors with high critical fields, such as NbTi films on sapphire substrates, are suitable for fabricating the microwave circuits.
- The device design must maintain threefold structural symmetry, featuring three identical waveguides inductively coupled to the ring resonator at $2\pi/3$ angular intervals.
- The resonant frequencies of the counterclockwise (ccw) and clockwise (cw) microwave modes are determined by the resonator geometry and are degenerate ($\omega_{\text{ccw}} = \omega_{\text{cw}}$) in the absence of magnon media.
- Intrinsic microwave loss ($\kappa_i/2\pi$) for both microwave modes is estimated to be several megahertz, based on literature values, when the static magnetic field is well below the NbTi critical field.
- The external coupling rate ($\kappa_e/2\pi$) for the ring resonator can be engineered by adjusting the impedance of the input waveguides; experimental values can range from tens of megahertz to several gigahertz.
- For exploring both strong and weak magnon-photon coupling regimes, a moderate external coupling rate ($\kappa_e/2\pi = 600$ MHz) is chosen, achievable by impedance design.
- Due to the structural symmetry, the external coupling rates for both ccw and cw modes are identical.
- The magnon-photon coupling strength g can be engineered by:
 - Modifying the microwave mode volume and frequency,
 - Engineering the spatial field overlap between microwave and magnon modes,
 - Selecting materials with different spin densities.
- During experiments, g can be dynamically tuned by:
 - Adjusting the gap between the ferrimagnetic disk and the microwave circuit to control the field overlap,
 - Adding a ground plate to alter the microwave field distribution,
 - Changing the system temperature.
- These engineering methods provide flexible approaches to optimize the system performance for various applications.

- They present the transmission spectra $|S_{21}|^2$ and $|S_{12}|^2$, respectively, mapped as functions of frequency detunings:

$$\Delta_m = \omega_m - \omega_{ccw/cw} \quad \text{and} \quad \Delta_c = \omega - \omega_{ccw/cw},$$

calculated from Eqs. (11) and (12) using parameters:

$$\frac{g}{2\pi} = 600 \text{ MHz}, \quad \frac{\kappa_e}{2\pi} = 600 \text{ MHz}, \quad \frac{\kappa_i}{2\pi} = 2 \text{ MHz}, \quad \frac{\kappa_m}{2\pi} = 3 \text{ MHz}.$$

- The observed avoided crossings in the transmission spectra indicate strong coupling between microwave photons and magnons, accompanied by clear asymmetric transmission under positive and negative magnon detuning Δ_m .

- They show that

$$|S_{21}(-|\Delta_m|)|^2 = |S_{12}(+|\Delta_m|)|^2,$$

reflecting symmetry between transmission in forward and reverse directions with opposite magnon detunings.

- Optimal circulation with minimal insertion loss (IL) and large isolation ratio (defined as $\text{iso.} = 20 \log_{10} \frac{|S_{21}|}{|S_{12}|}$) can be achieved by tuning Δ_m .
- Specifically, when

$$\Delta_m = \pm 0.87 \text{ GHz},$$

insertion loss can be as low as 0.05 dB, while isolation ratio reaches up to 56 dB.

- The signal propagation directionality can be switched simply by changing Δ_m , without physically reorienting the external magnetic field.
- The maximal isolation ratio achievable is limited by intrinsic losses κ_i and κ_m . Optimizing fabrication and/or operating at ultralow temperatures to reduce

$$\frac{\kappa_i}{2\pi} = \frac{\kappa_m}{2\pi} = 0.5 \text{ MHz}$$

can enhance isolation ratio to 63 dB and reduce insertion loss to 0.02 dB.

- They analyze the 20 dB isolation bandwidth as a function of the magnon-photon coupling strength g and the total microwave dissipation rate

$$\kappa = \kappa_e + \kappa_i,$$

with the microwave resonant frequency fixed at 10 GHz.

- Contour lines denote circulation bandwidths of 20, 50, 100, and 200 MHz for $|\text{iso.}| = 20 \text{ dB}$.
- Parameter space is divided into four regions:
 - *Region I (Weak coupling)*: $g \ll \kappa$, coupling strength too small to induce significant phase modulation at the output ports.
 - *Region IV (Ultrastrong coupling)*: $g \gg \kappa$, ccw and cw microwave mode splitting is much larger than linewidth, resulting in minimal spectral overlap and thus weak effective circulation.

- *Region III (Strong coupling)*: $g \sim \kappa$, isolation bandwidth increases linearly with both g and κ . Optimal circulation occurs when the magnon mode is detuned away from microwave modes.
- *Region II (Intermediate coupling)*: $g < \kappa$, isolation bandwidth also increases linearly with g and κ . Here, magnon mode is tuned close to microwave mode to achieve sufficient phase modulation for broadband circulation.
- These observations demonstrate that different isolation bandwidths can be engineered by tuning:
 - The magnon resonant frequency,
 - The magnon-photon coupling strength g ,
 - The total microwave dissipation rate κ .
- This tunability allows the system to be optimized for a variety of application requirements regarding isolation bandwidth and signal circulation.
- Another key performance metric for a circulator is the insertion loss ($|IL|$), which quantifies signal attenuation through the device.
- They investigate how the microwave intrinsic loss κ_i influences the insertion loss at various external coupling rates κ_e , with the magnon-photon coupling fixed at

$$\frac{g}{2\pi} = 1200 \text{ MHz}.$$

- As $\kappa_e/2\pi$ is varied from 300 MHz to 1200 MHz, increasing intrinsic loss κ_i causes a corresponding increase in insertion loss $|IL|$.
- This demonstrates that a high-quality-factor (high-Q) superconducting microwave resonator, which has low intrinsic loss κ_i , is ideal for achieving ultralow insertion loss in circulation.
- They also show that for a fixed intrinsic loss κ_i , increasing the external coupling rate κ_e reduces the insertion loss.
- The reason is that when κ_e dominates the total microwave dissipation rate $\kappa = \kappa_e + \kappa_i$, a smaller fraction of the microwave signal is lost through intrinsic dissipation channels, resulting in lower insertion loss.
- These analyses confirm that it is possible to simultaneously achieve:
 - High isolation ratio,
 - Low insertion loss,
 - Tunable bandwidth,

which are all critical features for high-performance microwave circulators.

Material Platforms ($\text{V}[\text{TCNE}]_2$ -based circulator):

- This section focuses on microwave circulation using the low-spin-density ferrimagnetic material $\text{V}[\text{TCNE}]_2$, an organic-based high-quality ferrimagnetic semiconductor characterized by:

- Bandgap $E_g = 0.5 \text{ eV}$,
- Electrical conductivity $\sigma = 0.01 \text{ S/m}$,
- Room temperature magnetic ordering with Curie temperature $T_c > 600 \text{ K}$.
- $\text{V}[\text{TCNE}]_2$ exhibits a very low Gilbert damping factor comparable to single-crystal YIG, both in continuous and micropatterned films.
- It can be integrated onto various substrates by chemical vapor deposition (CVD), maintaining excellent magnetic properties, unlike YIG which requires lattice-matched substrates for high-quality growth.
- The saturation magnetization of $\text{V}[\text{TCNE}]_2$ is approximately 90 G, which is over an order of magnitude smaller than YIG's 1750 G. This leads to:
 - Smaller demagnetizing fields,
 - Significantly reduced bias magnetic fields,
- However, the lower spin density results in weaker magnon-photon coupling strength g for the same microwave circuit design and magnon mode volume, making it difficult to reach strong coupling regimes.
- The microwave circulation is studied in the weak coupling regime where

$$\kappa_m < g < \kappa,$$

with parameters chosen as:

$$\frac{g}{2\pi} = 100 \text{ MHz}, \quad \frac{\kappa_i}{2\pi} = 2 \text{ MHz}, \quad \frac{\kappa_e}{2\pi} = 600 \text{ MHz}, \quad \frac{\kappa_m}{2\pi} = 2 \text{ MHz}.$$

- They map the transmission scattering parameters $|S_{12}|^2$ and $|S_{21}|^2$, showing similar directional transmission features with a Lorentzian-shaped transparency window indicating the magnon resonance.
- The optimized detuning $\Delta_m = 0.16 \text{ GHz}$ yields:
 - Low insertion loss of about 0.09 dB,
 - High isolation ratio reaching 77 dB.
- Due to the relatively small magnon-photon coupling strength, the 20 dB isolation bandwidth is approximately 0.5 MHz, much narrower than that of YIG-based devices.
- Despite the narrow bandwidth, this circulator is promising for circuit quantum electrodynamics (cQED) systems, where it can act as a filter for multiplexed superconducting qubits and resonators.
- The circulation performance of the $\text{V}[\text{TCNE}]_2$ -based narrow-band circulator is studied under conditions where the ideal threefold geometrical symmetry is not fully achieved experimentally.
- The same magnon-photon coupling strength, microwave intrinsic dissipation rates, and magnon linewidth are maintained while investigating two types of symmetry-breaking conditions:
- **Condition 1: Nondegenerate microwave dissipation rates for counterclockwise (ccw) and clockwise (cw) modes**

- The external coupling rates differ by 20%:

$$\kappa_{ccw,e}/2\pi = 600\text{MHz}, \quad \kappa_{cw,e}/2\pi = 720\text{MHz}.$$

- At the original magnon detuning $\Delta_m = 0.16\text{GHz}$, the isolation ratio reduces significantly from the symmetric case to 27 dB.
- By optimizing the magnon detuning to $\Delta_m = 0.21\text{GHz}$, the device performance is restored, achieving:

- * High isolation ratio of 61 dB,
- * Low insertion loss of 0.16 dB,
- * This occurs at a slightly shifted frequency.

• **Condition 2: Unequal external coupling rates to the three waveguides due to fabrication imperfections**

- Although the total external coupling rates for ccw and cw modes are designed to be equal:

$$\kappa_{ccw,e} = \kappa_{cw,e} = \kappa_e = 2\pi \times 600\text{MHz},$$

the coupling rates to individual waveguides differ:

$$\frac{1.2}{3}\kappa_e, \quad \frac{1}{3}\kappa_e, \quad \frac{0.8}{3}\kappa_e,$$

reflecting imperfections in device patterning and packaging.

- At the original magnon detuning $\Delta_m = 0.16\text{GHz}$, the isolation ratio decreases sharply to 19 dB (dashed lines in Fig. 5(b)).
- Tuning the magnon resonance recovers high-performance circulation:
 - * Isolation ratio improves to 67 dB,
 - * Insertion loss reduces to 0.15 dB,
- Due to the loss of perfect threefold symmetry, relations such as $S_{21} = S_{13} = S_{32}$ no longer hold, so magnon detuning optimization benefits only one connection type at a time.
- Despite these symmetry-breaking imperfections in experimental implementations, dynamic tuning of the magnon resonance enables optimization of signal interference among different ports.
- This dynamic tuning allows the system to recover high-performance microwave circulation even when geometric symmetry is not perfect.
- In the undercoupled scenario, the magnon resonance must be tuned *close* to the microwave resonance frequency to achieve circulation.
 - This makes the overall insertion loss ($|IL|$) more *sensitive* to the magnon linewidth κ_m than in the strongly coupled regime.
- Operating at cryogenic temperatures can reduce the magnon linewidth to $\kappa_m/2\pi = 0.5\text{MHz}$.
 - Under this condition, the insertion loss can be further decreased:

- * To as low as 0.04 dB for an *ideal symmetric device*.
- * To about 0.06 dB for a *nonsymmetric device*.
- Meanwhile, the isolation ratio $|iso.|$ can be maintained above 50 dB.
- Recent advancements in micropatterning techniques for $V[TCNE]_2$ thin films enable:
 - Creation of high-fidelity lithographically defined structures.
 - No observable deterioration in the magnetic properties post-patterning.
- Further improvements in the device design include:
 - Enhancing the magnon-microwave *mode overlap factor*.
 - Reducing the magnon mode volume.
- These improvements can enable achieving a higher magnon-photon coupling strength even with the low-spin-density $V[TCNE]_2$ material.
 - As a result, the circulator can operate with a *broader bandwidth*.

Conclusion:

They theoretically studied high-performance microwave circulation using a multiport magnon-photon system. Nonreciprocity arises from interference between counterpropagating microwave modes introduced by chirality dependent coupling with the magnon excitation. Using realistic parameters for superconducting circuits and low Gilbert damping materials (YIG and $V[TCNE]_2$), they show low insertion loss (~ 0.05 dB) and high isolation (~ 56 dB) are achievable. While $V[TCNE]_2$ offers narrower bandwidth than YIG, it suits frequency-selective applications in superconducting quantum computing to isolate the qubits from the environment and to curtail Purcell decay. The system also provides tunability and directional control. The theory is general and applicable to other hybrid systems exhibiting multimode nonreciprocity.

Appendix:

They denote the external and intrinsic dissipation matrices for the microwave signal as follows:

$$\Gamma_e = \begin{pmatrix} \kappa_{ccw,e} & 0 & 0 \\ 0 & \kappa_{cw,e} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \kappa_{ccw,i} & 0 & 0 \\ 0 & \kappa_{cw,i} & 0 \\ 0 & 0 & \kappa_m \end{pmatrix}.$$

Consider a special case with no incident wave $\mathbf{s}_{in} = 0$ and $\mathbf{i} = 0$ (no intrinsic loss). Due to energy conservation, the decay of the intracavity energy should equal the output power, i.e.,

$$-\frac{d}{dt}(\mathbf{a}^\dagger \mathbf{a}) = \mathbf{s}_{out}^\dagger \mathbf{s}_{out}.$$

The time derivative of the intracavity energy is

$$\frac{d}{dt}(\mathbf{a}^\dagger \mathbf{a}) = \frac{d\mathbf{a}^\dagger}{dt} \mathbf{a} + \mathbf{a}^\dagger \frac{d\mathbf{a}}{dt}.$$

Using the equation of motion $\dot{\mathbf{a}} = \mathbf{M}_1 \mathbf{a}$ (since $\mathbf{s}_{in} = 0$) and its Hermitian conjugate, we get

$$\frac{d}{dt} (\mathbf{a}^\dagger \mathbf{a}) = \mathbf{a}^\dagger (\mathbf{M}_1^\dagger + \mathbf{M}_1) \mathbf{a}.$$

By the definition of the external dissipation matrix Γ_e , this must equal

$$\frac{d}{dt} (\mathbf{a}^\dagger \mathbf{a}) = -\mathbf{a}^\dagger \Gamma_e \mathbf{a}. \quad (\text{A1})$$

On the other hand, the output power is given by

$$\mathbf{s}_{out}^\dagger \mathbf{s}_{out} = \mathbf{a}^\dagger \mathbf{M}_2^\dagger \mathbf{M}_2 \mathbf{a}. \quad (\text{A2})$$

Comparing Eqs. (A1) and (A2), we find

$$\mathbf{M}_2^\dagger \mathbf{M}_2 = \Gamma_e. \quad (\text{A3})$$

Proof of $-\mathbf{C}^\dagger \mathbf{M}_2 = \mathbf{K}^\dagger$

From energy conservation, the net input power minus output power equals the change of intracavity energy plus intrinsic loss:

$$\begin{aligned} \mathbf{s}_{in}^\dagger \mathbf{s}_{in} - \mathbf{s}_{out}^\dagger \mathbf{s}_{out} &= \frac{d}{dt} (\mathbf{a}^\dagger \mathbf{a}) + \mathbf{a}^\dagger \Gamma_i \mathbf{a}. \\ \mathbf{s}_{in}^\dagger \mathbf{C}^\dagger \mathbf{M}_2 \mathbf{a} + \mathbf{a}^\dagger \mathbf{M}_2^\dagger \mathbf{C} \mathbf{s}_{in} &= -\mathbf{s}_{in}^\dagger \mathbf{K}^\dagger \mathbf{a} - \mathbf{a}^\dagger \mathbf{K} \mathbf{s}_{in}. \end{aligned} \quad (\text{A4})$$

Because $\mathbf{s}_{in}$ and $\mathbf{a}$ are arbitrary, Eq. (A4) implies the matrix relation

$$-\mathbf{C}^\dagger \mathbf{M}_2 = \mathbf{K}^\dagger.$$

Doubts:

1. The output power is given by

$$\mathbf{s}_{out}^\dagger \mathbf{s}_{out} = \mathbf{a}^\dagger \mathbf{M}_2^\dagger \mathbf{M}_2 \mathbf{a}.$$

Is there a proof for this one?

2. I don't understand how this equation is derived in the appendix?

$$\mathbf{s}_{in}^\dagger \mathbf{C}^\dagger \mathbf{M}_2 \mathbf{a} + \mathbf{a}^\dagger \mathbf{M}_2^\dagger \mathbf{C} \mathbf{s}_{in} = -\mathbf{s}_{in}^\dagger \mathbf{K}^\dagger \mathbf{a} - \mathbf{a}^\dagger \mathbf{K} \mathbf{s}_{in}.$$

3. I want to know how they derived the scattering matrix elements here.

9 Fast readout and reset of a superconducting qubit coupled to a resonator with an Intrinsic Purcell Filter

- **Challenges presented:**

- Coupling a resonator to a superconducting qubit allows essential operations such as dispersive readout and unconditional reset.
- The speed of these qubit operations is fundamentally limited by the external decay rate of the resonator. For fast readout, the probe signal must quickly leave the resonator into the output transmission line for amplification and detection. Achieving fast signal extraction requires increasing the resonator's external decay rate.
- Increasing the resonator's decay rate to speed up operations inadvertently increases the qubit's decay rate via the resonator.
- This enhanced qubit decay reduces the qubit lifetime, which is detrimental for quantum information processing.

- **What they do:**

- Exploit the distributed-element, multimode nature of the resonator to suppress the resonator-mediated qubit decay.
- Demonstrate suppression of qubit decay by more than two orders of magnitude over a wide bandwidth of 600 MHz. They use a low Q resonator and optimize the position where output line couples to resonator (intrinsic purcell filter).
- Utilize intrinsic Purcell filter effect to achieve faster qubit operations without compromising qubit lifetime.
- Achieve a 40-ns qubit readout with 99.1% fidelity.
- Realize a 100-ns unconditional reset with residual qubit excitation below 1.7%.
- Previous research utilized band stop or band pass filters between resonator and output lines.
 - * Measured external qubit decay time: $T_1^{\text{ex}} = 130 \mu\text{s}$.
 - * This is $\sim 30,000$ times longer than the resonator's external decay time.
 - * Without the filter, the decay time ratio would be limited to $(\Delta/g)^2 \approx 96$. Here, Δ is the qubit–resonator detuning and g is the coupling strength.

- **Device Design : Intrinsic Purcell Filter**

- The device consists of a transmon qubit and the inner conductor of a resonator fabricated on a silicon substrate. They are made from a lithographically patterned niobium film and an Al/AlO_x/Al Josephson junction. The substrate is clamped inside an aluminum block (outer conductor of the resonator), with an SMA connector output coupler. The system is cooled to ~ 40 mK in a dilution refrigerator. Measured parameters yield a qubit–resonator coupling strength of $g/2\pi = 224$ MHz.
- The resonator is a transmission line with open termination at both ends. Conventionally, the output line couples capacitively to one end of the resonator, where the electric field is largest, enabling strong coupling.

- In this design, an intrinsic Purcell filter is implemented by optimizing the position of the output coupler along the resonator. The coupler position is chosen to minimize resonator-mediated qubit decay.
- The decay is minimized when the coupler is placed at a distance from the open end (opposite to the qubit) equal to a quarter wavelength at the qubit frequency.
- When a qubit and a resonator are coupled, they no longer behave as independent systems. Their interaction leads to hybrid modes, which are combinations of the qubit excitation and the resonator field pattern.
- The mode that contains most of the qubit's character is called the dressed-qubit mode. The spatial distribution of voltage along the resonator changes from its uncoupled form because the qubit acts as an additional load.
- For analysis, the transmon qubit can be approximated as a linear harmonic oscillator. In this linearized model, the eigenmodes of the coupled system can be calculated.
- The dressed-qubit mode is primarily qubit-like but still has some resonator field component. The voltage distribution for this mode can be plotted along the resonator length.
- The qubit loses energy to the outside world mainly through the output transmission line connected to the resonator. This energy loss occurs because the dressed-qubit mode has some voltage at the coupler location, which can excite the output line.
- The decay rate into the output line is proportional to the square of the mode voltage at the coupler. Small voltage at the coupler means weak coupling and longer qubit lifetime.
- The section of the resonator between the coupler and the far open end behaves like a quarter-wavelength stub at the qubit's frequency. This stub transforms the open circuit at its end into a short circuit at the coupler location.
- A short circuit enforces zero voltage at the coupler, creating a voltage node for the dressed-qubit mode at that point. With a voltage node at the coupler, energy leakage from the qubit into the output line is strongly suppressed, effectively decoupling the qubit from the line at its own frequency.
- This configuration functions as an intrinsic Purcell filter, suppressing qubit decay without the need for external filtering components.
- In the fundamental mode of the resonator, the voltage distribution has a node (zero voltage) at the center of the resonator.
- This voltage node does not coincide with the position of the coupler in the implemented design.
- As a result, the coupler is positioned at a location where the voltage amplitude of the fundamental mode is relatively high.
- Because of this higher voltage at the coupler location, the output transmission line couples much more strongly to the fundamental mode of the resonator than to the dressed-qubit mode.
- This strong coupling to the fundamental mode allows efficient extraction or injection of signals at frequencies corresponding to the fundamental mode.
- The coupler position and resonator design form a filter with a notchlike transmission spectrum.
- This filter has a stop band precisely at the qubit frequency, meaning signals at the qubit frequency are strongly suppressed from entering or leaving through the coupler.

- Most other frequencies are transmitted efficiently, since they lie outside the stop band.
- Because of this property, the coupler can be used to apply strong off-resonant drive tones to the qubit.
- An off-resonant drive is useful in all-microwave reset schemes, which allow rapid qubit initialization without requiring additional hardware.
- An alternative explanation for the suppression of qubit decay is based on the higher-harmonic modes that naturally exist in a distributed-element resonator.
- The spontaneous emission of the qubit can occur via multiple resonator modes, and these processes can interfere strongly with each other.
- Each resonator mode has its own unique field distribution along the length of the resonator.
- By adjusting the position of the output coupler, one can change the relative contribution (weight) of each mode to the interference.
- At a specific optimal coupler position, the interference between these modes becomes destructive, resulting in a minimal total spontaneous emission rate from the qubit.
- To demonstrate this filter concept, the authors implement it using a coaxial-transmission-line resonator device architecture.
- The coaxial-transmission-line architecture is mechanically simpler but less versatile than the conventional coplanar waveguide architecture.
- The optimal coupler position can still be determined by calculating the electric field distribution of the dressed-qubit mode using finite-element electromagnetic simulations.
- Given device parameters (as measured):
 - * Resonator frequency: $\omega_r/2\pi = 10.5106$ GHz.
 - * Resonator external linewidth: $\kappa_{\text{ex}}/2\pi = 45.7$ MHz.
 - * Qubit frequency: $\omega_{eg}/2\pi = 8.319$ GHz.
 - * Qubit–resonator coupling: $g/2\pi = 224$ MHz.
- Define the qubit–resonator detuning:

$$\Delta := \omega_{eg} - \omega_r.$$

Numerically,

$$\frac{\Delta}{2\pi} = 8.319 - 10.5106 = -2.1916 \text{ GHz},$$

so $|\Delta|/2\pi \approx 2.1916$ GHz.

- Purcell (resonator-mediated) external decay rate in the dispersive limit:

$$\Gamma'_{\text{ex}} = \frac{g^2}{\Delta^2} \kappa_{\text{ex}}.$$

κ_{ex} . This is the standard expression for the qubit spontaneous-emission rate into the resonator output channel when $|\Delta| \gg g$.

- Convert the quoted numbers to angular frequencies (rad/s) for a direct numerical evaluation:

$$\begin{aligned} g_{\text{rad}} &= 2\pi \times 224 \times 10^6 \text{ s}^{-1} \approx 1.407 \times 10^9 \text{ rad/s}, \\ \Delta_{\text{rad}} &= 2\pi \times (-2.1916 \times 10^9) \text{ s}^{-1} \approx -1.3774 \times 10^{10} \text{ rad/s}, \\ \kappa_{\text{ex,rad}} &= 2\pi \times 45.7 \times 10^6 \text{ s}^{-1} \approx 2.8729 \times 10^8 \text{ rad/s}. \end{aligned}$$

- Evaluate the Purcell rate and the corresponding lifetime predicted from the single-mode formula:

$$\begin{aligned} \Gamma_{\text{ex}} &= \left(\frac{g_{\text{rad}}}{\Delta_{\text{rad}}} \right)^2 \kappa_{\text{ex,rad}} \approx \left(\frac{1.407 \times 10^9}{1.3774 \times 10^{10}} \right)^2 \times 2.8729 \times 10^8 \text{ s}^{-1} \\ &\approx 2.9996 \times 10^6 \text{ s}^{-1}, \end{aligned}$$

hence the predicted external-limited energy relaxation time would be

$$T_{1,\text{ex}}^{(\text{predicted})} = \frac{1}{\Gamma_{\text{ex}}} \approx 3.33 \times 10^{-7} \text{ s} = 0.33 \mu\text{s}.$$

- Measured values and comparison with the single-mode prediction:

- * Measured total qubit energy relaxation time: $T_1^{\text{meas}} = 17 \mu\text{s}$.
- * This is about $17/0.33 \approx 51$ times longer than the single-mode Purcell-limited prediction, i.e., a $\sim 50\times$ enhancement in the actual lifetime compared to what the simple single-mode formula would predict.
- * Measured resonator-mediated external decay time (extracted from continuous-wave reflection spectroscopy): $T_{1,\text{ex}}^{\text{meas}} = 130 \mu\text{s}$.
- * Comparing the measured external-limited lifetime to the unfiltered prediction:

$$\frac{T_{1,\text{ex}}^{\text{meas}}}{T_{1,\text{ex}}^{(\text{predicted})}} \approx \frac{130 \mu\text{s}}{0.33 \mu\text{s}} \approx 393,$$

i.e., the intrinsic Purcell filter extends the external-limited lifetime by a factor $\gtrsim 300$.

Transmission Spectrum:

Limitations of the conventional characterization method:

- In earlier studies, the transmission spectrum of a Purcell filter was often determined by varying the frequency of a flux-tunable qubit and measuring its lifetime.
- This method works because the Purcell decay rate depends on how strongly the qubit couples to the output line at each frequency, which directly reflects the filter's transmission spectrum.
- In the present case, this approach is ineffective because, near the notch frequency of the filter, the qubit's lifetime is dominated by its internal losses rather than by Purcell decay via the output line.

Alternative characterization approach:

- Instead of tuning the qubit frequency, the method drives the qubit through the Purcell filter from the output line at various drive frequencies ω_d .

- When the drive frequency is resonant with the qubit transition, it induces Rabi oscillations.
- When the drive frequency is off-resonant, it induces an ac Stark shift of the qubit transition.
- The efficiency of this drive reveals how much of the signal passes through the filter and reaches the qubit.

Drive amplitude and Hamiltonian representation:

- The strength of the applied drive is described by the drive Hamiltonian

$$\hat{H}_d(t)/\hbar = \Omega(\hat{b} + \hat{b}^\dagger)\cos(\omega_d t),$$

where $\hat{b}$ is the annihilation operator for the transmon qubit and Ω is the drive amplitude.

- The amplitude Ω is determined experimentally by measuring the qubit response to the applied drive.

Calculation of the external coupling rate:

- Once Ω is known, the external coupling rate of the qubit to the output line at the drive frequency is calculated as

$$\Gamma_{\text{ex}}(\omega) = \frac{\Omega\hbar\omega_d}{4P},$$

- Here, P is the drive power applied to the device.
- The power P is calibrated using the qubit's reflection spectrum.

Physical interpretation:

- The function $\Gamma_{\text{ex}}(\omega)$ represents the external decay rate of a qubit if it were tuned to frequency ω .
- This is equivalent to measuring how strongly the output line couples to a hypothetical qubit at frequency ω .
- Therefore, $\Gamma_{\text{ex}}(\omega)$ effectively maps out the outward transmission spectrum of the Purcell filter.

Experimental evaluation of coupling rates at specific frequencies:

- The goal is to determine $\Gamma_{\text{ex}}(\omega_d)$ for two important drive frequencies: $\omega_d = \omega_{fe}$ (transition from $|e\rangle$ to $|f\rangle$) and $\omega_d = \omega_{eg}$ (transition from $|g\rangle$ to $|e\rangle$).
- Rabi oscillations for each transition are observed.
- The qubit states are labeled:
 - $|g\rangle$ = ground state
 - $|e\rangle$ = first excited state
 - $|f\rangle$ = second excited state
- The observed Rabi frequencies are:

$$\sqrt{2}\Omega \quad \text{for the } |e\rangle \rightarrow |f\rangle \text{ transition,}$$

$$\Omega \quad \text{for the } |g\rangle \rightarrow |e\rangle \text{ transition.}$$

AC Stark shift measurement for drive amplitude extraction:

- For $\omega_d \neq \omega_{fe}$ and $\omega_d \neq \omega_{eg}$, the drive-induced ac Stark shift of the $|e\rangle \rightarrow |g\rangle$ transition frequency is measured.
- The ac Stark shift $\Delta\omega_{eg}$ is extracted from Ramsey fringe measurements.
- For each drive frequency, the drive power P is chosen such that:

$$|\omega_{fe} - \omega_d| \quad \text{and} \quad |\omega_{eg} - \omega_d|$$

are much larger than Ω , allowing the use of a perturbative formula for the Stark shift

- The drive amplitude Ω is then calculated from:

$$\Omega = \sqrt{\frac{2(\omega_{fe} - \omega_d)(\omega_{eg} - \omega_d)}{\omega_{fe} - \omega_{eg}}} \Delta\omega_{eg}.$$

The perturbative formula for the AC Stark shift describes the change in the qubit transition frequency due to photons in a coupled resonator, assuming the system is in the dispersive regime. In the Jaynes–Cummings model, the Hamiltonian is given by

$$H = \omega_r a^\dagger a + \frac{\omega_q}{2} \sigma_z + g (a^\dagger \sigma^- + a \sigma^+),$$

where ω_r is the resonator frequency, ω_q is the qubit frequency, g is the coupling strength, $a, a^\dagger$ are the photon annihilation and creation operators, and $\sigma_z, \sigma^\pm$ are the Pauli and raising/lowering operators for the qubit. In the dispersive limit, with detuning $\Delta = \omega_q - \omega_r$ and $|\Delta| \gg g$, a perturbative expansion in g/Δ yields the effective Hamiltonian

$$H_{\text{eff}} \approx (\omega_r + \chi \sigma_z) a^\dagger a + \frac{\omega_q + \chi}{2} \sigma_z,$$

where, for a two-level qubit,

$$\chi \approx \frac{g^2}{\Delta},$$

and for a transmon with anharmonicity α ,

$$\chi \approx -\frac{g^2 \alpha}{\Delta(\Delta + \alpha)}.$$

The AC Stark shift of the qubit frequency is then

$$\delta\omega_q \approx 2\chi \langle n \rangle,$$

where $\langle n \rangle$ is the average photon number in the resonator. This perturbative result is valid only when $\langle n \rangle \ll n_{\text{crit}}$, ensuring the dispersive approximation remains accurate. The *Stark shift* refers to the change in the energy levels of a quantum system due to the presence of an external electromagnetic field. In the case of a transmon qubit coupled to a resonator, the *AC Stark shift* arises from the oscillating electric field of the resonator photons, leading to a shift in the qubit transition frequency without causing excitation. In the dispersive regime, the shift is given by

$$\delta\omega_q \approx 2\chi \langle n \rangle,$$

where $\delta\omega_q$ is the change in the qubit frequency, χ is the dispersive shift per photon, and $\langle n \rangle$ is the average photon number in the resonator.

Comparison with the unfiltered case:

- This single-mode resonator is assumed to have the same parameters as the fundamental mode of the actual resonator.
- The measured external coupling rate for the filtered qubit is suppressed:
 - By more than two orders of magnitude over a 600 MHz bandwidth.
 - By a factor exceeding one thousand over a 100 MHz bandwidth.

Implications for qubit lifetime:

- The results indicate that, if the qubit frequency is aligned with the filter notch, the lifetime limit T_1 could exceed 10 ms.
- This demonstrates the effectiveness of the Purcell filter in suppressing qubit decay into the output line.

Ramsey Fringe Measurements:

1. Purpose:

- Measure how quickly a qubit loses phase coherence when in a superposition state.
- Detect frequency detuning between the qubit and the drive signal, which appears as oscillations in the measured signal.
- Extract the inhomogeneous dephasing time T_2^* and identify small frequency shifts (e.g., the ac Stark shift).

2. Basic Principle:

- Initialize the qubit in the ground state $|g\rangle$.
- Apply a $\pi/2$ pulse to create a superposition $\frac{|g\rangle + |e\rangle}{\sqrt{2}}$.
- Let the qubit freely evolve for a time τ , accumulating a relative phase depending on the detuning $\delta = \omega_{\text{drive}} - \omega_{\text{qubit}}$.
- Apply a second $\pi/2$ pulse to map the phase information into population.
- Measure the probability of finding the qubit in $|e\rangle$.
- Vary τ to observe oscillations (Ramsey fringes).

3. Step-by-Step Procedure:

1. Initialization: Prepare the qubit in $|g\rangle$ by reset.
2. First $\pi/2$ pulse: Apply a resonant microwave pulse to rotate $|g\rangle$ to the equator of the Bloch sphere:

$$|g\rangle \xrightarrow{\pi/2} \frac{|g\rangle + |e\rangle}{\sqrt{2}}.$$

3. Free evolution: Let the state evolve for time τ . If detuning δ exists, the qubit state vector rotates around the Z-axis at frequency δ .
4. Second $\pi/2$ pulse: Converts accumulated phase into population difference.
5. Measurement: Measure excited-state probability $P_e(\tau)$, which follows:

$$P_e(\tau) \propto \cos^2\left(\frac{\delta\tau}{2}\right) e^{-\tau/T_2^*}.$$

4. Key Observations:

- Oscillation frequency $\rightarrow$ gives detuning δ .
- Decay envelope $\rightarrow$ yields T_2^* , the dephasing time.
- Off-resonant drive $\rightarrow$ causes an ac Stark shift, changing fringe frequency.

Fast Dispersive Readout:

- The experiment uses a low-Q resonator to perform fast dispersive readout of a transmon qubit.
- A 40-ns square probe pulse is applied to the resonator to measure the qubit state.
- A lumped-element Josephson parametric amplifier (JPA) is used in phase-sensitive mode to amplify the reflected probe signal.
- The amplitude and phase of both the probe pulse and the JPA pump are optimized to maximize readout fidelity.
- In this measurement:
 - The first readout prepares the qubit in the ground state $|g\rangle$ via postselection (Postselection is basically throwing away the “bad” runs of an experiment and only keeping the ones where you know the system started (or ended) in the desired state).
 - The second readout is performed without the JPA pump.
- After a readout pulse, the resonator field decays rapidly due to the large external decay rate κ_{ex} :

$$\text{Decay factor} \approx e^{-\kappa_{\text{ex}}\tau} \sim 10^{-7} \quad \text{for } \tau = 60 \text{ ns.}$$

- This rapid decay allows the resonator to reset quickly, enabling fast resumption of qubit operations after readout.
- The amplified signal is delayed because the JPA bandwidth is smaller than that of the probe.
- Integration of each single-shot signal is performed after weighting by the difference between the averaged waveforms.
 - Perform many readouts of the qubit in the ground state $|g\rangle$ and average them to obtain the *average waveform for $|g\rangle$* .

- Perform many readouts of the qubit in the excited state $|e\rangle$ and average them to obtain the *average waveform for $|e\rangle$* .
- Subtract the two averaged waveforms to obtain the *difference waveform*:

$$w_{\text{diff}}(t) = \langle s_e(t) \rangle - \langle s_g(t) \rangle,$$

which highlights the time regions most sensitive to the qubit state.

- For each single-shot readout signal $s_{\text{single}}(t)$, multiply it by $w_{\text{diff}}(t)$ before integrating over time:

$$S_{\text{int}} = \int s_{\text{single}}(t) \cdot w_{\text{diff}}(t) dt.$$

- This weighting functions as a *matched filter*, amplifying signal components that differ between $|g\rangle$ and $|e\rangle$ while suppressing noise-dominated parts.
- The resulting single integrated value S_{int} is then compared to a threshold to decide whether the qubit was in $|g\rangle$ or $|e\rangle$.
- The discrimination threshold for readout is set at zero, allowing clear separation of the two states $|g\rangle$ and $|e\rangle$.

Analysis of QND and Readout Fidelities:

- Let $P(y|x)$ denote the conditional probability that the outcome of the second readout is “y” given that the outcome of the first readout is “x”.
- For the pulse sequences the measured conditional probabilities are:

$$P_a(e|g) = 0.7\%, \quad P_b(g|e) = 3.0\%, \quad P_c(g|g) = 1.0\%.$$

- The QND (quantum non-demolition) fidelity is defined as:

$$Q := 1 - \frac{P_a(e|g) + P_b(g|e)}{2} = 98.1\%.$$

- The readout fidelity is defined as:

$$F := 1 - \frac{P_a(e|g) + P_c(g|g)}{2} = 99.1\%.$$

- Error budget analysis:
 - QND infidelity: $1 - Q = 1.9\%$ consists of:
 - * Back action of the readout: 1.2%.
 - * Internal loss: 0.3%.
 - * Separation error: 0.3%.
 - * External decay: 0.1%.
 - Readout infidelity: $1 - F = 0.9\%$ has only upper bounds for individual error contributions:
 - * State-preparation error: $\leq 0.6\%$.

- * Back action: $\leq 0.6\%$.
- * Internal loss: $\leq 0.3\%$.
- * Separation error: 0.3% .
- * External decay: $\leq 0.1\%$.
- The external decay of the qubit is strongly suppressed in this device, contributing only 0.1% to both QND and readout infidelities.
- The relatively large back-action error is attributed to the strength of the probe pulse used during readout.
- Using the measurement-induced dephasing model, the probe power corresponds to a steady-state resonator photon number of approximately 35 photons.
- This photon number is about 1.5 times the critical photon number:

$$n_{\text{crit}} = \frac{\Delta^2}{4g^2} = 24,$$

where Δ is the qubit–resonator detuning and g is the coupling strength. The *critical photon number* n_{crit} tells you the maximum average number of photons that can be present in the resonator before the dispersive approximation begins to break down.

- For $n < n_{\text{crit}}$: The qubit–resonator interaction is sufficiently linear such that the qubit frequency shift per photon (χ) is constant, enabling high-fidelity dispersive readout.
- For $n \gtrsim n_{\text{crit}}$:
 - * The resonator field strongly dresses the qubit.
 - * The qubit and resonator states begin to mix significantly, breaking the dispersive regime.
 - * Measurement can induce qubit transitions, reducing quantum nondemolition (QND) fidelity.

Fast Unconditional Reset:

- The protocol utilizes the low- Q resonator to rapidly reset the transmon qubit.
- An unconditional all-microwave reset protocol is implemented, which uses only the resonator already present for dispersive readout (no extra hardware).
- This reset protocol targets both:
 - The first excited state $|e\rangle$.
 - The second excited state $|f\rangle$, which may be populated due to leakage during qubit operations.
- Reset is performed by simultaneously driving the following transitions:
 - $|f0\rangle \rightarrow |g1\rangle$
 - $|e0\rangle \rightarrow |f0\rangle$

where the numbers denote the resonator’s Fock states.

- These drives transfer excitations from $|e0\rangle$ and $|f0\rangle$ into $|g1\rangle$.
- The state $|g1\rangle$ rapidly decays into the ground state $|g0\rangle$ due to resonator photon loss.
- The $|f0\rangle \rightarrow |g1\rangle$ transition is a second-order process, requiring a stronger drive compared to the $|e0\rangle \rightarrow |f0\rangle$ transition.
- A strong drive can be applied to the $|f0\rangle \rightarrow |g1\rangle$ transition without heating the setup because its frequency:

$$\omega_{f0g1}/2\pi = 5.611 \text{ GHz}$$

lies outside the stop band of the filter.

- The transmission of this drive signal through the filter is quantified by the external coupling rate at the drive frequency:

$$\kappa_{\text{ex}}(\omega_{f0g1})/2\pi = 11 \text{ kHz}.$$

- Driving through the filter is more efficient than adding a dedicated drive line to the qubit, because in the latter case the coupling rate would need to be less than:

$$\kappa_{\text{ex}}(\omega_{eg})/2\pi = 1.3 \text{ kHz}$$

to avoid reducing the qubit relaxation time T_1 .

- Initial state preparation for qubit reset:
 - The transmon is first prepared in the ground state $|g\rangle$ using a postselecting readout.
 - A π_{ge} pulse is applied to excite the qubit from $|g\rangle$ to $|e\rangle$.
 - Optionally, a π_{ef} pulse is applied to further excite the qubit from $|e\rangle$ to $|f\rangle$.
- After state preparation, the reset drives are applied according to the all-microwave reset protocol.
- The residual excitation is measured as:

$$1 - P_g$$

where P_g is the probability of being in the ground state $|g\rangle$.

- The readout detects any excited state (either $|e\rangle$ or $|f\rangle$) as “e”.
- Accounting for the readout error of 0.5%–0.7%, the following is observed:
 - Residual excitation drops below 1.7% within 100 ns in the worst case (initial state $|f\rangle$).
 - If the transmon starts in the $|g\rangle$ – $|e\rangle$ subspace, residual excitation is less than 0.9%.
- The measured residual excitation remains above the upper bound of the readout error for reset drives longer than 100 ns.
- This excess can be explained by leakage errors:
 - Leakage from π_{ef} pulse and $|e0\rangle \rightarrow |f0\rangle$ reset drive can excite $|f0\rangle$ to $|h0\rangle$.
 - $|h\rangle$ denotes the third excited state of the transmon, which cannot be reset by the current protocol.

- Leakage to $|h\rangle$ can also be caused by the strong $|f0\rangle \rightarrow |g1\rangle$ reset drive, with a coupling strength of 1.2 GHz and detuning of -1.9 GHz from the $|f\rangle \rightarrow |h\rangle$ transition.
- It is expected that these leakage errors can be reduced by optimizing the waveforms of the reset drive pulses.

Conclusion:

Intrinsic Purcell Filter Concept:

- The decay of a qubit through a readout resonator can be suppressed by shifting the output coupler position of the resonator.
- This approach is called an intrinsic Purcell filter. Unlike the conventional band-pass Purcell filter, it achieves stronger suppression without introducing any additional resonance.
- Demonstrated using a coaxial-transmission-line resonator. The coaxial resonator can be combined with a stub cavity (A stub cavity is basically a short section of transmission line or waveguide that is intentionally left open-circuited or short-circuited at one end, creating a resonant cavity for certain frequencies) for bosonic quantum error correction. The same concept can be applied to a coplanar waveguide resonator, which is favorable for large-scale integration.

Advantages of the Low-Q Resonator coupled to long lived qubit:

- Enables fast dispersive readout of a long-lived qubit.
- Allows fast unconditional reset of the qubit.
- The band-stop property of the filter permits efficient off-resonant driving of the qubit.
- Low-Q resonators and efficient off-resonant driving can be used for rapid generation and absorption of itinerant microwave photons. Such photons can mediate internode communication in a quantum network.

Relevance to Quantum Error Correction:

- Fast readout of syndrome qubits reduces idle-time decoherence in data qubits.
 - *What are syndrome qubits?*
 - * In quantum error correction (QEC), there are two types of qubits:
 - **Data qubits** – store the actual quantum information you want to protect.
 - **Syndrome qubits** (or ancilla qubits) – measure error syndromes without directly measuring the data qubits.
 - * Syndrome qubits interact with the data qubits in such a way that errors can be detected without disturbing the quantum state of the data qubits.
 - *Why do we need to read out syndrome qubits quickly?*
 - * The readout of syndrome qubits reveals what kind of error occurred (if any), allowing the appropriate correction to be applied.
 - * If the readout takes too long, the data qubits remain idle during that time.

- * Even when idle (no gates applied), qubits suffer from decoherence:
 - T_1 decay – energy relaxation.
 - T_2 dephasing – loss of phase coherence.
- * In large-scale QEC, idle-time errors can accumulate faster than the error-correction cycle can correct them.
- *How fast readout helps*
 - * Shorter idle time for the data qubits – they can resume useful operations sooner.
 - * More frequent correction cycles – reducing the time errors have to accumulate before being corrected.
 - * Higher effective logical qubit fidelity – errors are detected and corrected before they compound.
- Rapid reset can remove leakage errors in the second excited state, preventing degradation of error-correcting performance.

Considerations for Qubit Control Pulses:

- The effect of the notch-like filter response on π pulses requires further study.
- Current π_{ge} and π_{ef} pulses are calibrated using the half-DRAG technique to minimize phase errors.
- The half-DRAG method can suppress leakage but may not be directly applicable when the drive is applied through a filter.
- Due to this, the qubit is intentionally detuned from the notch frequency in experiments. More advanced pulse-shaping techniques may be necessary if the qubit frequency is aligned closer to the notch.
 - The system uses a filter with a sharp dip (notch) in transmission at a certain frequency. If the qubit frequency is near this notch, control pulses are attenuated and spectrally distorted before reaching the qubit.
 - The π_{ge} pulse flips the qubit between the ground state $|g\rangle$ and the first excited state $|e\rangle$, while the π_{ef} pulse flips between $|e\rangle$ and the second excited state $|f\rangle$.
 - These pulses require precise amplitude, frequency, and shape to function correctly.
 - When the drive passes through a notch filter, the frequency-dependent attenuation and phase shift distort the pulse, making the DRAG calibration less effective.
 - To avoid these issues, the qubit frequency is intentionally detuned from the notch frequency in the experiments.
 - If the qubit frequency must be close to the notch in future experiments, more advanced pulse-shaping or optimal-control techniques will be required to compensate for the filter-induced distortions.
 - The half-DRAG technique (Derivative Removal by Adiabatic Gate) is a pulse-shaping method that minimizes phase errors and suppresses leakage into higher energy states. In unfiltered setups, it performs effectively. In a transmon qubit, the energy levels are not perfectly two-level. There exists a third level $|f\rangle$ close in frequency to the $|e\rangle$ state. A standard microwave pulse tuned to the $|g\rangle \leftrightarrow |e\rangle$ transition has spectral components that can unintentionally drive transitions to $|f\rangle$, causing *leakage*. The DRAG (Derivative Removal by Adiabatic Gate) technique

mitigates this effect by adding a quadrature component to the control pulse.

Full DRAG Technique:

In the IQ representation of the control field:

- * The *in-phase* component $I(t)$ drives the desired $|g\rangle \leftrightarrow |e\rangle$ transition.
- * The *quadrature* component $Q(t)$ is added, proportional to the time derivative of $I(t)$:

$$Q(t) = -\alpha \frac{dI(t)}{dt}$$

where α is a scaling parameter.

- * This quadrature correction shifts the instantaneous drive frequency, suppressing leakage to $|f\rangle$.

Half-DRAG Variant:

The *half-DRAG* method applies only half of the derivative correction compared to the full DRAG value:

$$Q(t) = -\frac{\alpha}{2} \frac{dI(t)}{dt}$$

This is useful when the full DRAG correction overcompensates, introducing additional phase errors. Half-DRAG balances leakage suppression with phase error minimization.

Experimental Implementation:

The typical calibration procedure for half-DRAG is:

1. Select a smooth pulse envelope, e.g., Gaussian or cosine.
2. Calibrate the π -pulse amplitude and duration without DRAG.
3. Measure the leakage probability P_f to $|f\rangle$.
4. Add the derivative quadrature $Q(t)$ with $\alpha/2$ scaling.
5. Fine-tune $\alpha/2$ to minimize both leakage and phase errors (measured via Ramsey experiments or randomized benchmarking).
6. Fix these parameters for repeated use in gate operations.

Doubt: In the first paragraph of fast unconditional reset section a reset is performed by simultaneously driving the following transitions:

- $|f0\rangle \rightarrow |g1\rangle$
- $|e0\rangle \rightarrow |f0\rangle$

Shouldn't this be $|e0\rangle \rightarrow |g1\rangle$. Because we want to transfer excitation to $|g0\rangle$ via photon loss from $|g1\rangle$?

Note : Appendix for this paper is yet to be read

10 Controlling the dynamic range of a JPA

- **Challenges presented:** A key difficulty is controlling the *dynamic range* in relation to the *gain* and *bandwidth*. This limitation means that quantum-limited amplification is usually possible only for signals containing a few photons per inverse bandwidth.
- **Proposed solution:** The authors address dynamic range control in *Josephson parametric amplifiers (JPAs)* by using *Josephson junction arrays*. These arrays help distribute the nonlinearity, improving performance. Both *transmission-line* based JPAs and *lumped-element* based JPAs are analyzed, focusing on:
 - Gain (how much the amplifier boosts the signal)
 - Bandwidth (frequency range over which gain is maintained)
 - Noise (how close the amplifier is to the quantum limit)
 - Dynamic range (range of input powers over which amplification is linear)

From these investigations, the authors derive practical *design criteria* for JPAs.

- Interest in low-noise amplifiers has surged due to advances in quantum optics and superconducting circuits.
- Josephson parametric amplifiers (JPAs) are the most successful quantum-limited detectors in the microwave regime.
- JPAs enable generation of squeezed radiation, quantum feedback, post-selection experiments, displacement measurements of nanomechanical oscillators, and photon correlation studies.
- Despite operating near the quantum limit, JPAs are limited by a small dynamic range, causing gain saturation for large signals.
- Using Josephson junction arrays in JPAs can improve dynamic range and performance.
- The paper reviews parametric amplification principles, bandwidth, noise constraints, and derives strategies for optimized circuit design.
- In quantum optics, *parametric* processes involve a nonlinear refractive medium mixing different light frequencies.
- A coherent pump field modulates the refractive index as a time-varying parameter, stimulating photon generation in modes detuned from the pump frequency.
- The energy for photon creation comes from the pump field.
- The refractive index in optics corresponds to the impedance in electrical circuits; thus, parametric processes at microwave frequencies modulate effective impedance.
- Modulation is done by varying parameters of capacitive or inductive elements over time.
- Josephson junctions, which are dissipationless, are preferred for this purpose due to their nonlinear inductance properties.

- For currents much smaller than the critical current I_C , the Josephson inductance approximates to

$$L \approx L_J \left(1 + \frac{1}{6} \left(\frac{I(t)}{I_C} \right)^2 \right),$$

and applying an AC current leads to four-wave mixing.

- Alternatively, varying magnetic flux through a SQUID loop modulates inductance linearly:

$$L \approx L_J \left(1 + \frac{I(t)}{I_0} \right),$$

resulting in three-wave mixing.

- To control amplification bandwidth and suppress unwanted frequencies, the modulated Josephson inductance is embedded in microwave resonators.
- Recent designs include spatial separation of signal and idler modes and traveling wave amplifiers where amplification occurs along the propagation path.
- Various pump drive schemes have been studied, including single, double, and magnetic flux pumps.
- This work focuses on a single-mode (degenerate) parametric amplifier driven by a single pump tone near its resonance frequency.
- The Josephson Parametric Amplifier (JPA) is a weakly nonlinear oscillator with nonlinearity provided by Josephson tunnel junctions.
- JPAs are typically realized as:
 - Transmission line resonators shunted by a SQUID (allowing tunability of resonance frequency).
 - Lumped element nonlinear oscillators.
- Tunability via SQUIDs is desirable to match the amplifier bandwidth with the signal frequency.
- The relevant Hamiltonian for the parametric amplifier is:

$$H_{\text{JPA}} = \tilde{\omega}_0 A^\dagger A + \frac{K}{2} A^{\dagger 2} A^2,$$

where A is the annihilation operator of the intra-resonator field, $\tilde{\omega}_0$ is the resonance frequency, and K is the Kerr nonlinearity.

- The system dynamics are studied using input-output formalism.
- Without internal losses, all incident power is reflected, and the reflected field's phase ϕ depends on both frequency and probe power for a non-linear oscillator.
- For low probe powers, ϕ remains constant (quasi-linear response).
- At higher powers, the phase ϕ varies significantly, leading to two regimes:

- **Bistable regime:** Two stable solutions for ϕ exist; useful for bifurcation amplifiers and nonlinear dispersive readout.
- **Unique solution regime:** Single stable phase response.
- Focus is on the unique solution regime for linear amplification.
- Amplification occurs because the reflected phase ϕ is highly sensitive to input power near a specific drive frequency and power.
- Small perturbations (e.g., an additional signal) cause large changes in the output field phase, resulting in amplification or deamplification depending on the relative phase with the pump.
- This nonlinear response to a monochromatic drive field qualitatively explains the amplification mechanism.
- A detailed analysis follows to characterize the amplifier response for input signals with finite bandwidth.
- The input-output formalism is used to calculate the nonlinear resonator response of the JPA.
- The nonlinear resonator couples to a transmission line with rate κ , allowing pump and signal propagation.
- The equation of motion for the intra-resonator field A is:

$$\dot{A} = -i\tilde{\omega}_0 A - iKA^\dagger AA - \frac{\kappa + \gamma}{2}A + \sqrt{\kappa}A_{\text{in}}(t) + \sqrt{\gamma}b_{\text{in}}(t),$$

where γ accounts for internal losses.

- Boundary condition for the output field:

$$A_{\text{out}}(t) = \sqrt{\kappa}A(t) - A_{\text{in}}(t).$$

- Input field $A_{\text{in}}(t)$ is decomposed into a strong classical pump $\alpha_{\text{in}}e^{-i\omega_p t}$ and a weak quantum signal $a_{\text{in}}(t)e^{-i\omega_p t}$.

$$A_{\text{in}}(t) = (a_{\text{in}}(t) + \alpha_{\text{in}})e^{-i\omega_p t}, \quad (4)$$

$$A_{\text{out}}(t) = (a_{\text{out}}(t) + \alpha_{\text{out}})e^{-i\omega_p t}, \quad (5)$$

$$A(t) = (a(t) + \alpha)e^{-i\omega_p t}. \quad (6)$$

- Working in a rotating frame at the pump frequency ω_p simplifies the analysis.
- The steady-state classical pump field α satisfies:

$$\left[i(\tilde{\omega}_0 - \omega_p) + \frac{\kappa + \gamma}{2} \right] \alpha + iK\alpha^2\alpha^* = \sqrt{\kappa}\alpha_{\text{in}}.$$

Substituting field modes into steady state solution and multiplying both sides with their complex conjugate one obtains

$$\frac{\kappa}{(\kappa + \gamma)^2} |\alpha_{\text{in}}|^2 = \left(\frac{\omega_p - \tilde{\omega}_0}{\kappa + \gamma} \right)^2 |\alpha|^2 + \frac{1}{4} |\alpha|^2 - \frac{2(\omega_p - \tilde{\omega}_0)K}{(\kappa + \gamma)^2} |\alpha|^4 + \frac{K^2}{(\kappa + \gamma)^2} |\alpha|^6, \quad (6)$$

which determines the average number of pump photons $|\alpha|^2$ in the resonator. The above equation reduces to

$$1 = \delta^2 + \frac{1}{4}n - 2\delta\xi n^2 + \xi^2 n^3, \quad (7)$$

by defining the scale-invariant quantities

$$\delta \equiv \frac{\omega_p - \tilde{\omega}_0}{\kappa + \gamma}, \quad \tilde{\alpha}_{\text{in}} \equiv \frac{\sqrt{\kappa}\alpha_{\text{in}}}{\kappa + \gamma}, \quad \xi \equiv \frac{|\tilde{\alpha}_{\text{in}}|^2 K}{\kappa + \gamma}, \quad n \equiv \frac{|\alpha|^2}{|\tilde{\alpha}_{\text{in}}|^2}. \quad (8)$$

Here, δ is the detuning between pump and resonator frequency in units of the total resonator linewidth, $\tilde{\alpha}_{\text{in}}$ is the dimensionless drive amplitude, and ξ is the product of drive power and nonlinearity in dimensionless units. Finally, n is the mean number of pump photons in the resonator relative to the incident pump power.

- The product of drive power and nonlinearity (ξ) determines dynamics; small nonlinearity can be offset by increasing drive power.
- Gain-bandwidth product does not depend on nonlinearity strength if pump power dominates fluctuations.
- For JPAs, ξ is negative, so analysis focuses on this case, when ξ is positive there is a sign change in δ .
- At the critical value $\xi_{\text{crit}} = -1/\sqrt{27}$, the derivative $\partial n/\partial \delta$ diverges, making the parametric amplifier response extremely sensitive.
- For effective drive powers $\xi/\xi_{\text{crit}} > 1$, there are three real solutions:
 - High and low photon number solutions are stable.
 - Intermediate solution is unstable.
- The system exhibits bifurcation in this regime.
- Critical detuning below which bistability occurs is $\delta_{\text{crit}} = -\sqrt{3}/2$.
- The critical point $(\xi_{\text{crit}}, \delta_{\text{crit}})$ satisfies $\partial \delta/\partial n = 0$ and $\partial^2 \delta/\partial n^2 = 0$.
- Maximal value of n in scale-invariant units is 4, reached at detuning $\delta = 4\xi$.
- Experimentally, system parameters are characterized by measuring the complex reflection coefficient $\Gamma \equiv \alpha_{\text{out}}/\alpha_{\text{in}}$.

- The reflection coefficient is:

$$\Gamma = \frac{\kappa}{\kappa + \gamma} \frac{1}{\frac{1}{2} - i\delta + i\xi n} - 1.$$

- At $\xi = \xi_{\text{crit}}$, plots of $|\Gamma|$ versus δ show:
 - For zero loss $\gamma = 0$, all incident power is reflected ($|\Gamma| = 1$).
 - For $\gamma \approx \kappa$, part of the radiation dissipates into loss modes.
 - At critical coupling $\gamma = \kappa$, all coherent power is transmitted into loss modes at resonance.
 - This corresponds to a symmetrically coupled $\lambda/2$ resonator with unit transmission at resonance.
- Assume photon flux of the signal $a^\dagger a_{\text{in}}$ is much smaller than pump photon flux $|\alpha_{\text{in}}|^2$.
- Thus it is possible to neglect small terms like $Ka^\dagger a\alpha$ compared to leading terms $Ka^\dagger \alpha^2$ and $Ka|\alpha|^2$.
- This yields a linearized equation of motion for the signal mode a in the presence of the pump field.
- To maintain validity for larger input signals, pump amplitude α must be increased, which can be done experimentally by reducing the nonlinearity strength K .
- Substituting into the main equation, the linearized equation of motion for a is:

$$\dot{a}(t) = i(\omega_p - \tilde{\omega}_0 - 2K|\alpha|^2)a(t) - \frac{\kappa + \gamma}{2}a(t) - iK\alpha^2 a^\dagger(t) + \sqrt{\kappa}a_{\text{in}}(t) + \sqrt{\gamma}b_{\text{in}}(t).$$

- Since the equation is linear, solve by Fourier decomposition of all modes, e.g.:

$$a(t) = \frac{\kappa + \gamma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\Delta e^{-i\Delta(\kappa + \gamma)t} a_\Delta$$

- Signal detuning $\tilde{\omega}$ is expressed in units of linewidth $\kappa + \gamma$.
- Fourier substitution leads to the coupled-mode equation:

$$0 = \left[i(\delta - 2\xi n + \Delta) - \frac{1}{2} \right] a_\Delta - i\xi n e^{2i\phi} a_{-\Delta}^\dagger + \tilde{c}_{\text{in},\Delta},$$

where $\tilde{c}_{\text{in},\Delta} = \frac{\sqrt{\kappa}a_{\text{in},\Delta} + \sqrt{\gamma}b_{\text{in},\Delta}}{\kappa + \gamma}$ is the sum of all field modes incident on resonator.

- ϕ is the phase of the intra-resonator pump field, $\alpha = |\alpha|e^{i\phi}$.
- The coupling of modes a_Δ and $a_{-\Delta}^\dagger$ indicates a wave mixing process.
- Rewrite as a matrix equation:

$$\begin{pmatrix} \tilde{c}_{\text{in},\Delta} \\ \tilde{c}_{\text{in},-\Delta}^\dagger \end{pmatrix} = \begin{pmatrix} i(-\delta + 2\xi n - \Delta) + \frac{1}{2} & i\xi n e^{2i\phi} \\ -i\xi n e^{-2i\phi} & i(\delta - 2\xi n - \Delta) + \frac{1}{2} \end{pmatrix} \begin{pmatrix} a_\Delta \\ a_{-\Delta}^\dagger \end{pmatrix}.$$

- Invert the matrix to express intra-resonator quantum field a_Δ in terms of input field $\tilde{c}_{\text{in},\Delta}$:

$$a_\Delta = \frac{i(\delta - 2\xi n - \Delta) + \frac{1}{2}}{(i\Delta - \lambda_-)(i\Delta - \lambda_+)} \tilde{c}_{\text{in},\Delta} - \frac{i\xi n e^{2i\phi}}{(i\Delta - \lambda_-)(i\Delta - \lambda_+)} \tilde{c}_{\text{in},-\Delta}^\dagger,$$

where $\lambda_\pm = \frac{1}{2} \pm \sqrt{(\xi n)^2 - (\delta - 2\xi n)^2}$.

- Using the boundary condition, the input-output relation is:

$$a_{\text{out},\Delta} = g_{S,\Delta} a_{\text{in},\Delta} + g_{I,\Delta} a_{\text{in},-\Delta}^\dagger + \sqrt{\frac{\gamma}{\kappa}} (g_{S,\Delta} + 1) b_{\text{in},\Delta} + \sqrt{\frac{\gamma}{\kappa}} g_{I,\Delta} b_{\text{in},-\Delta}^\dagger.$$

- In the ideal lossless limit $\gamma/\kappa \rightarrow 0$:

$$a_{\text{out},\Delta} = g_{S,\Delta} a_{\text{in},\Delta} + g_{I,\Delta} a_{\text{in},-\Delta}^\dagger.$$

- The gain coefficients are:

$$g_{S,\Delta} = -1 + \frac{\kappa}{\kappa + \gamma} \frac{i(\delta - 2\xi n - \Delta) + \frac{1}{2}}{(i\Delta - \lambda_-)(i\Delta - \lambda_+)},$$

$$g_{I,\Delta} = \frac{\kappa}{\kappa + \gamma} \frac{-i\xi n e^{2i\phi}}{(i\Delta - \lambda_-)(i\Delta - \lambda_+)}.$$

- The output field at a detuning Δ from the pump frequency ω_p is a sum of the input fields at frequencies Δ and $-\Delta$, multiplied by the signal gain factor $g_{S,\Delta}$ and the idler gain factor $g_{I,\Delta}$, respectively. The additional noise contributions introduced via the loss modes b_{in} vanish in the limit $\gamma/\kappa \rightarrow 0$. In the ideal case $\gamma = 0$, the coefficients g_S and g_I satisfy the relation

$$G_\Delta \equiv |g_{S,\Delta}|^2 = |g_{I,\Delta}|^2 + 1,$$

- Consider the lossless case $\gamma = 0$.
- The input signal at detuning Δ is amplified by the power gain $G = |g_{S,\Delta}|^2$ and mixed with frequency components at the opposite detuning from the pump.
- The gain g_S depends on pump-resonator detuning δ , effective drive strength ξ , and signal detuning Δ .
- They show gain G_0 for zero signal detuning as a function of δ and ξ :
 - Maximal gain increases with ξ .
 - Optimal δ shifts approximately linearly with ξ .
 - Gain diverges as $\xi \rightarrow \xi_{\text{crit}}$, but gain is practically limited by stiff pump approximation.
- Gain can be tuned by adjusting ξ and δ , typically to about 20 dB.
- Finite internal losses $\gamma > 0$ can reduce gain below unity.
- Bandwidth is characterized by analyzing gain versus signal detuning Δ :
 - Gain curves approximate Lorentzian lines.

- Increasing gain narrows the amplification bandwidth.
- Gain-bandwidth relation: $\sqrt{G_0}B \approx 1$, where B is the detuning for which gain reaches half its maximum value and is expressed in units of resonator linewidth $\kappa + \gamma$.
- Amplifier bandwidth is roughly the resonator linewidth divided by $\sqrt{\text{gain}}$.
- Noise behavior:
 - Ideal lossless amplifier ($\gamma = 0$) reaches the quantum vacuum noise limit when input modes are cooled to vacuum.
 - Finite losses γ increase added noise by factor $\frac{\kappa + \gamma}{\kappa}$ due to amplified noise from loss modes $b_{\text{in},\Delta}$.
 - Resonance frequency fluctuations (e.g., magnetic flux noise in SQUID loop) cause fluctuations in effective gain and additional noise.
- The derivation assumes the *stiff pump approximation*: the classical pump field is unaffected by additional input signals or quantum fluctuations.
- This approximation assumes that the pump power at the output equals the pump power at the input. This holds when the pump power is much larger than the total output power of amplified signals and vacuum fluctuations.
- To analyze pump depletion, additional terms $2iKa^\dagger a\alpha$ and $iKa^2\alpha^*$ are added.
- The incoming signal is modeled as white noise with average photon number n_{th} per unit time and bandwidth.
- The gain decreases once signal strength exceeds a threshold. The number of input photons n_{th} at which this happens becomes smaller for decreasing ratio $\kappa/|K|$.
- Near the bifurcation point, the pump power scales with $\kappa/|K|$, providing the energy for amplification.
- For small $\kappa/|K|$, gain compression occurs even at $n_{\text{th}} = 0$ due to amplified vacuum fluctuations.
- The *1 dB compression point* $n_{1\text{dB}}$ is defined as the input photon number where the gain G_0 decreases by 1 dB compared to the stiff pump gain.
- 1dB compression point increases roughly proportionally to $\kappa/|K|$ in the limit of $n_{1\text{dB}} \gg 1$.
- Vacuum fluctuations cause gain compression greater than 1 dB for $n_{\text{th}} = 0$ when $\kappa/|K|$ becomes smaller.
- Comparing pump power $P_p = \hbar\omega_p|\alpha_{\text{in}}|^2$ with the power of amplified signal and vacuum fields.

$$P_{\text{out}}^{\gamma=0} = \hbar\omega_p\kappa(2n_{\text{th}} + 1) \int \frac{d\Delta}{2\pi} (G_\Delta - 1),$$

yields the scaling

$$\frac{P_{\text{out}}}{P_p} \propto (2n_{\text{th}} + 1) \frac{|K|}{\kappa} \sqrt{G_0}.$$

- Validity of the stiff pump approximation depends on the ratio P_{out}/P_p .

- Dynamic range increases by reducing the nonlinearity ratio $|K|/\kappa$, which can be engineered using multiple SQUIDs in series.
- The nonlinear resonator with parameters resonance frequency $\tilde{\omega}_0$, Kerr nonlinearity K , and decay rate κ is derived from a full distributed circuit model.
- The system consists of a $\lambda/4$ transmission line resonator terminated by a SQUID loop at the short-circuited end and capacitively coupled to a transmission line.
- The goal is to determine effective parameters $\tilde{\omega}_0$, K , and κ from distributed circuit parameters and compare with lumped-element parametric amplifier models.
- To find the normal mode structure, the capacitive coupling to the transmission line is initially neglected.
- The total Lagrangian L is expressed in terms of the magnetic flux field $\Phi(x)$:

$$L = \int_0^d dx \left(\frac{c}{2} (\partial_t \Phi(x))^2 - \frac{1}{2l} (\partial_x \Phi(x))^2 \right) + E_J \cos \left(\frac{\Phi(d)}{\phi_0} \right),$$

where $\phi_0 = \hbar/(2e)$ is the reduced flux quantum.

- The SQUID self-capacitance is neglected since its plasma frequency is much higher than the resonator frequencies of interest.
- The SQUID is modeled as a single Josephson junction with tunable effective Josephson energy E_J .
- In the linear regime, the SQUID cosine potential is approximated by a quadratic potential:

$$E_J \cos \left(\frac{\Phi(d)}{\phi_0} \right) \approx \text{const} - \frac{1}{2} \frac{\Phi(d)^2}{\phi_0^2}.$$

- The Euler-Lagrange equation applied to the Lagrangian yields the wave equation for the flux field:

$$v^2 \partial_x^2 \Phi(x) - \partial_t^2 \Phi(x) = 0,$$

with phase velocity $v = 1/\sqrt{cl}$.

- The general solution is a sum over normal modes:

$$\Phi(x) = \sum_{j=0}^{\infty} \phi_j \cos(k_j x).$$

- The wavevectors k_j are set by the boundary conditions of the $\lambda/4$ transmission line.
- At the open end ($x = 0$), the current $\partial_x \Phi(x)/l$ must vanish. This is automatically satisfied by using the cosine ansatz.
- At the shorted end ($x = d$), the boundary condition is altered by the Josephson junction.
- To find this boundary condition, they:

1. Discretize the Lagrangian:

$$L = \lim_{n \rightarrow \infty} \sum_{j=1}^n \left[\frac{\Delta x c}{2} (\partial_t \phi_j)^2 - \frac{1}{l} \frac{(\phi_j - \phi_{j-1})^2}{\Delta x^2} - \frac{1}{2} E_J \left(\frac{\phi_n}{\phi_0} \right)^2 \right],$$

where $\phi_n \equiv \phi(x = d)$ and $\Delta x = d/n$.

2. Apply the Euler–Lagrange equation at $x = d$:

$$\frac{1}{l} \partial_x \phi(d) + E_J \frac{\phi(d)}{\phi_0^2} = 0.$$

- Substituting the ansatz into the above equation and comparing coefficients gives the transcendental equation:

$$k_j d \tan(k_j d) = \frac{ld}{L_J}, \quad L_J = \frac{\phi_0^2}{E_J}.$$

- The solutions k_j determine the normal mode structure in the linear regime.
- In the limit $L_J \rightarrow 0$ (SQUID inductance vanishes), the equation reduces to the poles of $\tan(k_j d)$, which corresponds to the modes of a $\lambda/4$ resonator.
- In the absence of the Josephson junction inductance ($L_J = 0$), the transcendental equation reduces to the standard $\lambda/4$ resonator modes:

$$k_j d = \frac{\pi}{2} (1 + 2j), \quad j \in \{0, 1, 2, 3, \dots\}.$$

- When L_J is small but finite ($L_J/(ld) \ll 1$), we can treat its effect as a perturbation.
- Expanding to first order in $(k_j^{(0)} - k_j)d$.
- This yields the relation:

$$k_j \frac{L_J}{ld} = (k_j^{(0)} - k_j),$$

or equivalently:

$$k_j \approx \frac{k_j^{(0)}}{1 + \frac{L_J}{ld}}.$$

- Physical meaning:
 - The Josephson inductance effectively increases the electrical length of the resonator, lowering k_j .
 - The correction factor $\frac{1}{1 + L_J/(ld)}$ scales all k_j values.
- Accuracy:
 - For the **fundamental mode** ($j = 0$), this linearized approximation is accurate even for $L_J/(ld) \approx 0.5$.

- For **higher harmonics** ($j > 0$), the approximation breaks down for much smaller $L_J/(ld)$ because higher modes are more sensitive to the junction impedance.
- Using the normal mode decomposition, the Lagrangian is rewritten as a sum of oscillators coupled only through the SQUID boundary condition.
- For parametric amplification, the phase drop across the junction $\varphi/\phi_0 < 1$ is small, implying the current is small compared to the critical current.
- The SQUID cosine potential is expanded including the first non-quadratic correction:

$$E_J \cos \frac{\varphi}{\phi_0} = \text{const} - \frac{1}{2} E_J \left(\frac{\varphi}{\phi_0} \right)^2 + \frac{1}{24} E_J \left(\frac{\varphi}{\phi_0} \right)^4 + \dots$$

- Substituting the normal mode decomposition into the Lagrangian's Taylor expansion yields:

$$L = \frac{1}{2} \sum_i (\dot{\phi}_i C_i \dot{\phi}_i - \phi_i L_i^{-1} \phi_i) + \sum_{i,j,k,l} N_{ijkl} \phi_i \phi_j \phi_k \phi_l,$$

where

$$C_i = c \int_0^d dx \cos^2(k_i x) = cd \frac{1}{2} \left(1 + \frac{\sin(2k_i d)}{2k_i d} \right),$$

$$L_i^{-1} = L_J^{-1} \cos^2(k_i d) + k_i^2 l \int_0^d dx \sin^2(k_i x) = \frac{(k_i d)^2}{2ld} \left(1 + \frac{\sin(2k_i d)}{2k_i d} \right),$$

and the nonlinear coefficients

$$N_{ijkl} = \frac{1}{24} E_J \phi_0^{-4} \prod_{m \in \{i,j,k,l\}} \cos(k_m d).$$

- The linear part of the Lagrangian is diagonal in the normal mode basis, describing uncoupled LC oscillators.
- The effective resonance frequencies are $\omega_j = k_j v = 1/\sqrt{L_j C_j}$.
- The Hamiltonian is derived by introducing conjugate charge variables $q_i = \partial L / \partial \dot{\phi}_i = C_i \dot{\phi}_i$ and performing a Legendre transform.
- Considering only self-interactions and two-mode interactions, the Hamiltonian is:

$$H = \frac{1}{2} \sum_i (q_i C_i^{-1} q_i + \phi_i L_i^{-1} \phi_i) - 3 \sum_{j \geq i} N_{ijjj} \phi_i^2 \phi_j^2 - \sum_i N_{iiii} \phi_i^4.$$

- In the quantum regime, the variables q_i and ϕ_i become operators with the commutation relation:

$$[\phi_j, q_k] = \delta_{kj}/i.$$

- The Hamiltonian is expressed using normal mode annihilation and creation operators:

$$\phi_j = i\phi_{\text{zpf},j}(a_j^\dagger - a_j), \quad q_j = q_{\text{zpf},j}(a_j + a_j^\dagger),$$

where zero point fluctuations are

$$q_{\text{zpf},i} = \sqrt{\frac{\omega_i C_i}{2}}, \quad \phi_{\text{zpf},i} = \sqrt{\frac{\hbar}{2\omega_i C_i}}.$$

- After performing the rotating wave approximation (removing all terms with an unequal number of creation and annihilation operators) and neglecting small photon number independent frequency shifts due to nonlinear terms (Lamb shifts), the Hamiltonian becomes:

$$H = \sum_i \omega_i a_i^\dagger a_i + \frac{K_{ii}}{2} a_i^\dagger a_i^\dagger a_i a_i + \sum_{j \geq i} K_{ij} a_i^\dagger a_i a_j^\dagger a_j.$$

- The nonlinear coefficients are:

$$K_{ij} = -E_J \frac{\phi_{\text{zpf}}^4}{2\phi_0^4} \cos^2(k_i d) \cos^2(k_j d).$$

- The fundamental mode Kerr nonlinearity $K = K_{00}$ governs the parametric amplification process.
- The terms with $i \neq j$ represent cross-Kerr interactions, coupling different modes.
- Cross-Kerr interactions enable photon number detection in one mode by probing another mode with a coherent field, analogous to dispersive qubit measurements.
- When using an array of SQUIDs instead of a single SQUID, the values of K_{ij} are divided by the square of the number of SQUIDs, reducing the nonlinearity.
- Parametric amplifier bandwidth is proportional to the decay rate κ .
- Devices are designed with a low external quality factor Q by increasing the coupling capacitance C_κ between the transmission line and the resonator.
- Coupling to the environment shifts the resonance frequency $\omega_j \rightarrow \tilde{\omega}_j$, which can be significant for large coupling rates.
- Using effective inductance and capacitances, the corrected resonance frequency and decay rate for the j -th mode are approximately:

$$\tilde{\omega}_j^2 \approx \frac{\omega_j^2}{1 + C_\kappa/C_j} = \frac{1}{(C_j + C_\kappa)L_j},$$

$$\kappa_j \approx \frac{\tilde{\omega}_j^2 C_\kappa^2 R}{C_\kappa + C_j}.$$

- The external quality factor is defined as:

$$Q_j \equiv \frac{\tilde{\omega}_j}{\kappa_j}.$$

- Lumped element JPAs can be realized by shunting a SQUID with a large capacitance C_J .
- Such a resonator is described by the transmon Hamiltonian, where in the deep transmon limit:

$$K \approx \frac{E_C}{\hbar}, \quad \text{and} \quad \tilde{\omega}_0 \approx \frac{1}{\sqrt{L_J C_J}},$$

with charging energy $E_C = e^2/(2C_J)$.

- Coupling rate κ to the transmission line can be independently designed from E_J and E_C by proper capacitive network design.
- The effective Hamiltonian description assumes small phase drop across the Josephson junctions.
- The validity of this small phase drop approximation near the bifurcation point, relevant for parametric amplification, is analyzed in the following section.
- Understanding this is crucial for realizing parametric amplifiers with large bandwidth and dynamic range.
- The dynamic range of the JPA scales with $\kappa/|K|$, and bandwidth increases with κ , making large κ desirable.
- There are limitations on the maximal κ value due to the validity of the quartic expansion of the SQUID cosine potential.
- The Hamiltonian derivation assumes a quartic expansion in the dimensionless flux variable Φ_n/ϕ_0 , where $n \equiv \varphi(x=d)$ is the phase drop across the SQUID.
- To maintain this approximation in the parametric amplification regime, n/ϕ_0 must remain small, i.e., the current through the SQUID should be small compared to its critical current.
- Define the maximal coherent field amplitude $\alpha_{\max,j}$ inside the resonator for mode j as the amplitude at which $\Phi_n = \phi_0$ (critical current reached):

$$\alpha_{\max,j} = \frac{\phi_0}{\phi_{\text{zpf},j}} \frac{1}{2 \cos(k_j d)} = \frac{\phi_0}{\phi_{\text{zpf},j}} \frac{ld}{2L_J k_j d \sin(k_j d)}.$$

- The quartic approximation is valid only if the coherent field amplitude α inside the resonator satisfies $\alpha < \alpha_{\max,j}$.
- The photon number at the bifurcation point is $N_{\text{crit}} = (\kappa + \gamma)/(\sqrt{3}K)$.
- Define the maximal photon number as $N_{\max,j} \equiv |\alpha_{\max,j}|^2$.
- The ratio between the critical photon number and maximal photon number is:

$$\frac{N_{\text{crit}}}{N_{\max,j}} = \frac{8\kappa}{\sqrt{3}} \frac{\hbar}{E_J \cos^2(k_j d)} \frac{\phi_0^2}{\phi_{\text{zpf}}^2}.$$

- To minimize the effect of higher-order nonlinearities, the ratio $\frac{N_{\text{crit}}}{N_{\max,j}}$ should be kept small.

- This ratio has two important limits:

$$\frac{N_{\text{crit}}}{N_{\text{max},j}} = \begin{cases} \sqrt{\frac{8}{3}} Q^{-1} \frac{ld}{L_J}, & \text{for } L_J \ll ld \text{ (distributed element JPA)} \\ \sqrt{\frac{16}{3}} Q^{-1}, & \text{for lumped element JPA} \end{cases}$$

- This result sets clear constraints on the maximally achievable bandwidth and dynamic range of the JPA.
- To keep $\frac{N_{\text{crit}}}{N_{\text{max},j}}$ small, κ (decay rate) must be sufficiently small.
- Increasing the dynamic range by reducing Kerr nonlinearity $|K|$ can be achieved by choosing a small ratio L_J/ld (Josephson to geometric inductance).
- However, increasing geometric inductance ld leads to a larger ratio $\frac{N_{\text{crit}}}{N_{\text{max},j}}$, which is detrimental.
- In lumped element JPAs, L_J and thus K can be made smaller without affecting $\frac{N_{\text{crit}}}{N_{\text{max},j}}$.
- Practically, small L_J must be compensated by a large capacitance C_J to maintain the resonance frequency, which introduces parasitic geometric inductances.
- Building a parametric amplifier with simultaneously large bandwidth and high dynamic range using a single SQUID is challenging.
- A solution is to replace the single SQUID with a series array of M SQUIDs, each having M -times larger Josephson energy, which keeps $\frac{N_{\text{crit}}}{N_{\text{max}}}$ constant while reducing nonlinearity and increasing dynamic range.
- Consider an array of M SQUIDs with each having effective Josephson energy ME_J .
- The array is treated as a lumped element since its spatial extent is small compared to resonance wavelengths.
- The total phase drop n across the array is assumed to be homogeneously distributed as Φ_n/M per SQUID.
- The quadratic term in the Lagrangian scales as:

$$E_J \frac{\Phi_n^2}{2} \rightarrow \sum_{i=1}^M ME_J \frac{(\Phi_n/M)^2}{2} = E_J \frac{\Phi_n^2}{2}$$

meaning it remains constant, preserving the total linear Josephson inductance.

- The quartic term scales as:

$$E_J \frac{n^4}{24} \rightarrow \sum_{i=1}^M ME_J \frac{(n/M)^4}{24} = \frac{1}{M^2} E_J \frac{n^4}{24}$$

leading to a quadratic decrease in Kerr nonlinearity: $K \rightarrow K/M^2$.

- Consequently, the critical photon number scales as $N_{\text{crit}} \propto M^2$.
- The maximal photon number also scales as $N_{\text{max}} \propto M^2$ because the critical current per junction increases by factor M .
- The ratio $\frac{N_{\text{crit}}}{N_{\text{max}}}$ depends only on total Josephson inductance and remains constant while bifurcation power increases quadratically in M .
- Thus, the dynamic range of the JPA can be increased quadratically with M without affecting bandwidth by using a SQUID array instead of a single SQUID.
- This conclusion applies to both transmission line and lumped element JPAs.
- In practice, Josephson energies in the array vary due to:
 - Inhomogeneous coupling to external magnetic flux.
 - Fabrication-induced variations in junction critical currents.
- Future studies could quantitatively analyze the impact of these variations on amplifier performance and tuning range.
- They presented a detailed analysis of Josephson junction based parametric amplifiers (JPAs), covering bandwidth, noise, and dynamic range.
- Established relations between fundamental JPA properties and designable circuit parameters.
- Derived two simple design strategies to optimize JPA performance:
 - Ensure the Josephson inductance contributes sufficiently to the total effective inductance of the resonant circuit.
 - Use SQUID arrays instead of single SQUIDs to increase the pump field strength near the bifurcation point.
- Enhancing the pump field strength via SQUID arrays improves the dynamic range of the JPA.

11 Characterizing quantum microwave radiation and its entanglement with superconducting qubits using linear detectors

What they present:

1. Advances in superconducting quantum circuits now let researchers create special types of microwave light that behave in ways impossible for ordinary classical waves, such as single photons, squeezed states, and quantum superpositions.
2. This work looks at ways to measure the amplitude and phase components (quadratures) of traveling microwave photons, and how those measurements can reveal quantum correlations, including entanglement, between the traveling photons and fixed qubits on a chip.
3. In particular, the authors explain how to fully reconstruct the combined quantum state of a single mode of the photon field together with a two-level qubit, a process known as joint state tomography.
4. The analysis includes realistic cases where the detectors are not perfect. It explains how the measured data connects to different types of quasiprobability functions (such as Wigner) and to statistical averages, for both single-quadrature and dual-quadrature measurement setups.
5. They describe how to use maximum-likelihood estimation to find the most probable quantum state that would produce the measured quadrature data, while ensuring the result obeys the physical rules of quantum mechanics.
6. Finally, they show real experimental results where microwave photon states more complex than just single photons or simple Gaussian shapes were created in the lab and reconstructed using the above methods.

7. First part of the paper:

- Discuss field quadrature detection schemes (measuring the amplitude/phase components of the field) at microwave frequencies.
- Compare them to their optical analogs to highlight the connection between microwave and optical techniques.
- Show how to determine the quantum state of a single mode (one frequency-temporal mode (One specific wavepacket of the electromagnetic field, defined by a fixed carrier frequency and a fixed pulse shape in time, treated as a single quantum harmonic oscillator.)) from these quadrature measurements.

8. Single-channel detection → quasiprobability link:

- Analyze single-channel detection (homodyne-like) at microwave frequencies.
- Show how the results are mathematically related to quasiprobability distributions (Wigner).

9. New tomography method:

- Take measured quadrature probability distributions.
- Use maximum-likelihood estimation (MLE) to directly reconstruct the Fock-space density matrix (matrix in photon-number basis).

- Bypass intermediate steps such as reconstructing the Wigner function first.

10. Experiments beyond single photons:

- Experimentally prepare microwave quantum states more complex than single photons — e.g., multi-photon Fock states, superpositions, and other non-Gaussian states.
- Reconstruct these states using the new tomography method.

11. Two-channel detection insight:

- Show that two-channel microwave detection (heterodyne or dual-homodyne) can be mathematically interpreted as a positive- P function measurement.
- Demonstrate that this holds even when extra classical noise is added by the detection chain.

12. Final part – qubit–field entanglement:

- Extend the tomography techniques to full joint tomography of a localized qubit and a traveling field mode.
- Characterize entanglement between the two by reconstructing the combined density matrix.

Maximum-Likelihood Estimation (MLE) in Quantum State Tomography:

1. Goal In quantum state tomography, the unknown parameters are the elements of the density matrix ρ of the quantum state. We perform many measurements on identically prepared systems and collect a dataset of outcomes. The aim of MLE is:

Find the density matrix ρ that makes the observed data *most probable* according to quantum mechanics.

2. Model space

- In classical statistics, a “model” might be a biased coin or a loaded die.
- In quantum tomography, the “model” is a candidate density matrix ρ .
- Physical constraints:

$$\rho \geq 0 \quad (\text{positive semidefinite}), \quad \text{Tr}(\rho) = 1 \quad (\text{normalized}).$$

3. Predicting measurement outcomes Quantum mechanics predicts the probability of an outcome x via the Born rule:

$$p_\rho(x) = \text{Tr}(\rho \Pi_x),$$

where Π_x is the POVM element (A POVM element (short for Positive Operator-Valued Measure element) is basically the mathematical object that represents one possible outcome of a quantum measurement) for outcome x . Examples:

- For a projective measurement in state $|\psi_x\rangle$, $\Pi_x = |\psi_x\rangle\langle\psi_x|$.
- For homodyne detection at phase ϕ , $\Pi_x = |x_\phi\rangle\langle x_\phi|$.

4. The likelihood function Suppose we perform N independent measurements and obtain:

$$\{x_1, x_2, \dots, x_N\}.$$

The likelihood of ρ given the data is:

$$\mathcal{L}(\rho) = \prod_{j=1}^N p_{\rho}(x_j).$$

5. Log-likelihood It is often easier to maximize the log-likelihood:

$$\log \mathcal{L}(\rho) = \sum_{j=1}^N \log p_{\rho}(x_j),$$

since it turns products into sums and avoids numerical underflow.

6. Optimization problem The MLE problem is:

$$\text{Maximize } \log \mathcal{L}(\rho) \quad \text{subject to} \quad \rho \geq 0, \quad \text{Tr}(\rho) = 1.$$

This is a constrained optimization over the space of physical density matrices.

7. Why constraints matter Without constraints, the solution could yield probabilities matching the data perfectly, but with an unphysical ρ (e.g., negative eigenvalues). The constraints guarantee:

- *Positivity:* All eigenvalues of ρ are non-negative.
- *Normalization:* $\text{Tr}(\rho) = 1$.

Optical and Microwave Frequency Field Detection:

- The radiation field is modeled as a single bosonic mode a arriving at the detector during a specific time window.
- The field normally consists of a continuum of modes.
- To isolate mode a , temporal mode matching is used.
- Temporal mode matching means shaping your measurement so that it is sensitive only to the mode that matches the photon's temporal profile. Mathematically, this is done by integrating the continuous-time field operator with a weighting function that matches the pulse shape:

$$a = \int_{-\infty}^{\infty} dt \psi^*(t) \hat{a}(t)$$

where:

- $\hat{a}(t)$ is the annihilation operator in the time domain (related to $\hat{a}_{\omega}$ by a Fourier transform).
- $\psi(t)$ is the temporal envelope (shape) of the photon pulse.
- The complex conjugate $\psi^*(t)$ ensures proper overlap in quantum field theory.

Physically:

- If your detection scheme integrates the signal using exactly $\psi(t)$, you filter out all other temporal shapes.
- You then obtain a single bosonic mode operator a that behaves like the single mode usually described in quantum optics textbooks.

Full Quantum State Reconstruction via Quadrature Measurements

Full reconstruction means determining the density operator ρ (or an equivalent representation such as the Wigner function) that completely describes the quantum state of the field mode. The density matrix in the Fock (number) basis can be written as:

$$\rho = \sum_{m,n} \rho_{mn} |m\rangle\langle n|.$$

The diagonal elements

$$\rho_{nn} = \langle n|\rho|n\rangle$$

represent the **photon number statistics**, i.e., the probability to find n photons. The off-diagonal elements ρ_{mn} with $m \neq n$ are the **coherences** — complex numbers encoding relative phases and quantum superpositions between different photon-number components. Without coherences, the state is only a classical mixture of Fock states. For example, to distinguish a coherent superposition such as $|0\rangle + |1\rangle$ from an incoherent mixture of $|0\rangle$ and $|1\rangle$, one must access these off-diagonal elements.

This can be achieved by measuring **generalized field quadrature components**:

$$\hat{X}^\phi \equiv \frac{1}{2} (ae^{-i\phi} + a^\dagger e^{i\phi})$$

instead of the photon number operator $a^\dagger a$.

$\hat{X}^\phi$ is a **quadrature operator** — a phase-dependent linear combination of annihilation and creation operators. It is the quantum analogue of a position or momentum coordinate in the optical phase plane. Different ϕ pick out different directions in phase space:

$$\phi = 0 \quad \Rightarrow \quad \hat{X}^0 = \frac{1}{2}(a + a^\dagger) \quad (\text{“position” quadrature}),$$

$$\phi = \frac{\pi}{2} \quad \Rightarrow \quad \hat{X}^{\pi/2} = \frac{1}{2i}(a - a^\dagger) \quad (\text{“momentum” quadrature}).$$

Measuring $\hat{X}^\phi$ requires a **phase reference** (typically a strong local oscillator) so that the detector is sensitive to a specific phase angle ϕ . In optics, this is implemented via **homodyne detection**: the signal is mixed with a local oscillator of phase ϕ , and the photocurrent difference is measured, which is proportional to the quadrature value. Quadrature phase can be tuned by changing the phase of the local oscillator. Operationally, the experiment must be repeated many times with identically prepared copies of the state to build the probability distribution $p(X, \phi)$ of quadrature outcomes for each phase ϕ .

By scanning ϕ , one explores the **full phase space**, gaining access to the off-diagonal elements of the density matrix in the number basis. Quadrature measurements are phase-sensitive and thus detect interference effects, making them sensitive to ρ_{mn} for $m \neq n$. This is why quadrature measurements — unlike photon

counting alone — enable full quantum state reconstruction.

A **local oscillator** (LO) is a reference signal — a strong, phase-controlled electromagnetic wave — that is mixed with a weak signal in order to measure it more precisely. Many detectors (especially for quadrature measurements) do not directly measure the tiny, noisy field of interest. Instead, they detect the *interference* between the signal and a strong reference. The LO provides:

- **Phase reference:** Allows one to choose which quadrature $\hat{X}^\phi$ is measured.
- **Signal amplification:** Through interference, the LO transforms a tiny quantum field into a large photocurrent that is easier to measure.
- In optics: The LO is a bright, coherent laser beam with:
 - The **same frequency** as the signal (or very close, for heterodyne detection).
 - A **well-defined phase** ϕ relative to the signal.
- The LO and signal are combined at a *beam splitter*, producing an interference pattern that carries the quadrature information.
- The difference of the photocurrents at the two outputs is

$$\Delta I \propto \alpha (ae^{-i\phi} + a^\dagger e^{i\phi}) \propto \hat{X}^\phi,$$

where α is the LO amplitude.

- In microwaves: The LO is a strong microwave tone (from a signal generator) at the same or near frequency as the signal.
- The LO is mixed with the weak signal using a *frequency mixer*:
 - If the LO frequency equals the signal frequency → **homodyne detection**.
 - If the LO frequency is slightly different → **heterodyne detection**.
- After mixing, the signal can be *down-converted* to a low frequency (baseband) where it is easier to digitize. A mixer multiplies the incoming microwave signal with a strong LO tone at (or near) the carrier frequency. That multiplication produces sum and difference frequency components. After low-pass or band-pass filtering the high-frequency components are removed and you are left with a baseband (or intermediate frequency) electrical signal whose envelope encodes the quadratures.
- If you use two mixers with LO phases differing by 90 degrees (cosine and sine LO), you obtain in-phase (I) and quadrature (Q) baseband signals — together these give the full complex envelope of the field. Sampling with ADCs: these I and Q analog voltages are digitized by ADCs for storage and further processing (histograms, tomography).
- But because single microwave photons are very weak you need a low-noise linear amplifier before digitization; the added amplifier noise is usually the dominant obstacle to high-fidelity detection and full quantum tomography.

Simultaneous Measurement of Conjugate Quadratures:

Measuring only a single field quadrature $\hat{X}_\phi$ for many ϕ (single-quadrature homodyne scanned in phase) is a standard tomographic method. However, there is an alternative: measure a pair of orthogonal quadratures simultaneously on different outputs of a beam splitter. This provides the complex field amplitude (both real and imaginary parts) in each trial, yielding the full phase-space information required for state reconstruction. **Why extra noise is necessary:**

Since $\hat{X}$ and $\hat{P}$ for the same mode do not commute:

$$[\hat{X}, \hat{P}] \neq 0,$$

any simultaneous measurement must introduce additional noise to satisfy the uncertainty principle. The measurement scheme accomplishes this by coupling the signal to an extra mode (vacuum fluctuations).

Beam-splitter implementation:

A physical beam splitter has two input ports. If only the signal a is injected into one port, the other port is filled by some mode h . In the ideal case, h is in the vacuum state $|0\rangle_h$. For a 50:50 beam splitter, one common convention for the transformation is:

$$a \rightarrow \frac{a+h}{\sqrt{2}}, \quad h \rightarrow \frac{a-h}{\sqrt{2}},$$

where b_1 and b_2 are the output modes sent to detectors. Because h is at least vacuum, it contributes quantum fluctuations to b_1 and b_2 .

Preserving the uncertainty principle:

The quadratures $\hat{X}_{b_1}$ and $\hat{P}_{b_2}$ act on different output modes and thus commute:

$$[\hat{X}_{b_1}, \hat{P}_{b_2}] = 0.$$

This allows simultaneous readout without direct violation of the uncertainty principle. The vacuum contribution from h supplies the extra fluctuations necessary to avoid paradoxes for the original mode a .

Quadrature definitions and complex amplitude:

We define:

$$\hat{X}_{b_1} = \frac{1}{2}(b_1 + b_1^\dagger), \quad \hat{P}_{b_2} = \frac{1}{2i}(b_2 - b_2^\dagger).$$

For the beam-splitter outputs:

$$\hat{X}_{b_1} + i\hat{P}_{b_2} = \frac{1}{\sqrt{2}}(a + h^\dagger).$$

If we scale by $\sqrt{2}$, we obtain exactly:

$$a + h^\dagger.$$

A physical (phase-insensitive) amplifier transforms the input annihilation operator into a larger-amplitude output while necessarily adding noise. That transformation must be included before the beam-splitter/mixing

stage in the microwave chain.

Effect of phase-insensitive amplification:

A generic *phase-insensitive* linear amplifier can be modeled using the standard Caves model as

$$a_{\text{out}} = \sqrt{G} a_{\text{in}} + \sqrt{G-1} h_{\text{amp}}^{\dagger},$$

where:

- $G \geq 1$ is the *power gain* of the amplifier.
- h_{amp} is an additional bosonic mode representing the internal degrees of freedom of the amplifier or the input from the amplifier's "idler" port.
- The term $\sqrt{G-1} h_{\text{amp}}^{\dagger}$ accounts for the *added noise* required by quantum mechanics for phase-insensitive amplification.

Preservation of commutation relations:

The coefficients in Caves equations are chosen so that the canonical commutator is preserved:

$$\begin{aligned} [a_{\text{out}}, a_{\text{out}}^{\dagger}] &= [\sqrt{G} a_{\text{in}} + \sqrt{G-1} h_{\text{amp}}^{\dagger}, \sqrt{G} a_{\text{in}}^{\dagger} + \sqrt{G-1} h_{\text{amp}}] \\ &= G [a_{\text{in}}, a_{\text{in}}^{\dagger}] - (G-1) [h_{\text{amp}}, h_{\text{amp}}^{\dagger}] \\ &= G - (G-1) = 1. \end{aligned}$$

Thus, bosonic commutation relations are maintained.

Physical meaning of h_{amp} :

- In the *quantum-limited* case, h_{amp} is in the vacuum state $|0\rangle$, and the amplifier adds the minimum possible noise allowed by quantum mechanics.
- In a more realistic scenario, h_{amp} may be in a thermal state or otherwise noisy, leading to additional noise beyond the quantum limit.

Combining the amplification transformation with the beam-splitting operation at the mixing stage, and then dividing the output by $\sqrt{G/2}$, we obtain:

$$\hat{S} \equiv a + h^{\dagger} = \hat{X} + i\hat{P}. \quad (7)$$

Here, $\hat{S}$ is the *complex amplitude operator* representing the two conjugate quadratures $\hat{X}$ and $\hat{P}$ as a single complex number.

- The total noise mode h is given by:

$$h = \frac{G-1}{G} h_{\text{amp}} + \frac{1}{G} h_{\text{mix}}, \quad (8)$$

where:

- h_{amp} is the bosonic mode associated with the noise added by the amplifier.
- h_{mix} is the bosonic mode entering from the *open port* of the beam splitter (often in vacuum).
- Physically, $\hat{S}$ encodes the measurement of *both* quadratures ($\hat{X}$ and $\hat{P}$) simultaneously, which can be thought of as measuring the full complex amplitude of the mode.
- In the limit of **large gain** $G \gg 1$:
 - The prefactor $\frac{G-1}{G} \rightarrow 1$ and $\frac{1}{G} \rightarrow 0$.
 - Therefore, the noise contribution from h_{mix} is negligible.
 - The total noise is dominated by the amplifier noise:

$$h \approx h_{\text{amp}}.$$

- Important consequence: once the field is phase-insensitively amplified, at least the *vacuum noise* from h_{amp} is always added, regardless of whether we detect one quadrature or both.

Why measuring both quadratures is natural after amplification:

When the amplifier-added noise dominates the total detection noise, splitting the amplified signal into two detectors (to read both $\hat{X}$ and $\hat{P}$) does not significantly worsen the signal-to-noise ratio (SNR). In the large-gain limit ($G \gg 1$), the penalty from beam-splitter vacuum noise is negligible compared to the amplifier noise. Thus, measuring both conjugate quadratures simultaneously often provides more information with nearly the same effective SNR.

- **Alternative detection scheme:** There exists a measurement approach using *linear amplifiers* that is *ideally noiseless* for *one* quadrature component. This is in contrast to phase-insensitive amplification, which always adds at least the vacuum noise to *both* quadratures.
- **Phase-sensitive amplifier:** The noiseless detection of one quadrature is achieved by using a *phase-sensitive amplifier* instead of a *phase-insensitive* one. In the quantum limit, such an amplifier can be modeled by the *squeezing transformation*:

$$a \longrightarrow \sqrt{G} e^{-i\phi} a + \sqrt{G-1} e^{i\phi} a^\dagger, \quad (9)$$

where:

- $G \geq 1$ is the *power gain* of the amplifier.
- ϕ is a tunable phase parameter controlling the amplified quadrature.
- The $a^\dagger$ term indicates that the device mixes the signal with its conjugate field, producing *squeezing*.
- **Effect on quadratures:** For a properly chosen ϕ , the quadrature operator $\hat{X}_\phi$ is *noiselessly amplified*, while its conjugate quadrature $\hat{X}_{\phi+\pi/2}$ is *deamplified* (squeezed). This selective amplification allows perfect measurement of one quadrature in principle.
- **Connection to optical detection:** This detection scheme is mathematically equivalent to *optical homodyne detection*, where a strong local oscillator is used to measure one quadrature with high precision. In optics it's harder to measure both quadratures at once, but in microwaves it's straightforward, so they naturally use it for reconstructing quantum states.

Quantum State Reconstruction Based on Single-Channel Field Quadrature Detection

- The method focuses on *quantum state tomography* using measurements of the complex field amplitude $\hat{S}$.
- The aim is to estimate the density matrix ρ_a describing the quantum state of the field mode a .
- Experimentally, the process involves:
 - Preparing the same quantum state repeatedly.
 - Performing a series of measurements on these prepared states.
- The measurements provide information about both the diagonal and off-diagonal elements of ρ_a .
- Different measurement sets correspond to specific representations of ρ_a .
- For field quadrature detection, these representations appear as phase-space distributions such as:
 - The Husimi Q function.
 - The Wigner function.
- The discussion connects these functions to statistical moments and the Fock basis representation of the density matrix.
- *Phase space picture of quantum states:* In quantum mechanics, a state can be represented not only by a wavefunction or density matrix, but also as a distribution over phase space (e.g., position and momentum, or in quantum optics, two field quadratures). Common examples include:
 - **Wigner function** – can take negative values.
 - **Husimi Q function** – always non-negative.
- *Definition:* The Husimi Q function for a state ρ is defined as:

$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \rho | \alpha \rangle$$

where $|\alpha\rangle$ is a *coherent state* labeled by the complex amplitude α .

- *Meaning:* $Q(\alpha)$ measures “how much the quantum state looks like” the coherent state $|\alpha\rangle$ for each point α in phase space. A larger $Q(\alpha)$ means the state has a stronger overlap with that coherent state.
- *Properties:*
 - Always positive and smooth.
 - Does not have negative “quantum interference” regions like the Wigner function.
 - Is a *blurred* version of the Wigner function due to convolution with vacuum noise.
- *Usefulness:* The Q function is easy to interpret experimentally and is often used when measurements are noisy or have limited resolution.

A. Phase-space distributions:

- **Coherent states are not orthogonal:** The overlap between two coherent states $|\alpha\rangle$ and $|\beta\rangle$ is

$$\langle\alpha|\beta\rangle = e^{-\frac{1}{2}|\alpha|^2 - \frac{1}{2}|\beta|^2} e^{\alpha^*\beta},$$

which is nonzero for $\alpha \neq \beta$. This means coherent states “blur” into each other rather than being orthogonal like Fock states $|n\rangle$.

- **Overcompleteness:** Despite their nonorthogonality, coherent states form an *overcomplete basis* for the Hilbert space of a single bosonic mode. This allows any state to be expressed as a weighted sum or integral over coherent states.
- **Expansion of the density matrix:** Any quantum state with density matrix ρ_a can be written as

$$\rho_a = \int_{\mathbb{C}} P_a(\alpha) |\alpha\rangle \langle\alpha| d^2\alpha,$$

where:

- $P_a(\alpha)$ is the **Glauber–Sudarshan P -function**, giving the weight of each coherent state. It uniquely represents the density matrix as a distribution in phase space.
 - $|\alpha\rangle \langle\alpha|$ is the **projector** onto the coherent state $|\alpha\rangle$.
 - $d^2\alpha$ denotes integration over the complex plane (phase space).
 - The density operator ρ_a is Hermitian, and the projectors $|\alpha\rangle \langle\alpha|$ are also Hermitian. Therefore, the coefficients $P_a(\alpha)$ must be real when interpreted as generalized coefficients. However, $P_a(\alpha)$ need not be nonnegative.
 - **Singularities:** For many strongly quantum (nonclassical) states, the P -function is not a regular function, but rather a distribution involving Dirac delta functions and their derivatives.
- **Why this works despite nonorthogonality:** The overcompleteness of coherent states ensures that such an expansion is always possible, even though the states are not orthogonal. The P -function is chosen so that it directly reproduces *normally ordered* expectation values of field operators, which is particularly important in quantum optics.
 - **Physical meaning:** This expansion represents ρ_a as a mixture of classical-like coherent states with weights given by $P_a(\alpha)$. If $P_a(\alpha)$ were a well-behaved positive probability distribution, the state would be classical. However, for nonclassical states $P_a(\alpha)$ can take negative values or even become singular (e.g., involving Dirac delta functions and their derivatives).

Conclusion

- The authors propose schemes to characterize a single radiation-field mode and its entanglement with a two-level system.
- These schemes are based on **linear amplification** and **quadrature detection**.
- They consider two types of detection:
 - **Single-channel detection:** measures one quadrature at a time.

- **Two-channel detection:** measures both quadratures simultaneously, allowing direct measurement of the **positive P function** even when the amplifier adds significant noise.
- For both photon-field tomography and joint qubit–field tomography, they use **maximum-likelihood estimation** that incorporates the full measured quasiprobability distributions.

Doubt:

- Two channel microwave detection can be interpreted as a positive P function measurement.

Note: This paper is yet to be read completely.

12 Hybrid quantum systems with circuit quantum electrodynamics

The review paper has been read